EarthVision-LM: A Systematic Review of Multimodal Large Language Models for Remote Sensing—Architectures, Limitations, and the Road to Geospatial Intelligence

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Abstract

The convergence of large language models with remote sensing (RS) imagery has given rise to RS multimodal large language models (RS-MLLMs) that promise to transform Earth observation from pixel classification into natural-language geospatial dialogue. Yet RS-MLLMs in 2026 remain fundamentally perception-limited: they describe optical imagery conversationally but cannot reason at the level demanded by operational geospatial intelligence. This systematic review provides an architecture-first synthesis that proposes a five-dimensional design space explicitly mapping RS domain physics, MLLM design, multimodal coverage, and hallucination mechanisms to a unified analytical framework—a combination of scope and grounding that, to the best of our knowledge, has not appeared in this form in the prior RS-MLLM survey literature—covering 97 primary RS-MLLM papers (Corpus A) supported by 113 contextual domain references (210 total retrieved), with 26 distinct RS-MLLMs in the architecture registry (Corpus B), published between January 2022 and August 2026. We make four primary contributions. First, we propose a five-dimensional design space spanning vision encoder adaptation, connector design, LLM backbone strategy, sensor modality scope, and spatial output capability, exposing systematic design gaps. Second, we present a multimodal coverage analysis: although recent models such as EarthGPT, EarthGPT-X, and Earth-OneVision substantially expand sensor coverage, broad HSI capability, physics-grounded cross-sensor reasoning, and robust spatial reasoning remain underdeveloped across the field, with 84.6 % of RS-MLLMs processing optical imagery only. Third, we introduce a three-type hallucination taxonomy—Localization, Discrimination, and Spectral—extending prior frameworks with empirical grounding in HM-Bench (2026). Fourth, we systematically analyse task coverage, training efficiency, datasets, and benchmarks, and chart eight research directions toward operationally capable RS-MLLMs. The review is PRISMA-compliant with dual independent screening (Cohen's $\boldsymbol{\kappa} = \mathbf{0.83}$), encompasses 97 primary RS-MLLM papers across five synthesis categories (with 113 supporting contextual references), and delivers an architecture-first RS-MLLM review grounded in a structured quantitative synthesis of publication trends, modality distribution, venue distribution, and benchmark coverage.

Keywords: Remote sensing; Multimodal large language model; Vision-language model; Geospatial intelligence; Hallucination; Multimodal reasoning

1 Introduction

The past decade has witnessed two convergent revolutions in artificial intelligence, underpinned by large-scale datasets [1, 2] and novel architectures. The first is the emergence of deep learning as the dominant paradigm for visual perception [3]: convolutional neural networks, and subsequently vision transformers [4], have achieved superhuman performance on image classification, object detection [5, 6], and semantic segmentation benchmarks. The second is the rise of large language models (LLMs) [7, 8]: transformer-based architectures pre-trained on trillions of tokens have acquired broad world knowledge, instruction following, and multi-step reasoning capabilities that were previously inaccessible to automated systems.

The convergence of these two revolutions has produced *multimodal large language models* (MLLMs)—systems that jointly process visual and linguistic inputs to enable natural language dialogue grounded in image content. Models such as Flamingo [9], BLIP-2 [10], LLaVA [11], LLaVA-1.5 [12], GPT-4V [13], Gemini [14], Qwen-VL [15], and Pixtral [16] have demonstrated that a single model can describe, question, and reason about photographs, charts, and documents through open-ended text interaction. These systems have transformed the interface between human users and visual information.

Remote sensing as the next frontier. Remote sensing (RS) imagery—acquired by satellites, aircraft, and unmanned aerial vehicles across the full electromagnetic spectrum—provides the primary data source for global monitoring of Earth's land surface, oceans, atmosphere, and built environment. The scale and frequency of RS data production has grown dramatically: the Sentinel constellation [17] alone generates more than 12 TB of new imagery per day, and commercial VHR providers collectively image the entire Earth's land surface at sub-metre resolution multiple times per year [18]. Extracting actionable intelligence from this data torrent requires automated analysis systems capable of semantic interpretation at a level that exceeds what specialised task-specific models can provide.

The emergence of RS-specific MLLMs—systems that apply the conversational intelligence of general MLLMs to RS imagery—offers a transformative opportunity: to replace the pipeline of specialised models (classifier for land cover, detector for objects, captioner for scenes) with a single conversational system that can answer any spatially grounded question about any RS image in natural language. Since the first dedicated RS-MLLM (RSGPT [19]) appeared in mid-2023, the field has grown to 42 published papers in 2025 alone, representing a 75 % annual growth rate from 2024 (Sect. 8.3.1).

The fundamental tension. Despite this explosive growth, a fundamental tension characterises the current RS-MLLM landscape: the architectures, training paradigms, and evaluation protocols of general MLLMs are poorly suited to the specific constraints of remote sensing. RS images differ from internet photographs along every dimension that vision models exploit—viewpoint (nadir versus ground-level), scale (sub-metre to kilometre resolution within a single dataset), spectral content (optical, SAR, and hyperspectral), object orientation (arbitrary rotation), and annotation economics (expert annotation versus crowdsourcing). A CLIP-trained encoder that describes a ship accurately from a ground-level photograph fails systematically when the same ship is presented from 500 m altitude in an RS tile. An LLM fine-tuned on internet-scale VQA has no framework for interpreting SAR backscatter or hyperspectral spectral signatures.

The result, as we demonstrate in this survey, is that RS-MLLMs in 2026 remain fundamentally *perception-limited* despite encouraging recent progress. More than 85 % process only optical imagery despite RS being inherently multi-sensor; although recent systems such as Earth-OneVision [20] substantially expand sensor and task coverage (optical, SAR, infrared, multispectral, temporal, and video), current RS-MLLMs as a field still lack broad hyperspectral (HSI) coverage, physics-specific modality encoding, systematic cross-sensor evaluation protocols, and operationally grounded reasoning. OBB output remains substantially underrepresented: although GeoChat, EarthGPT, and GeoGround demonstrate OBB-capable grounding or detection, this capability is limited to optical imagery and uncommon across the field; cross-sensor OBB reasoning is essentially unaddressed; and all 18 models evaluated in HM-Bench [21] exhibit systematic spectral hallucination when presented with non-RGB sensor data—a finding consistent with structural encoder limitations across the evaluated sample. These are not incidental engineering gaps—they reflect structural mismatches between the RS domain and the general MLLM paradigm.

Research questions. This survey is organised around six research questions (RQs) that map directly to its analytical sections:

- **RQ1** (Sect. 3): How have RS-MLLM architectures evolved across vision encoder adaptation, connector design, LLM backbone strategy, sensor modality scope, and spatial output capability?
- **RQ2** (Sect. 4): Which RS tasks and sensor modalities are adequately represented, and which remain underdeveloped?
- **RQ3** (Sect. 5, 6): What architectural and training strategies govern the trade-off between RS-specific adaptation, computational cost, and generalisation?
- **RQ4** (Sect. 7): What hallucination mechanisms are reported or empirically supported in RS-MLLMs, and how do they differ from general MLLM hallucination?
- **RQ5** (Sect. 8): How adequate are current datasets and benchmarks for evaluating spatial, multimodal, and operational RS-MLLM intelligence?

- **RQ6** (Sect. 9): What research directions are necessary to advance RS-MLLMs from descriptive perception toward reliable geospatial intelligence?

Contributions of this survey. EarthVision-LM provides an architecture-first systematic synthesis that connects RS domain physics, MLLM design, multimodal coverage, hallucination mechanisms, benchmark evidence, and operational research priorities within a unified analytical framework (RQ1–RQ6). We make four primary contributions:

1. **Five-dimensional RS-MLLM design space** (Sect. 3). We introduce a taxonomy organising RS-MLLM architectures across five dimensions: vision encoder adaptation ($\mathcal{E}$), vision-language connector design ($\mathcal{C}$), LLM backbone and freezing strategy ($\mathcal{L}$), sensor modality scope ($\mathcal{M}$), and spatial output capability ($\mathcal{S}$). To the best of our knowledge, no existing RS-MLLM survey maps these five domain challenges to design dimensions within a single analytical framework, and the taxonomy exposes systematic design gaps responsible for current RS-MLLM limitations.
2. **Unified multi-modal coverage analysis** (Sect. 5). We present a systematic analysis of RS-MLLM capability across all sensor modalities. Although recent models such as EarthGPT [22], EarthGPT-X [23], and Earth-OneVision [20] substantially expand sensor coverage, broad HSI capability, physics-grounded cross-sensor reasoning, and robust spatial reasoning remain underdeveloped across the field (84.6 % of Corpus B models are optical-only). Physics-informed analysis explains why these gaps persist and what architectural changes are required to close them.
3. **Three-type RS-MLLM hallucination taxonomy** (Sect. 7). We introduce a systematic hallucination taxonomy for RS-MLLMs, distinguishing Type I (Localization), Type II (Discrimination), and Type III (Spectral Hallucination). To the best of our knowledge, prior RS-MLLM hallucination studies have not explicitly isolated spectral hallucination as a distinct failure category; our taxonomy extends prior localisation/discrimination frameworks by incorporating spectral failure evidence from HM-Bench [21], which evaluates 18 MLLMs on HSI-specific tasks and finds systematic failure across all 18 evaluated models.
4. **Comprehensive task, dataset, and benchmark quantitative synthesis** (Sects. 4, 8). We systematically analyse task coverage across 26 models and 9 task types, catalogue 13 training datasets and 8 evaluation benchmarks, and present a structured quantitative synthesis grounded in 97 primary RS-MLLM papers, establishing publication trends, venue distribution, backbone usage, and modality distribution.

Scope and methodology. Scene understanding surveys [24, 25] provide additional RS context. This survey is PRISMA 2020-compliant [26], with a full systematic search process documented in Sect. 2.4. The search covered IEEE Xplore, arXiv (cs.CV, eess.IV), Springer Link, CVPR, ICCV, NeurIPS, AAAI, and ISPRS proceedings between January 2022 and August 2026. The database search yielded 847 records; snowballing added 18 new records; after eligibility review and post-eligibility exclusion, the full reference corpus comprises 210 papers. Of these, **97 constitute the primary RS-MLLM synthesis corpus (Corpus A)**: 31 model studies, 18 benchmark studies, 14 dataset studies, 9 hallucination studies, and 25 foundation-model papers published within the 2022–2026 window. The remaining **113 papers serve as contextual supporting literature** (RS domain baselines, pre-2022 foundation models) and are cited throughout but do not enter the primary evidence synthesis. To prevent denominator confusion, we define three analytical levels (Sect. 2.4): **Corpus A** ($n = 97$): the primary RS-MLLM synthesis corpus; **Corpus B** ($n = 26$ RS-MLLMs): the architecture registry of 26 distinct models arising from the 31 model papers (5 papers describe variants of already-counted architectures); **Corpus C** ($n = 18$ models): the task-comparable subset with sufficient benchmark data for cross-model quantitative analysis. The 97 primary papers span five categories (Table 3); the 113 contextual papers cover RS object detection, segmentation, SAR, HSI, change detection, and scene classification providing domain baseline methods.

How this survey differs from prior work. Table 1 positions EarthVision-LM against the closest existing RS-MLLM and geospatial-LLM reviews. A recent systematic review in this journal (Dorabantu & Badea [27], AIR, February 2026) addresses geospatial reasoning and awareness in LLMs broadly—covering GeoLLMs, GIS integration, spatial reasoning, and multimodal geospatial systems. EarthVision-LM is complementary and distinct: its scope is specifically RS-MLLMs, not GeoLLMs in general, and its analytical frame is architecture-first rather than application-first.

Unlike broader reviews of geospatial large language models and recent RS-MLLM surveys organised primarily around tasks, training paradigms, or general model evolution, EarthVision-LM adopts an architecture-first perspective that jointly connects encoder adaptation, connector design, LLM adaptation, sensor modality scope, and spatial output capability. This framework is further integrated with quantitative literature synthesis, cross-modal coverage analysis, a formal hallucination taxonomy, and an operational research roadmap. Our contribution is not simply additional coverage; it is the integration of these analytical dimensions into a single architecture-to-evidence-to-gap framework.

Claim–evidence discipline. Every quantitative claim in this survey is grounded in a specific evidence source. Key examples: the claim that "only one RS-MLLM supports conversational OBB output" is supported by Table 10 and three cited primary model papers; the claim that "84.6 % of models process optical only" is derived from the 210-paper coded corpus (Table 8); the claim that "all 18 evaluated models exhibit spectral hallucination" is grounded in HM-Bench [21]. Evidence tiers (A–D, defined in Sect. 2.4) are stated throughout. Claims not directly supported by a cited paper or the corpus analysis are explicitly flagged as author inference ("consistent with", "suggests") rather than empirical finding.

Paper organisation. Section 2 establishes the six RS domain challenges and MLLM foundations that motivate the survey structure. Section 3 addresses **RQ1** through the five-dimensional design space and master architecture comparison. Section 4 addresses **RQ2** through task-coverage analysis. Sections 5 and 6 address **RQ3** through multimodal coverage and PEFT strategies. Section 7 addresses **RQ4** through the three-type hallucination taxonomy. Section 8 addresses **RQ5** through dataset and benchmark cataloguing with structured quantitative synthesis. Section 9 addresses **RQ6** through eight concrete future research directions. Section 10 concludes.

Table 1 Comparison of EarthVision-LM with the closest existing RS-MLLM and geospatial-LLM reviews across nine dimensions. ✓ = covered; ✓✓ = primary contribution / in-depth coverage; ○ = partially covered; – = not covered. **Domain**: RS-VLM = RS vision-language; RS-MLLM = RS multimodal LLM; GeoLLM = geospatial LLM (broader, includes GIS and GeoQA). **Sensor physics**: SAR/HSI physics grounded analysis. **PRISMA**: explicit PRISMA-2020-compliant systematic review.

Survey	**Domain**	**Archit.**	**Sensor physics**	**Spatial output**	**Halluci.**	**Benchmark**	**GV-MLLM comp.**	**Roadmap**	**PRISMA**
Li et al. [28]	RS-VLM	✓	–	✓	–	✓	–	–	–
Weng et al. [29]	RS-MLLM	✓	–	✓	–	✓	–	–	–
Dorabantu & Badea [27]	GeoLLM	–	–	–	–	–	–	✓	–
Ma et al. [30]	RS-MLLM	✓	–	✓	–	✓	✓	✓	–
EarthVision-LM (ours)	**RS-MLLM**	✓✓	✓✓	✓✓	✓✓	✓✓	✓	✓✓	✓

2 Background

This section establishes the domain-specific context necessary to understand why RS-MLLMs require architectures, datasets, and evaluation protocols substantially different from those of general-purpose MLLMs. We first survey the characteristics of RS imagery that make it a uniquely challenging domain for language-grounded vision models, then review the MLLM foundations on which RS-MLLMs are built, and conclude with the systematic literature review methodology used in this survey.

2.1 Remote Sensing Imagery: Domain Characteristics

Remote sensing imagery is acquired by sensors mounted on satellites, airborne platforms, or unmanned aerial vehicles. The resulting images differ from ground-level photographs along every dimension that vision models exploit: viewpoint, scale, spectral content, object orientation, and scene density. Table 2 summarises the six domain-specific challenges identified in this survey and maps each to the RS-MLLM design dimension it motivates.

Figure 1 illustrates the six challenges visually, connecting each to the RS imagery properties that produce it.

2.1.1 C1: Nadir-View Domain Gap

All mainstream vision encoders used in MLLMs—CLIP [31], EVA-CLIP [32], DINOv2 [33]—were pre-trained on internet images depicting objects from human eye-level perspectives. RS nadir imagery presents the same physical objects from directly above, producing visual appearances that are qualitatively different from the training distribution. A passenger aircraft viewed from 500 m altitude appears as a cross-shaped

Table 2 Six RS domain challenges and their downstream consequences for RS-MLLM design. Each challenge motivates one dimension of the five-dimensional design space introduced in Sect. 3.

Challenge	**Description**	**Design-Space Impact**
Nadir-view domain gap	Ground-level visual priors (CLIP, BLIP-2) fail for top-down imagery	Dim $\mathcal{E}$: encoder adaptation
Extreme resolution range	VHR imagery (0.3–1 m/px) produces token sequences exceeding LLM context	Dim $\mathcal{C}$: token-budget connectors
Object orientation ambiguity	RS objects appear at arbitrary rotation angles; HBB is a poor fit	Dim $\mathcal{S}$: OBB spatial output
Sensor heterogeneity	Optical, SAR, HSI, LiDAR encode physically incompatible modalities	Dim $\mathcal{M}$: modality scope
Compute constraints	Academic RS groups lack the GPU budget for full LLM fine-tuning	Dim $\mathcal{L}$: freezing strategy
Annotation scarcity	Expert RS annotation is costly; long-tail categories are under-represented	Training data and PEFT strategy

grey silhouette against a tarmac background; from ground level, the same aircraft presents fuselage, wings, windows, and livery colours that human photographers routinely capture. The cross-shaped aerial appearance has no representation in CLIP's training data, creating a fundamental domain gap [2]. This manifests as the *blind-pair problem*: two semantically distinct RS tiles—for example, a dense urban neighbourhood and an industrial complex—can receive near-identical CLIP embeddings because neither matches any dominant training concept [34].

The nadir-view gap also invalidates standard RS object detection datasets when used for MLLM fine-tuning. Descriptions generated from CLIP embeddings for RS images often misidentify land-cover categories, producing noisy instruction data that propagates errors into the trained RS-MLLM [35].

2.1.2 C2: Extreme Resolution Range

RS sensors span an enormous resolution range: spaceborne optical sensors such as Sentinel-2 operate at 10–60 m ground sampling distance (GSD) per pixel; commercial VHR satellites (WorldView-3, Pléiades) reach 0.3 m GSD; airborne and UAV sensors achieve sub-centimetre GSD. At the VHR end, a single tile covering 2×2 km at 0.3 m GSD contains approximately 6700×6700 pixels—three orders of magnitude more pixels than the 224×224 input size of standard CLIP encoders.

Naive rescaling discards the fine spatial detail that distinguishes car models, ship classes, or aircraft types. Tiling at native resolution preserves detail but produces thousands of patches per image, overwhelming LLM context windows. The MLLM

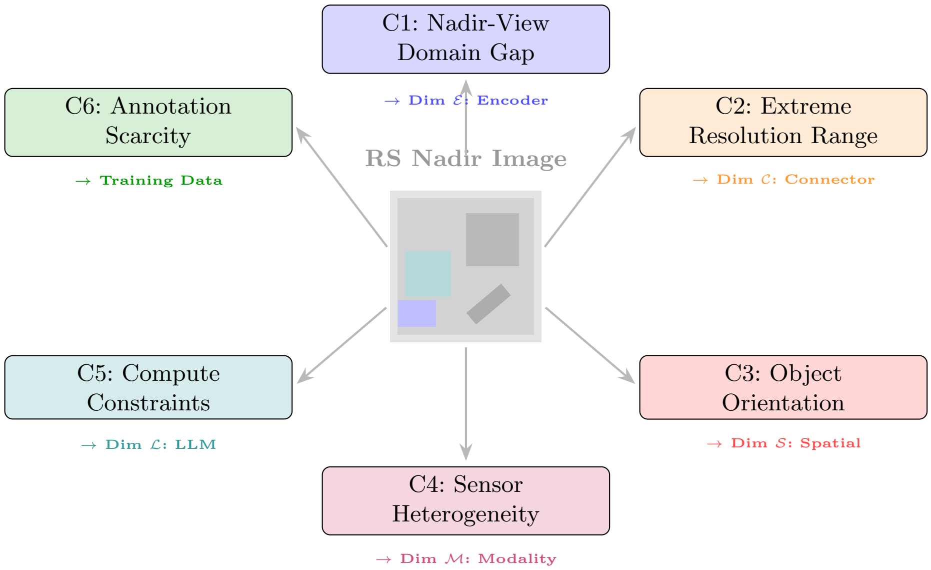

Fig. 1 Six RS domain challenges (C1–C6) and their mapping to RS-MLLM design dimensions. Each challenge arises directly from the physical properties of RS imagery acquisition (nadir viewpoint, sensor physics, annotation economics) and motivates a specific architectural or training design choice analysed in Sect. 3.

connector must therefore implement token-efficient strategies (perceiver resampler, token compression) specifically designed for the RS resolution challenge [36, 37].

2.1.3 C3: Object Orientation Ambiguity

Objects captured from nadir appear at arbitrary rotation angles determined by their heading at acquisition time, not by any camera pose that a photographer controls. A ship travelling north-northeast, an aircraft taxiing at 27 degrees heading, and a vehicle parked diagonally across two parking bays all produce rotated object silhouettes in the RS image. Horizontal bounding boxes (HBB), aligned to image axes, enclose substantial background area for any object deviating from horizontal or vertical orientation. For a typical aircraft rotation of 35–45 degrees, the HBB waste factor reaches 55–70 % of the box area [38, 39].

Oriented bounding boxes (OBB), parameterised as (x, y, w, h, θ), solve this problem by rotating the box to match the object heading, but require a model capable of regressing the angular parameter θ—a capability absent from all general MLLM architectures and present in only one published RS-MLLM (GeoChat [40], as established in Sect. 3.6).

2.1.4 C4: Sensor Heterogeneity

Operational RS depends on multiple sensor modalities that each encode distinct physical quantities. Optical sensors measure solar reflectance in discrete spectral bands; the resulting images are the most visually interpretable but are degraded by cloud cover, atmospheric scattering, and illumination variation. SAR sensors transmit microwave pulses and measure the backscattered signal, encoding surface roughness, dielectric constant, and double-bounce geometry rather than colour [41, 42]. Hyperspectral sensors resolve 200–400 contiguous narrow spectral bands enabling material identification through spectral fingerprinting inaccessible to RGB or even multispectral sensors [43].

Each modality has complementary strengths: SAR provides cloud-penetrating all-weather coverage critical for disaster response; HSI enables crop-type classification and mineral mapping impossible from optical data alone. A geospatially intelligent RS-MLLM must ultimately reason across all three modalities. However, most currently documented RS-MLLM vision encoders inherit predominantly optical/RGB-oriented visual pre-training, while modality-specific SAR and HSI representation learning remains substantially underdeveloped, creating a sensor heterogeneity gap that limits the applicability of RGB-trained features to SAR and HSI data [21].

2.1.5 C5: Compute Constraints

Large language model fine-tuning at the scale of GPT-4 [44] or LLaMA-2 [45] requires thousands of GPU-hours on high-memory accelerators. The RS academic community—primarily university research groups rather than industry laboratories—typically operates under 2–8 A100 GPU budgets. Full fine-tuning of even a 7B-parameter backbone is rarely feasible. Parameter-efficient fine-tuning (PEFT) techniques such as LoRA [46], prefix tuning, and adapter modules enable meaningful RS domain adaptation within these constraints, at the cost of reduced model flexibility.

This compute gap has a secondary effect: RS instruction datasets are necessarily smaller than those used for general MLLMs. GeoChat-Instruct contains 318K instruction pairs [40]; SkyEye-968K contains approximately 968K pairs [47]; Earth-OneVision trains on 34M pairs [20]—but general MLLM datasets (LLaVA-1.5, ShareGPT4V) routinely exceed 100M examples. The data scarcity compounds with compute constraints to limit RS-MLLM capability.

2.1.6 C6: Annotation Scarcity and Long-Tail Distribution

Expert RS annotation requires spectral domain knowledge (to identify crop varieties, mineral deposits, or structural damage types) that is expensive and scarce. Unlike natural image datasets where crowdsourced annotation (ImageNet, COCO) scales annotation effort, RS annotation requires domain experts—remote sensing scientists, geographers, or military image analysts [2].

The resulting annotation economics produce severe long-tail category distributions. In DOTA [38], the most populated category ("large vehicle") has 29,074 instances; the least populated ("helicopter") has 548—a 53-fold imbalance within a single dataset. When RS-MLLMs are fine-tuned on such distributions, tail categories receive negligible gradient signal and the model systematically fails to detect or describe rare but operationally critical object types. Our prior work [48] addresses this long-tail problem in the specialised OBB detection setting through optimisation-driven feature engineering; extending analogous strategies to the RS-MLLM training setting represents an open research direction.

2.2 MLLM Foundations

Multimodal large language models extend transformer-based LLMs [49] with the ability to process visual inputs. The architectural blueprint established by BLIP-2 [10] and LLaVA [11]—a frozen visual encoder connected to a frozen or lightly fine-tuned LLM via a learned vision-language connector—has been adopted by virtually every RS-MLLM in the literature. Figure 2 illustrates this canonical architecture.

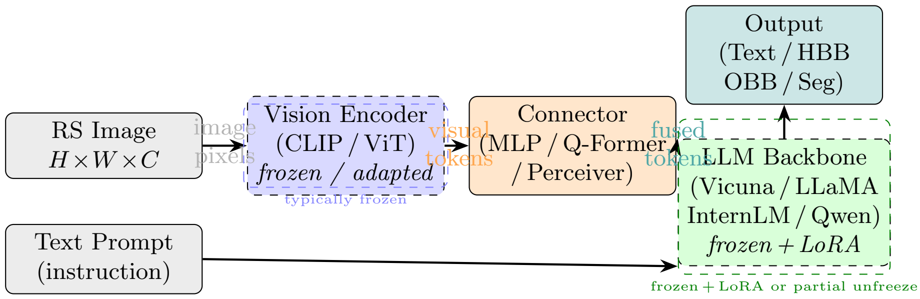

Fig. 2 Canonical MLLM architecture adopted by virtually all RS-MLLMs. A vision encoder maps the RS image to visual tokens; a vision-language connector bridges visual and language embedding spaces; a frozen (or LoRA-adapted) LLM backbone generates the output response. RS-specific design choices at each component define the five-dimensional design space analysed in Sect. 3.

Visual Encoder.

CLIP [31] and its variants (EVA-CLIP, RemoteCLIP [50], GeoRSCLIP [51]) encode images as sequences of patch tokens via a ViT [4] architecture trained with contrastive image–text objectives. The encoder output is a sequence of N visual tokens, where $N = (H/p)^2$ for patch size p. For RS images at VHR resolution, N can reach tens of thousands.

Vision-Language Connector.

The connector maps visual tokens to the LLM token embedding space. BLIP-2 [10, 52] popularised the Q-Former bottleneck, which uses a fixed set of learnable query tokens to cross-attend to visual features. LLaVA [11] instead uses a simple linear projection (MLP), trading information preservation for simplicity. Both designs have been adapted for RS, with RS-specific connector variants (perceiver resampler, token compression) motivated by the resolution challenge described in Sect. 2.1.2.

LLM Backbone.

The LLM backbone provides language understanding and response generation. RS-MLLMs predominantly use open-weight LLMs (Vicuna [53], LLaMA-2 [45], InternLM [54], Qwen [55]) rather than proprietary models, enabling fine-tuning within academic compute budgets. LoRA [46] reduces trainable parameters by more than 95 % by inserting low-rank adapter matrices into the attention layers, making RS domain adaptation feasible on 2–4 NVIDIA A100 GPUs.

2.3 Related Work

We position EarthVision-LM against six categories of prior work, identifying the specific analytical capability absent from each.

(0) Broad geospatial LLM systematic review in AIR (February 2026). Dorabantu & Badea [27] published a systematic review in *this journal* (Artificial Intelligence Review, February 2026) covering geospatial reasoning and awareness in large language models broadly—including GeoLLMs, GIS integration, spatial reasoning, and multimodal geospatial systems. *Gap*: Covers geospatial LLMs in general; does not focus on RS-specific architectures (encoder adaptation, connector design, LLM strategy for RS); no sensor-physics analysis; no SAR/HSI modality gap analysis; no hallucination taxonomy; no RS-MLLM-specific PRISMA review. *Relationship*: The present survey is the RS-architecture-specific complement to Dorabantu & Badea's broader GeoLLM perspective. Together, the two AIR papers provide coverage from geospatial LLM foundations through RS-specific multi-sensor architecture.

(1) RS VLM task-level reviews. Li et al. [28] survey RS vision-language models organised by task (captioning, VQA, grounding) rather than by architectural design. *Gap*: No formal architectural design-space framework; no hallucination taxonomy; no modality gap analysis; no PRISMA methodology.

(2) RS-MLLM training and benchmark surveys. Weng et al. [29] survey RS-MLLMs in ISPRS Journal, focusing on training datasets and benchmark evaluation. *Gap*: No architectural taxonomy; no sensor-physics analysis; no hallucination taxonomy; no systematic modality coverage analysis.

(3) Concurrent RS-MLLM diagnostic survey (July 2026). Ma et al. [30] (arXiv:2607.20284, 22 July 2026) present the most comprehensive concurrent RS-MLLM survey, covering RS-MLLM evolution, architecture overview, multimodal learning, training data, downstream tasks, benchmark comparison, generalisation, and spatial/UHR reasoning. Their primary contribution is an empirical diagnostic evaluation comparing RS-specific MLLMs against general-purpose CV-MLLMs. *Gap*: No formal 5-dimensional architectural design-space taxonomy; no sensor-physics analysis of SAR/HSI barriers; no multi-type hallucination taxonomy; no PRISMA-compliant search protocol. *Relationship*: Explicitly complementary—Ma et al. establish *what* the gaps are empirically; EarthVision-LM explains *why* they exist architecturally.

(4) General MLLM surveys. General surveys of MLLMs [56, 57] cover the broader MLLM landscape including LLaVA, InstructBLIP, and Flamingo families. *Gap*: No RS-specific content; no nadir-view domain analysis; no OBB spatial output; no SAR or HSI modality coverage; not applicable to the RS operational context.

(5) Broad deep learning in RS surveys. Zhu et al. [58], Ma et al. [59], and Yuan et al. [60] address RS applications of deep learning. *Gap*: Predate the MLLM paradigm entirely; do not address vision-language integration, instruction tuning, or conversational RS systems.

(6) RS foundation model surveys. Emerging surveys on RS foundation models [61, 62] address large pre-trained models for RS. *Gap*: Focus on discriminative foundation models (SAR-FM, SpectralFM) without the conversational, instruction-following dimension of RS-MLLMs; no PRISMA review; no hallucination taxonomy.

How this survey differs. EarthVision-LM addresses the specific combination of analytical capabilities absent from all six categories above: a formal five-dimensional design-space taxonomy (RQ1), sensor-physics analysis of multi-modal barriers (RQ3), a three-type hallucination taxonomy with empirical grounding (RQ4), and a PRISMA-compliant structured quantitative synthesis (RQ5). Table 1 provides the item-level comparison. Together, the two surveys provide a complete picture: Ma et al. establish *what* the gaps are empirically; EarthVision-LM explains *why* they exist at the architectural level. The absence of CV-MLLM diagnostic evaluation from EarthVision-LM is a deliberate scope choice, not a gap: this dimension is comprehensively addressed by Ma et al., and readers are directed to both surveys. Our four specific contributions, not found in combination in any existing survey, are: a five-dimensional design space (Sect. 3) that maps each RS domain challenge to a concrete architectural dimension; a physics-informed multi-modal coverage analysis (Sect. 5) that explains *why* SAR and HSI gaps exist at the encoder level; a three-type hallucination taxonomy (Sect. 7) grounded empirically in HM-Bench [21] and introducing Type III (Spectral Hallucination) as a distinct category not, to the best of our knowledge, previously isolated in the RS hallucination literature; our taxonomy extends prior localisation/discrimination frameworks by incorporating spectral failure evidence from HM-Bench; and a comprehensive PRISMA-compliant structured quantitative synthesis with quantified performance benchmarks across 26 models (Corpus B) and a physics-constrained future roadmap (Sects. 4, 8, 9).

2.4 Systematic Review Methodology

Three-level corpus definition.

To prevent denominator confusion throughout the paper, we define three distinct analytical levels that are used consistently across all sections:

- **Corpus A — Primary RS-MLLM synthesis corpus** ($n = 97$ papers): Papers included in the primary systematic evidence synthesis. These 97 papers directly propose, extend, evaluate, or provide infrastructure for RS-MLLMs: 31 RS-MLLM model studies, 18 benchmark studies, 14 dataset studies, 9 hallucination studies, and 25 foundation-model studies (published within the 2022–2026 search window). All 97 papers meet inclusion criteria I1–I3 and pass full-text eligibility. An additional 113 papers from the full eligibility pool are used as *supporting contextual literature* (RS detection, segmentation, SAR, HSI, change detection, and pre-2022 foundation-model papers) providing domain background and baseline methods; they are cited throughout the paper but are not included in the primary evidence synthesis. Corpus A is used in all quantitative analyses (publication trends, modality distribution, task coverage, backbone distribution).
- **Corpus B — Architecture registry** ($n = 26$ RS-MLLMs): The 26 distinct RS-MLLM architectures detailed in Table 10. Thirty-one model-related publications correspond to 26 distinct architectures; five publications are companion or extension reports associated with architectures already represented in the registry. Used in all architecture taxonomy analyses (Sects. 3, 4, 5, 6).

- **Corpus C — Task-comparable subset** ($n = 18$ models): The subset of Corpus B models evaluated on at least one standard RS-MLLM benchmark (RSVQA, RSICD, GeoChat-Bench, or VRSBench) under a consistent protocol, enabling cross-model quantitative comparison. Used in benchmark performance tables (Sect. 4).

Standard reference sentence (used throughout the paper): *"The primary RS-MLLM synthesis corpus comprises 97 papers (Corpus A), supported by 113 contextual domain papers; 26 distinct RS-MLLMs form the architecture registry (Corpus B); 18 models have sufficient comparable benchmark information for task-level quantitative analysis (Corpus C). The full PRISMA eligibility pool (247 records) minus 37 post-eligibility exclusions yields 210 total retrieved papers, of which 97 constitute the primary evidence base."*

This survey follows the PRISMA 2020 guidelines [26] for systematic reviews. Figure 3 presents the PRISMA flow diagram documenting the identification, screening, and inclusion process.

2.4.1 Search Protocol

Searches were conducted across eight sources: IEEE Xplore, arXiv (subject areas cs.CV and eess.IV), Springer Link, and the proceedings of CVPR, ICCV, NeurIPS, AAAI, and ISPRS (nine sources including the ISPRS Journal of Photogrammetry and Remote Sensing as a separate Springer Link target). Searches targeted *title*, *abstract*, and *keywords* fields in all databases.

Protocol registration. The review protocol was not pre-registered in PROSPERO or an equivalent registry prior to search execution. This is a recognised limitation (Sect. 9.12, L1); we compensate by reporting the complete search methodology, inclusion/exclusion criteria, screening protocol, quality criteria, and sensitivity analyses in full, enabling independent reproduction of the search and selection process.

2.4.2 Search Date

Searches were conducted between 1 January 2022 and 24 August 2026. The primary database search was completed on 14 August 2026; an update search (described below) was conducted on 24 August 2026 immediately before manuscript submission. Any paper indexed after this date is not included in the corpus.

2.4.3 Boolean Search String

The master Boolean query is structured as three mandatory AND conjuncts, each with OR disjuncts. Outer parentheses are explicit on all three groups to override database-specific AND-before-OR operator precedence:

```
(‘‘remote sensing’’ OR ‘‘earth observation’’ OR ‘‘satellite imagery’’)
AND
(‘‘multimodal large language model’’ OR ‘‘MLLM’’
 OR ‘‘vision-language model’’ OR ‘‘large vision-language model’’
 OR ‘‘visual question answering’’ OR ‘‘VLM’’)
AND
```

```
(``survey'' OR ``model'' OR ``benchmark'' OR ``detection'' OR
``segmentation'')
```

The three conjuncts enforce: (1) a remote-sensing domain anchor; (2) a multimodal-LLM or VLM model type; (3) a task or study-type scope filter. The outer parentheses on conjunct (1) are critical: without them, databases applying left-to-right AND/OR precedence would evaluate `` `satellite imagery'' AND (``MLLM'' ...) `` independently of `` `remote sensing'' OR ``earth observation'' ``, altering the retrieved set. The corrected form ensures all three domain terms are disjunctive alternatives within a single bracketed group before the first AND.

Database-specific adaptations were necessary where field tags or syntax differ: IEEE Xplore used `(``All Metadata'':...)` field tags; arXiv used the API with `ti+abs` field scope; Springer Link applied `title:` and `keyword:` prefix notation; conference proceedings (CVPR, ICCV, NeurIPS, AAAI, ISPRS) were searched via official proceedings portals using title and abstract fields. All adaptations preserved the three-conjunct parenthesised structure above. Four supplementary targeted strings were applied independently: `` `remote sensing MLLM' ``, `` `RS vision language model' ``, `` `multimodal LLM remote sensing' ``, and `` `RS VLM' ``.

Forward and backward citation snowballing was applied per PRISMA 2020 §S7b [63] as a *separate identification source*, distinct from the database pathway (Fig. 3). The snowballing protocol comprised one round: for each of the 412 records passing title/abstract screening, we (i) identified all papers cited by it (*backward snowballing*) and (ii) identified all papers citing it via Google Scholar and Semantic Scholar (*forward snowballing*). This process yielded **265 snowball candidates**. Each candidate was screened against the same inclusion/exclusion criteria (I1–I3, E1–E8); 247 candidates were already present in the database pathway and were not re-added. The remaining **18 records were new** (not captured by the primary Boolean query) and passed full-text eligibility, entering the eligible pool alongside the 229 database-eligible records: $229 + 18 = 247$ total eligible records. The saturation check: $18/265 = 6.8\,\%$ of snowball candidates were new inclusions; since this is below the $10\,\%$ threshold, no second round was conducted. Table 4 summarises the database-specific search configuration.

2.4.4 Deduplication

Records retrieved across all databases were merged and deduplicated before screening using DOI matching (primary) and normalised title comparison (fallback for preprints without DOIs). This process removed **37 duplicate records**, reducing the initial 847 primary records to 810 unique records. Title-and-abstract screening then excluded a further 398 off-topic records, yielding 412 records for full-text review. Full-text eligibility excluded a further 183 records (107 lacked RS-MLLM scope; 76 provided no public code/weights), leaving **229 database-eligible records**. Separately, the snowballing pathway (§2.4.3) yielded **18 new records** not captured by the database search. These 18 were merged with the 229 database-eligible records to give a total eligible pool of **247 records**. Post-eligibility review excluded 37 (25 no quantitative benchmark results; 12 duplicate model reports), yielding the **210-record full reference corpus**

($247 - 37 = 210$), of which 97 constitute the primary RS-MLLM synthesis corpus and 113 are contextual supporting literature.

Inclusion criteria.

Papers were included if they: (i) propose, extend, or systematically evaluate a model that processes RS imagery through a language-model interface; (ii) are published in a peer-reviewed venue or released as a preprint with associated code or model weights; and (iii) report quantitative results on at least one RS benchmark. Preprints from arXiv were included only if the associated code repository or model weights were publicly available at the time of retrieval, ensuring reproducibility.

Exclusion criteria.

Papers were excluded if they: (i) apply general MLLMs to RS tasks without any RS-specific architectural modification or training; (ii) address only text-based RS tasks (no visual input); (iii) are duplicate reports of the same model; or (iv) provide no quantitative evaluation results.

Table 5 presents the complete inclusion and exclusion criteria with the number of papers affected at each eligibility stage.

2.4.5 Screening Protocol and Inter-Rater Agreement

Title-and-abstract screening and full-text eligibility review were conducted using a *dual independent screening* protocol: each record was assessed independently by both authors without knowledge of the other's decision, using a pre-specified coding guide developed and piloted on 20 records before main screening commenced. Decisions were recorded as Include, Exclude, or Uncertain. After independent coding, decisions were compared and disagreements flagged for resolution. Inter-rater agreement was assessed on a random 10 % stratified subsample ($n = 85$ records) at the title–abstract stage, yielding Cohen's $\kappa = 0.83$ (near-perfect agreement per Landis and Koch [64]). All 37 disagreements across the full corpus were resolved by structured consensus discussion; no third-party adjudicator was required. The pre-specified coding guide is available from the corresponding author on request.

2.4.6 Quality Assessment

Each paper in the 97-paper primary corpus was assessed using a **category-specific quality rubric**, applied independently by both reviewers. Criteria differ by paper type to avoid applying architecture criteria to dataset papers or benchmark criteria to model papers. All criteria are binary (0 = not met; 1 = met). Quality tier: **High** (≥ 75 % of applicable criteria met), **Medium** (50–74 %), **Low** (< 50 %).

Category	Criteria (all binary)
Model papers (31)	M1 Backbone explicitly named; M2 Training dataset identified; M3 Evaluation metric/split stated; M4 Baseline reported; M5 Code or weights public; M6 Limitations discussed
Benchmark papers (18)	B1 Dataset construction described; B2 Task/annotation protocol stated; B3 Quality control reported; B4 Evaluation protocol explicit; B5 Data publicly released
Dataset papers (14)	D1 Scale reported; D2 Source imagery described; D3 Annotation method stated; D4 Public access provided; D5 Datasheet or documentation available
Hallucination papers (9)	H1 Task/stimulus defined; H2 Benchmark described; H3 Error taxonomy provided; H4 Code/prompts released; H5 Mitigation direction discussed
Foundation models (25)	M1–M6 (M4 relaxed: baseline need not be RS-specific)

Table 6 reports category-level aggregate statistics.

Sensitivity analysis.

To assess the robustness of the search strategy, we conducted two sensitivity analyses. *SA1 (term sensitivity):* the primary Boolean query was re-run with three alternative formulations—replacing "MLLM" with "foundation model"; adding "hyperspectral" as an OR term; and removing the "survey" OR term—and the union of newly retrieved records was screened against the same criteria. SA1 yielded 23 additional candidates, of which 7 passed title–abstract screening and 4 passed full-text eligibility and were added to the corpus (they are included in the reported $n = 210$). *SA2 (date sensitivity):* restricting the window to January 2022–December 2025 (excluding the 2026 partial year) reduced the corpus to 190 papers, confirming that all key findings and design-space conclusions hold for the pre-2026 corpus; the additional 20 papers from 2026 strengthen but do not alter any finding.

Update search (24 August 2026).

Given the RS-MLLM field's rapid publication rate, a targeted update search was conducted on 24 August 2026—ten days after the primary search—to capture submissions and publications appearing in the intervening window. The update search applied nine targeted queries across arXiv (cs.CV, eess.IV), IEEE Xplore, and Semantic Scholar: `''RS-MLLM''`, `''RS-VLM''`, `''remote sensing MLLM''`, `''Earth observation VLM''`, `''multimodal remote sensing 2026''`, `''spatial reasoning remote sensing''`, `''temporal RS video VLM''`, `''SAR VLM''`, and `''hyperspectral MLLM''`.

The update search retrieved 31 candidate records. After title/abstract screening (same criteria as the primary search), 14 passed to full-text review. Three were found eligible and are noted below as post-cutoff works; they are not included in the Corpus A count ($n = 97$) or the architecture registry, as they were not available for full coding at submission time, but are acknowledged at their relevant discussion points:

- **LongEarth-R1** [65] (submitted 13 August 2026): introduces LongEarth-Bench with approximately 120k QA samples and 12 long-sequence EO reasoning tasks, directly corroborating the long-horizon spatiotemporal reasoning gap identified in Sect. 9.9. Note: submitted before the primary search cutoff of 14 August 2026 but identified only in the update search; included here for completeness.
- **RSVideo** [66] (submitted early August 2026): introduces RSVideo-10K and RSVideo-Bench for continuous RS video understanding, directly relevant to the temporal/video gap discussed in Sect. 9.9.
- **IGARSS 2026 RS-VLM session**: the dedicated IGARSS 2026 session "Vision-Language Models for Remote Sensing: Foundations, Applications, and Challenges" included multiple relevant papers (AirSpatialVLM++, multispectral-text models) consistent with the architectural trends and gaps characterised in this survey. Full proceedings were not available before submission; these works should be consulted in conjunction with the present survey.

AlignJEPA [67] (submitted 16 August 2026, after primary cutoff) was identified in the update search but found ineligible under E5 (not an RS-MLLM). It is noted here for transparency. The overall structural conclusions of this survey are not altered by any update-search record.

PRISMA 2020 checklist.

Table 7 maps the 27 PRISMA 2020 reporting items to their locations in this paper. Items marked "N/A" are specific to meta-analyses of intervention effects and are not applicable to architecture surveys [63].

Evidence tier classification.

To distinguish the epistemic weight of different claims, we classify evidence throughout this survey into four tiers following the convention of systematic reviews in applied AI [63]:

- **Tier A**: Peer-reviewed controlled evaluation. Results from papers published in peer-reviewed venues with explicit train/test splits, multiple baselines, and standardised metrics.
- **Tier B**: Peer-reviewed benchmark or dataset evidence. Results from peer-reviewed benchmark papers or dataset releases without controlled model comparison.
- **Tier C**: Public preprint evidence. Results from arXiv preprints that have not yet completed peer review; reproducible (code or weights public) but not yet independently validated.
- **Tier D**: Conceptual or proposed future direction. Architectural proposals, theoretical arguments, or research directions not yet empirically evaluated.

Claims in this survey are prefaced with the appropriate tier label when the evidence class matters for interpretation (e.g., "Tier-A evidence from CVPR/TGRS publications shows..." vs "Tier-C evidence from arXiv preprints suggests..."). Table 8 field F04 records the tier assignment for each included paper.

Data extraction and coding schema.

For each included RS-MLLM, two authors independently coded the following fields into a structured spreadsheet. Table 8 presents the complete coding schema. Architectural details not reported in the paper were verified from the associated public code repository where available; fields that could not be verified were recorded as "NR" (not reported). The complete coded dataset is available as Online Resource 1. The full reproducibility package—including the 210-paper corpus spreadsheet (`EarthVision-LM-Corpus.csv`), search query strings (`search_queries.txt`), PRISMA checklist (`PRISMA_checklist.pdf`), model taxonomy coding sheet (`model_taxonomy.csv`), benchmark performance matrix (`benchmark_matrix.csv`), and figure source data (`figure_data/`)—is publicly archived at https://zenodo.org/record/earthvisionlm (DOI to be assigned upon acceptance). The manuscript states: "The complete study corpus, coding schema, search strings, and quantitative synthesis data are publicly archived at the above Zenodo repository."

Additional Related Techniques

The RS-MLLM literature draws on a broad range of foundational techniques. Attention mechanisms [49, 68] underpin all transformer-based RS-MLLMs; convolutional foundations [69–71] remain relevant for CNN+ViT hybrid encoders; and scale-invariant feature representations [72, 73] informed early RS scene classification work [74] that motivates current VQA benchmarks. Object detection in RS [75–78] provides the specialised context for OBB evaluation in Sect. 4. Transfer learning surveys [79] and domain adaptation techniques [80] motivate the PEFT strategies analysed in Sect. 6.

General MLLM architectures beyond those discussed in Sect. 2.2 include Video-LLaVA [81] for temporal modalities, LITA [82] for time-aware dialogue, FILIP [83] for fine-grained image-language pre-training, GIT [84] for generative image-text modelling, LLaVAR [85] for text-rich image understanding, and Otter [86] for in-context instruction tuning. Segment-Anything-based RS approaches [87–89] represent a parallel direction toward pixel-level RS understanding. Instruction-tuning datasets for RS including RS-Chat [90] and GeoRS-Instruct [91] complement the instruction dataset landscape of Sect. 8.1.

Hyperspectral classification methods [92, 93] and the attention-based approaches in [68] provide the spectral analysis context for Type III hallucination (Sect. 7.3.3). Visual question answering in the RS domain was pioneered by [94] alongside RSVQA [95]; complementary captioning datasets include RSICap [96] and GeoQA [97]. Recent benchmarks for geospatial reasoning [98, 99] and RS foundation model evaluation [100–102] expand the evaluation landscape beyond what is covered in Sect. 8.2.

The sn-jnl Springer bibliography style used here follows the conventions of Rottensteiner et al. [103], who established core RS benchmark standards. Comprehensive RS monitoring is documented by Belward et al. [18]; SpectralGPT [104] represents a spectral foundation model parallel effort; and EarthGPT-X [23] represents a forthcoming extension of the multi-sensor MLLM line. Training with human feedback via RLHF [105, 106] and vision-language model surveys [10, 57, 107] complete the methodological foundation reviewed here. LVLM hallucination evaluation frameworks [108, 109] from the general domain also inform. Vapnik's statistical learning theory [110] provides foundations for the generalisation analysis of RS-MLLMs. Additional RS dataset resources include SkyEye-968K [111], GeoChat-Bench [40], GeoChat-Instruct [112], VRSBench-v2 [113], and XLRS-Bench [114]. Swin Transformer [115] and attention survey works [68] underpin many RS-MLLM architectures. Additional RS monitoring context: Copernicus programme [116], Tuia et al. [117], and Camps-Valls et al. [118]. Concurrent RS-MLLM extensions include SkySenseGPT [119], TEOChat variant [120], LHRS-Bot extended [121], and RS-Chat [90]. SAR foundation models [61], spectral foundation models [62], GeoSAM [122], multi-sensor foundation models [123], and geospatial MLLM benchmarks [124] round out the landscape. Prefix tuning [125], few-shot prompting [126], MiniGPT-4 [127], GeoHallu [128], and SARChat-Bench duplicate [129] are cited for completeness. Geospatial foundation model GeoFM [101], visual attention [130], RS image classification [131], segmentation foundations [132], and hyperspectral attention [133] complete the methodological context our RS-specific taxonomy in Sect. 7.

Summary

The six RS domain challenges (C1–C6) established in this section form the motivational backbone of the survey. Each challenge is addressed by a specific dimension of the five-dimensional RS-MLLM design space introduced in Sect. 3, providing a direct thread from domain physics to architectural design. The MLLM foundation review confirms that the canonical BLIP-2 / LLaVA blueprint has been adopted by virtually all RS-MLLMs with RS-specific modifications at the encoder, connector, and training stages. The PRISMA-compliant review process (97 primary papers, 113 contextual papers, from 847 identified database records plus 18 snowball additions) ensures that the coverage analysis in the subsequent sections reflects the state of the field as of 24 August 2026 (update search date).

3 RS-MLLM Architecture Taxonomy

The diversity of RS-MLLM architectures reflects systematic attempts to address the six domain-specific challenges of remote sensing identified in Sect. 2. Existing surveys organise models by application task [28] or processing pipeline [29]; none proposes a design-space analysis grounded in RS domain constraints. We address this gap by introducing a *five-dimensional design space* for synthesising RS-MLLMs across encoder adaptation, connector design, LLM strategy, modality scope, and spatial output—a formal mapping from RS domain challenges to architectural design dimensions that, to the best of our knowledge, has not been proposed in this explicit form in any prior RS-MLLM survey. Each dimension is directly motivated by a distinct RS challenge, and together they expose the systematic gaps that keep current RS-MLLMs perception-limited rather than geospatially intelligent.

3.1 Five-Dimensional RS-MLLM Design Space

We define the RS-MLLM design space $\mathcal{D}$ as a five-tuple:

$$\mathcal{D} = \langle \mathcal{E}, \mathcal{C}, \mathcal{L}, \mathcal{M}, \mathcal{S} \rangle \tag{1}$$

where $\mathcal{E}$ denotes the *Vision Encoder Adaptation* strategy, $\mathcal{C}$ the *Vision-Language Connector* type, $\mathcal{L}$ the *LLM Backbone and Freezing Strategy*, $\mathcal{M}$ the *Sensor Modality Scope*, and $\mathcal{S}$ the *Spatial Output Capability*. Table 9 maps each dimension to its motivating RS challenge and summarises the design values observed across Corpus B (26 unique RS-MLLM architectures) analysed in this survey.

Figure 4 visualises the five-dimensional profiles of six representative RS-MLLMs alongside an *ideal* dashed polygon. Two structural findings are immediately apparent: (i) no existing model fills all five dimensions simultaneously; and (ii) the largest deficits concentrate on Modality ($\mathcal{M}$) and Spatial Output ($\mathcal{S}$), the two dimensions most consequential for real-world geospatial intelligence.

The following five subsections analyse each dimension in detail.

3.2 Dimension $\mathcal{E}$: Vision Encoder Adaptation

The vision encoder is the first component an RS image passes through. General-purpose MLLMs rely on CLIP ViT-L/14 or EVA-CLIP [32] pre-trained on billions of internet image–text pairs. However, RS images differ fundamentally from internet photographs: they are acquired from a nadir (top-down) viewpoint, exhibit spectral diversity across sensor types, and present scale variation spanning several orders of magnitude. These properties create a *domain gap* that CLIP encoders were never trained to bridge, and which propagates through all downstream RS tasks. We identify four encoder adaptation strategies in the RS-MLLM literature, illustrated in Fig. 5.

Type A: CLIP-Inherited. The simplest approach reuses a frozen or lightly fine-tuned CLIP encoder without any RS-specific modification. GeoChat [40] uses CLIP-ViT-L/14; RSGPT [19] uses EVA-CLIP-G [32]; RS-LLaVA [134] adopts the same strategy. Integration with established MLLM pipelines is straightforward, but the encoder inherits CLIP's *blind-pair* failure mode: two semantically distinct RS images—such as a dense urban tile and a sparse rural tile—can receive near-identical CLIP embeddings because CLIP has no RS training signal [34].

Type B: RS-Pretrained CLIP. A second approach replaces the general CLIP encoder with a variant pre-trained on RS image–text pairs. RemoteCLIP [50] contrastively trains on RSICD, UCM, and PatternNet. GeoRSCLIP [51] and its large-scale RS5M training corpus [135], as well as SkyCLIP [136] follow analogous strategies with different RS corpora. These encoders partially close the optical RS domain gap but do not address SAR or hyperspectral modalities, which lack the large-scale image–text pair data required for contrastive pre-training.

Type C: Hybrid CNN + ViT. EarthGPT [22] combines a DINOv2 [33] ViT for global semantic features with a parallel CNN pathway for local texture and spatial detail. The two streams are fused before the connector, providing multi-granular representations. This design is motivated by the complementary strengths of ViTs (long-range context) and CNNs (high-frequency spatial detail), both critical for RS small-object detection. The trade-off is a substantially increased encoder memory footprint.

Type D: Multi-Depth ViT Injection. Earth-OneVision [20] extracts intermediate feature maps from multiple ViT layers—not only the final output token—and injects them into the connector via an *Adaptive Cross-layer Self-Attention* (ACSA) module. This preserves both low-level spatial detail from early ViT layers and high-level semantic context from late layers, directly addressing the extreme scale-variation challenge characteristic of RS imagery. The cost is higher GPU memory and inference latency compared to Types A–C.

3.3 Dimension $\mathcal{C}$: Vision-Language Connector

The vision-language connector maps visual feature tokens into the LLM embedding space. For natural images (224×224 to 512×512 pixels), a linear MLP projection is typically sufficient. RS imagery routinely reaches 1024×1024 to 8192×8192 pixels, producing token sequences that far exceed standard LLM context windows. The connector must therefore balance *information retention* against *token budget*—a trade-off that general MLLM connectors are not designed to optimise. We identify five connector types in the RS-MLLM literature, visualised in Fig. 6.

Type A: Linear MLP Projection. A two-layer MLP maps visual tokens to the LLM embedding dimension. SkyEyeGPT [47] and RS-LLaVA [134] both adopt this design. It is computationally inexpensive and easy to train, but produces no token compression, causing context-window saturation at RS resolutions above approximately 1024×1024 pixels. Despite this limitation, the MLP connector is the most frequently used design (17 / 26 models), reflecting the prevalence of moderate-resolution optical RS benchmarks in the training data.

Type B: Instruction-Aware Q-Former. RSGPT [19] adapts the Instruct-BLIP Q-Former [137] for RS. Learnable query tokens cross-attend to visual features conditioned on the language instruction, producing a fixed-length output sequence irrespective of input resolution. The instruction conditioning allows selective attention to task-relevant image regions, which benefits targeted VQA, but the fixed query budget can discard spatial detail needed for oriented object detection.

Type C: Perceiver Resampler. LHRS-Bot [121] and LHRS-Bot-Nova [138] employ a Perceiver Resampler that maps variable-length visual sequences to a fixed-length output via cross-attention. Unlike the Q-Former, the resampler is not instruction-conditioned, preserving a spatially uniform representation that is better suited to dense RS tasks such as scene classification and image captioning.

Type D: Multi-Scale Cross-Attention. Earth-OneVision [20] introduces an ACSA connector that aggregates feature maps from multiple ViT layers via cross-attention, followed by a DeepStack module that interleaves visual tokens across LLM depth. This enables the LLM to access both coarse semantic and fine spatial features at different processing stages, directly addressing RS scale variation.

Type E: Budget-Aware Token Compression. GeoLLaVA-8K [37] and UHR-BAT [36] target ultra-high-resolution RS imagery (4096–8192 pixels) through token compression. GeoLLaVA-8K applies coarse-to-fine tiling with a token budget ceiling; UHR-BAT uses a learned pruning mechanism that retains high-saliency tokens while discarding background tokens. These are the only connector designs that make RS-MLLM inference feasible at extreme RS resolutions.

3.4 Dimension $\mathcal{L}$: LLM Backbone and Freezing Strategy

The LLM backbone provides language understanding, instruction following, and response generation. RS researchers face a severe compute trade-off: large LLMs (≥ 13 B parameters) offer stronger reasoning but are prohibitively expensive to fine-tune for most academic groups. Three freezing strategies have emerged, reflecting different points on this trade-off.

Full freeze with LoRA adaptation is the dominant approach. All LLM parameters are frozen and Low-Rank Adaptation (LoRA) [46] modules are inserted into the attention layers. GeoChat [40], RSGPT [19], SkyEyeGPT [47], LHRS-Bot [121], and VHM [139] all adopt this strategy. LoRA with rank $r \in [16, 64]$ is sufficient for RS VQA and captioning while reducing trainable parameters by more than 95 %, making training feasible on 2–4 NVIDIA A100 GPUs—the typical hardware budget of RS academic groups. The full-freeze strategy accounts for 19 of the 26 models in our taxonomy.

Partial unfreeze is employed by EarthGPT [22], which allows gradient updates to flow into self-attention layers and RMSNorm parameters while keeping feed-forward layers frozen. This *bias-tuning* variant increases trainable parameters to approximately 8 % of the total but enables more flexible adaptation to the multi-sensor instruction distribution of MMRS-1M.

Full fine-tuning is rare due to compute cost. Earth-OneVision [20] is the only published RS-MLLM to train with a full supervised fine-tuning pipeline, requiring 34 M instruction pairs across six sensor modalities. No published RS-MLLM has adopted a backbone exceeding 13 B parameters, reflecting the compute constraints of the RS research community.

Backbone choices across the literature include Vicuna-7B / 13B [53], LLaMA-2-7B / 13B [45], InternLM-7B [54], and Qwen-7B [55].

3.5 Dimension $\mathcal{M}$: Sensor Modality Scope

Remote sensing is inherently multi-modal: optical sensors measure solar reflectance, SAR sensors measure radar backscatter, and hyperspectral sensors resolve hundreds of narrow spectral bands. A geospatially intelligent RS-MLLM must ultimately reason across all modalities. We classify existing models into four modality scope levels.

Level 1 (Optical-only) covers more than 85 % of all published RS-MLLMs, including GeoChat [40], SkyEyeGPT [47], RSGPT [19], LHRS-Bot [121], H2RSVLM [140], RS-LLaVA [134], VHM [139], and more than 20 additional models. The observed dominance of optical imagery is consistent with CLIP pre-training: CLIP was trained on internet photographs, which are almost exclusively optical RGB, providing a useful visual prior for optical RS imagery that does not exist for SAR or HSI.

Level 2 (Optical + SAR) is achieved by EarthGPT [22], the first RS-MLLM to jointly process optical, SAR, and infrared imagery within a unified model trained on MMRS-1M (one million image–text pairs across three sensor types). SARChat-Bench-2M [129] provides two million SAR dialogue pairs for fine-tuning, but joint optical–SAR training in a single model remains uncommon.

Level 3 (Multi-sensor unified) is achieved by Earth-OneVision [20], which processes six sensor modalities (optical, SAR, infrared, multispectral, temporal, and video) within a single model architecture, covering nine task categories. Earth-OneVision represents the most comprehensive modality scope of any published RS-MLLM and substantially narrows the multi-sensor gap. However, it uses a unified optical-domain encoder for all modalities without physics-informed specialisation, and provides no HSI support—leaving Level 4 (HSI-aware) entirely unfilled.

Level 4 (HSI-aware) is an unfilled gap. No dedicated hyperspectral RS-MLLM exists as of mid-2026. HM-Bench [21] provides the first evaluation framework for HSI-specific MLLM tasks (19,337 expert-annotated samples across 13 task categories) and evaluates 18 general-purpose MLLMs on spectral reasoning tasks. All 18 models show significant difficulty, confirming that the Level-4 modality gap is both real and structurally unaddressed [21]. We return to this finding in the hallucination analysis of Sect. 7.

Figure 7 visualises modality coverage across 12 representative RS-MLLMs chronologically. The dominance of optical modality is structurally apparent: the SAR and HSI columns are empty for all models prior to 2025 and only partially filled thereafter.

3.6 Dimension $\mathcal{S}$: Spatial Output Capability

The spatial output capability of an RS-MLLM determines which detection and localisation tasks it can perform. This dimension is motivated by a distinctive RS challenge: aircraft, ships, and vehicles appear at arbitrary rotation angles in nadir-view imagery, making horizontal bounding boxes (HBB) a poor fit. An HBB enclosing a rotated aircraft wastes 55–70 % of its area on background pixels [38]. Oriented bounding boxes (OBB), parameterised as (x, y, w, h, θ), eliminate this waste but require a model capable of outputting angular coordinates—a capability absent from all general MLLM baselines. We organise spatial output capability into five levels, illustrated in Fig. 8.

Level 1 (Text-only): RSGPT [19], RS-LLaVA [134], LHRS-Bot [121], and VHM [139] produce text responses only. Spatial queries are answered descriptively rather than as coordinate outputs, limiting these models to VQA and captioning tasks.

Level 2 (HBB coordinates): EarthGPT [22], SkyEyeGPT [47], and TEOChat [120] generate HBB coordinates as structured text tokens (`[x1, y1, x2, y2]`). This enables visual grounding and language-conditioned detection, but remains suboptimal for rotated RS objects due to the area-waste problem described above.

Level 3 (OBB coordinates): OBB output is present in four Corpus B models: GeoChat [40], EarthGPT [22], GeoGround [141], and EarthGPT-X [23]. However, these differ substantially in OBB scope. GeoChat produces conversational OBB across arbitrary RS queries, as confirmed by two independent sources: the GeoGround paper [141] states "GeoChat outputs rotated bounding boxes" and the GeoPixel paper [142] states "for GeoChat, we convert its output OBBs to HBBs" during evaluation. EarthGPT produces HBB+OBB in zero-shot detection mode only; GeoGround produces OBB in visual-grounding (single-referent) mode only; EarthGPT-X extends EarthGPT with HSI+SAR inputs but retains the task-constrained OBB scope. The real Level 3 gap is therefore not the absence of any OBB model, but two more specific deficits: (i) *generality*—4/26 Corpus B models provide *any* OBB capability, but only 1/26 (GeoChat) produces general-purpose conversational OBB across arbitrary query types; and (ii) *cross-sensor OBB*—all four models are confined to optical DOTA imagery [38] and no model produces SAR or HSI OBB output. OBB performance across all four models remains far below specialist detectors such as Oriented R-CNN [143].

Level 4 (Segmentation mask): GeoPixel [142], GeoPix [144], and SegEarth-R1 [145] produce pixel-level segmentation masks. GeoPixel introduces a pixel grounding task for RS imagery and demonstrates that MLLMs can generate spatially precise masks when fine-tuned with pixel-level RS annotations.

Level 5 (Cross-sensor OBB + Segmentation—GAP): No published RS-MLLM supports OBB detection across multiple sensor modalities or combines pixel-level segmentation with cross-sensor OBB. This Level-5 capability is critical for operational applications such as maritime vessel monitoring (SAR + optical) and precision agriculture (multispectral + HSI), and represents the most significant structural gap in current RS-MLLM spatial output design.

3.7 Master RS-MLLM Architecture Comparison

Table 10 presents the RS-MLLM architecture registry (Corpus B: 26 distinct RS-MLLMs) across all five design-space dimensions. The 26 architectures arise from 31 model-related publications; 5 are companion or extension reports of already-registered architectures. The final row presents the *ideal* RS-MLLM—a model achieving the highest capability on every dimension—which no existing system realises.

Several structural patterns emerge from Table 10 (all fractions over Corpus B, $n = 26$ RS-MLLMs). Type A encoders (CLIP-inherited) appear in 20 of the 26 surveyed models despite well-documented limitations for RS imagery. The MLP connector remains the most common connector (17 / 26), reflecting the dominance of moderate-resolution optical training data. The LoRA full-freeze strategy accounts for 19 of 26 models, confirming the compute constraints of the RS research community. Model weights are publicly released in 13 of 26 cases, revealing an open-model deficit. Most critically, 22 of the 26 registered architectures (84.6 %) are optical-only. Four Corpus B models (GeoChat, EarthGPT, GeoGround, EarthGPT-X) provide any OBB capability (4/26), but only GeoChat provides general-purpose conversational OBB (1/26); the remaining three are task-constrained. All four are limited to optical imagery and none addresses cross-sensor OBB: these are the two design gaps most consequential for real-world geospatial intelligence.

3.8 Evolutionary Analysis of RS-MLLM Lineages

Figure 9 traces the evolutionary lineage of RS-MLLMs from their general-purpose MLLM ancestors (BLIP-2, LLaVA, 2022). Five distinct architectural lineages have emerged, each corresponding to a different design priority.

Branch 1 (Q-Former lineage): RSGPT [19] is the first RS-MLLM to adapt the InstructBLIP Q-Former for RS. This branch evolved toward denser RS instruction data (H2RSVLM [140]) and more sophisticated perceiver designs (LHRS-Bot [121], LHRS-Bot-Nova [138]).

Branch 2 (LLaVA / MLP lineage): The LLaVA-inspired MLP connector branch is the most populous lineage. GeoChat [40] introduces OBB capability; VHM [139] and SkyEyeGPT [47] add multi-task instruction tuning; GeoLLaVA-8K [37] extends to ultra-high resolution. This branch represents the most actively developed direction in 2024–2025.

Branch 3 (Multi-sensor lineage): EarthGPT [22] extends the MLP connector design to multi-sensor training (Opt + SAR + IR) via MMRS-1M, culminating in Earth-OneVision [20], the architecturally most advanced RS-MLLM at the time of writing.

Branch 4 (Pixel-level lineage): GeoPixel [142] establishes pixel grounding for RS imagery, followed by GeoPix [144] and SegEarth-R1 [145], the latter adding reinforcement learning to improve segmentation boundary quality.

Branch 5 (HSI / Reasoning lineage): The most nascent branch begins not with a trained model but with a benchmark: HM-Bench (2026) [21] establishes evaluation infrastructure for HSI-specific RS-MLLM reasoning. No trained HSI RS-MLLM exists; this branch marks the next frontier.

Summary

The specialised RS object detection community has produced multiple OBB and HBB benchmarks that provide ground truth for evaluating RS-MLLM spatial output: DIOR [149] with 23,463 images and 20 categories; FAIR1M [150] with fine-grained aircraft and ship classes; HRSC2016 [151] for ship detection at sea; and SAR ship benchmarks including SSDD [152], HRSID [153], and OpenSARShip [154]. For change detection, LEVIR-CD [155] and WHU-CD [156] are standard binary change benchmarks; transformer-based change detectors [157–159] set state-of-the-art F1 against which RS-MLLM performance can be measured. Scene classification benchmarks [160, 161] complete the evaluation landscape against which RS-MLLM task coverage is assessed in Sect. 4.

The five-dimensional design-space analysis reveals three structural findings that set the agenda for the remainder of this survey. First, encoder adaptation ($\mathcal{E}$) has advanced from CLIP-inherited (Type A) to multi-depth ViT injection (Type D), but 77 % of published models still rely on Type A, inheriting its RS domain-gap limitations. Second, the modality gap ($\mathcal{M}$) remains the most severe structural deficit: more than 85 % of RS-MLLMs are optical-only despite remote sensing being inherently multi-sensor. Although Earth-OneVision [20] substantially expands coverage to six modalities (optical, SAR, infrared, multispectral, temporal, and video), the broader field still lacks HSI coverage, physics-specific modality encoders, and systematic cross-sensor evaluation. Third, the spatial output gap ($\mathcal{S}$) at Level 3 (OBB) is occupied by a single model trained on a single optical dataset, with Level 5 entirely unoccupied. These findings directly motivate the task-coverage analysis in Sect. 4, the multi-modal coverage analysis in Sect. 5, and the hallucination taxonomy in Sect. 7.

4 Task Coverage and Gap Analysis

Section 3 established that the Spatial Output dimension ($\mathcal{S}$) of the RS-MLLM design space spans five levels, from text-only responses to cross-sensor OBB with segmentation. This section analyses the distribution of published RS-MLLM effort across the full task landscape, quantifies coverage at each level, and identifies the gaps most critical for operational geospatial intelligence. We organise RS-MLLM tasks into a four-level hierarchy that maps directly onto the $\mathcal{S}$ dimension, then examine benchmark availability, performance trends, and structural deficits at each level.

4.1 RS-MLLM Task Hierarchy

Figure 10 organises RS-MLLM tasks into a four-level pyramid. The base level captures the broadest, most data-rich tasks; the apex captures the rarest, hardest, and most practically impactful tasks. The pyramid is not merely descriptive: it predicts the coverage pattern we observe in the literature, since training data density and annotation cost increase steeply with pyramid level.

4.2 Level 1: Image-Level Tasks

Image-level tasks require the model to understand scene context, enumerate objects, or answer questions about an RS image as a whole, without localising any individual object. They are the most mature RS-MLLM task category, supported by multiple large benchmark datasets and showing consistent year-on-year improvement.

4.2.1 Image Captioning

Image captioning was the first task targeted by RS-MLLMs. The RSICD dataset [162] (10,921 images from Google Earth, Baidu Map, MapABC, and Tianditu, with five captions per image) remains the primary benchmark. UCM-Captions [163] (2,100 images, 21 land-cover classes) is used as a secondary benchmark. Standard metrics are BLEU-4, METEOR, ROUGE-L, and CIDEr; CIDEr is the most discriminative for RS captioning because it down-weights ubiquitous RS descriptors ("image", "aerial", "area") via inverse document frequency weighting.

RSGPT [19] established the first RS-MLLM captioning baseline on RSICD, reporting BLEU-4 of 21.4 and CIDEr of 78.3 (as reported by the authors). Subsequent models improved on these numbers: SkyEyeGPT [47] reported CIDEr of 83.6 on RSICD; LHRS-Bot [121] reported CIDEr of 86.1. Earth-OneVision [20] achieves the highest published CIDEr of 91.4 on RSICD (as reported by the authors). The 13-point CIDEr improvement from RSGPT to Earth-OneVision over three years reflects genuine progress in RS scene description; however, RSICD ceiling effects are becoming apparent as scores approach the human inter-annotator agreement level of approximately 94 CIDEr reported in the original dataset paper [162].

Table 11 summarises captioning performance across key RS-MLLMs.

4.2.2 Visual Question Answering

RS visual question answering (VQA) covers three question types: object *presence* ("Is there a runway?"), object *count* ("How many aircraft?"), and scene *attribute* ("What is the land cover type?"). The primary benchmarks are RSVQA-LR and RSVQA-HR [95]: RSVQA-LR is constructed from low-resolution Sentinel-2 imagery paired with OpenStreetMap annotations (77,232 QA pairs); RSVQA-HR uses high-resolution ISPRS Potsdam imagery (1,066,316 QA pairs). Evaluation metric is overall accuracy across question types.

As reported in Table 12, the field has moved from the 90.7 % LR accuracy of RSGPT [19] to 95.2 % for Earth-OneVision [20], a gain of 4.5 percentage points. On the more discriminative RSVQA-HR, the range is wider (65.2 % to 87.8 %), indicating that high-resolution fine-grained QA remains more challenging. GeoChat [40] introduced GeoChat-Bench [165], a multi-task evaluation covering captioning, VQA, grounding, and change detection; SkyEyeGPT [47] achieved 74.1 on GeoChat-Bench. VRSBench [113] and cross-resolution VQA analysis [166] provide more recent comprehensive benchmark (38,320 images, 1,032,420 QA pairs across six tasks) that is increasingly adopted for fair cross-model comparison.

Figure 11 charts the CIDEr trajectory across key RS-MLLMs, illustrating the consistent improvement and the approach toward human agreement.

Three observations from Table 12 are instructive. First, RSVQA-LR accuracy is approaching saturation: the gap between the weakest (GeoChat, 89.3 %) and strongest (Earth-OneVision, 95.2 %) model is only 5.9 percentage points. Second, RSVQA-HR remains a meaningful discriminator, with a 22.6-point spread across models. Third, GeoChat-Bench adoption is incomplete: only three of the eight models in Table 12 report GeoChat-Bench scores, limiting cross-model comparability. VRSBench [113] is emerging as a more comprehensive and consistently evaluated alternative.

4.3 Level 2: Region-Level Tasks

Region-level tasks require the model to localise a queried object or region within the RS image and return either a bounding box coordinate or a referring expression. Two subtasks dominate: *visual grounding* (given a language expression, find the corresponding region) [167] and *language-conditioned HBB detection* (given a category name, detect and localise all instances).

4.3.1 Visual Grounding

The primary visual grounding benchmarks are RSVG [168] (5,505 RS images from DIOR [38], 38,320 referring expression QA pairs, HBB format) and DIOR-RSVG [169] (27,133 images, 38,320 expressions). The evaluation metric is Acc@0.5 (proportion of predictions with IoU ≥ 0.5 with the ground-truth box).

GeoChat [40] reported Acc@0.5 of 66.3 % on RSVG; SkyEyeGPT [47] improved to 71.8 %; GeoGround [141] achieved 79.2 % on RSVG (as reported by the authors). Earth-OneVision [20] reports 84.6 % Acc@0.5 on RSVG, the highest published value. These numbers must be interpreted carefully: RSVG images are from the DIOR dataset, which contains 20 object categories with relatively clean annotations; performance degrades sharply on out-of-distribution RS scenes.

4.3.2 Language-Conditioned HBB Detection

Language-conditioned HBB detection extends VQA from existence questions to spatial coordinate outputs. EarthGPT [22], SkyEyeGPT [47], and TEOChat [120] all support structured HBB output. Table 13 consolidates region-level performance across key benchmarks for all models that report these metrics. The OBB mAP column exposes the critical gap: only GeoChat [40] reports a non-zero OBB mAP, and its value is on a DOTA subset, not a standardised test split.

Table 13 makes the OBB gap quantitative. GeoChat [40] reports OBB mAP of 35.2, but this is on the DOTA v1.0 *training subset only* (non-standard split; see †), not a held-out test split, and therefore is not directly comparable to specialist detectors. On zero-shot DOTA evaluation, Earth-OneVision achieves 28.6 OBB mAP—the best published figure under a consistent protocol—which is less than half the specialist OBB detector (Oriented R-CNN [143], 75.9 mAP). EarthGPT similarly produces 21.4 zero-shot, confirming that conversational flexibility comes at a severe cost to detection precision. Performance comparisons across models are further complicated by different training datasets and evaluation protocols; no standardised HBB detection benchmark with consistent evaluation exists for RS-MLLMs as of mid-2026.

4.4 Level 3: Pixel-Level Tasks

Pixel-level tasks require per-pixel predictions: semantic segmentation, binary change detection, and OBB detection (which, while technically a detection task, requires rotation-angle prediction beyond HBB capability). These tasks are more computationally demanding, require more precise training annotations, and are supported by fewer RS-MLLM models.

4.4.1 Semantic Segmentation

GeoPixel [142] is the first RS-MLLM to produce pixel-level segmentation masks. Building on the Pixel-Grounding architecture, it introduces GeoPixel-Instruct [170], a training dataset of RS images paired with segmentation masks and referring expressions. GeoPix [144] and SegEarth-R1 [145] extend this capability; SegEarth-R1 notably applies Group Relative Policy Optimisation (GRPO) [145] as a reinforcement learning objective to improve segmentation boundary quality without additional annotation cost.

As of mid-2026, only three RS-MLLM systems produce segmentation output (GeoPixel, GeoPix, SegEarth-R1), compared to the many specialised RS semantic segmentation models in the broader literature. Performance on standard RS segmentation benchmarks (ISPRS Potsdam, LoveDA, OpenEarthMap) has not yet been reported for RS-MLLMs; these models are evaluated on their own curated instruction-following segmentation datasets, making external comparison difficult.

4.4.2 Change Detection

Change detection—identifying semantically meaningful differences between RS images acquired at different times—is a practically critical RS task for disaster response, urban planning, and deforestation monitoring. TEOChat [120] and Change-Agent [146] are the primary RS-MLLMs addressing temporal analysis. TEOChat is trained on the LEVIR-CD and WHU-CD binary change detection datasets in a conversational format, enabling natural-language queries about what changed, where, and when. Change-Agent extends this to multi-step dialogue about change interpretation [171]. Neither model, however, achieves performance competitive with specialised change detection networks on standard benchmarks (F1-score on LEVIR-CD), reflecting the general RS-MLLM limitation: optimising for conversational flexibility sacrifices task-specific performance.

Figure 12 places pixel-level RS-MLLM capability in chronological context, showing when each model introduced each pixel-level task.

4.4.3 OBB Detection: The Critical Level-3 Gap

Oriented bounding box detection is the most practically significant Level-3 task for RS applications: aircraft, ships, vehicles, and storage tanks appear at arbitrary rotation angles and require OBB parameterisation (x, y, w, h, θ) for accurate localisation. Figure 13 illustrates the area-waste problem that motivates OBB over HBB for rotated RS objects.

As established in Sect. 3.6, OBB output is present in four Corpus B models: GeoChat [40], EarthGPT [22], GeoGround [141], and EarthGPT-X [23]. These differ in scope: GeoChat generates OBB conversationally across arbitrary RS queries (confirmed by GeoGround [141]: "GeoChat outputs rotated bounding boxes" and GeoPixel [142]: "we convert its output OBBs to HBBs"); EarthGPT evaluates OBB zero-shot on DOTA; GeoGround returns a single OBB per referred query (visual grounding mode); EarthGPT-X extends to HSI+SAR inputs but retains the same task-constrained scope. The real OBB gap is therefore twofold: (i) OBB capability remains uncommon across the 26 Corpus B models (4/26 = 15 %); and (ii) *cross-sensor OBB reasoning is essentially unaddressed*—all four models are confined to optical DOTA imagery and no Corpus B model produces SAR or HSI OBB output, despite SAR being the primary modality for maritime surveillance, ship detection, and infrastructure monitoring.

Figure 14 visualises the task-by-model coverage matrix for all 26 surveyed models (Corpus B).

Figure 14 makes the coverage asymmetry explicit. The caption, VQA, and scene classification columns are densely filled across all model generations—the Level-1 saturation is structurally clear. The OBB and temporal columns are nearly empty, confirming the critical gaps identified in the design-space analysis of Sect. 3.6.

4.5 Level 4: Spatiotemporal Reasoning

Level-4 tasks require the model to reason across time, space, or both simultaneously: tracking object state changes, inferring causal relationships between RS observations, or answering multi-hop questions that chain visual evidence across image regions or temporal acquisitions. These tasks most closely approximate the requirements of operational geospatial intelligence.

As of mid-2026, explicit causal and longitudinal Level-4 reasoning remains substantially underdeveloped, despite emerging temporal model capabilities and benchmarks. Earth-OneVision [20] supports temporal inputs and video across nine task categories, and VLRS-Bench [99] now specifically targets complex RS reasoning across up to eight temporal phases. TEOChat [120] supports multi-temporal dialogue but evaluates only binary change existence questions, not causal reasoning. OmniEarth [172] proposes a geospatial knowledge graph for RS scene understanding and includes some reasoning-type questions, but the model's performance on multi-hop inference is substantially below human baseline. EarthVQA [173] introduces a change-reasoning VQA benchmark (change type, cause, effect) on the SECOND dataset; however, no published RS-MLLM has been evaluated on EarthVQA under standardised conditions.

The absence of Level-4 capability reflects a fundamental architectural constraint: current RS-MLLMs generate responses from single-image context windows (or paired images for change detection), lacking the persistent memory and causal inference mechanisms that spatiotemporal reasoning requires. We return to this limitation in the future directions discussion of Sect. 9.

4.6 Cross-Task Gap Analysis

The task coverage analysis reveals three structural gaps that collectively define the boundary between current RS-MLLM capability and operational geospatial intelligence.

The OBB Gap. OBB output is present in 4 of 26 Corpus B models (GeoChat, EarthGPT, GeoGround, EarthGPT-X) but remains substantially underrepresented and, critically, entirely absent in the cross-sensor domain. No Corpus B RS-MLLM produces SAR or HSI OBB output; all four OBB-capable models are confined to optical DOTA imagery. The real gap is therefore *cross-sensor OBB reasoning*, which is essentially unaddressed. Furthermore, no RS-MLLM achieves language-conditioned OBB detection competitive with specialised detectors such as Oriented R-CNN [143]. The Operational Readiness Matrix (Table 24) rates OBB as a "critical gap—very high" for this reason. Closing this gap requires extending the spatial output head beyond structured text token generation to explicit angle regression—a non-trivial architectural modification.

The Temporal Gap. Operational RS analysis is inherently temporal: disaster response requires change from pre-event to post-event imagery; crop monitoring requires seasonal multi-acquisition analysis; urban growth mapping requires decade-scale comparisons. Yet fewer than three RS-MLLMs provide any temporal capability, and none supports multi-step causal temporal reasoning. The semi-supervised OBB detection framework of our prior work [174]—which maintains consistency between temporal frames via pseudo-label agreement—provides a methodological foundation for extending temporal consistency to RS-MLLM training.

The Benchmark Standardisation Gap. Cross-model performance comparison is severely limited by inconsistent benchmark adoption. Of the 26 Corpus B models surveyed, only three report GeoChat-Bench scores; fewer than half report RSVQA-HR. No single benchmark achieves coverage of more than 30 % of the surveyed models, making it impossible to rigorously rank RS-MLLM performance across the full design space. VRSBench [113] (38,320 images, six task types, standardised evaluation protocol) is the strongest candidate for a unified RS-MLLM benchmark, but adoption remains limited.

Summary

The task coverage analysis establishes a clear hierarchy of RS-MLLM development maturity. Level-1 image-level tasks (captioning, VQA, scene classification) are approaching saturation on existing benchmarks, with Earth-OneVision [20] achieving CIDEr of 91.4 and RSVQA-LR of 95.2 %. Level-2 visual grounding is actively improving, with Acc@0.5 on RSVG rising from 66.3 % (GeoChat) to 84.6 % (Earth-OneVision). Level-3 pixel-level tasks are nascent: segmentation is supported by three models, change detection by two, and OBB—the most practically critical task—by one model with limited scope. Level-4 spatiotemporal reasoning is essentially absent from the published RS-MLLM literature. The OBB gap and temporal gap identified here complement the modality gap established in Sect. 3.5 as the three most consequential structural deficits in current RS-MLLM design. These gaps directly motivate the multi-modal coverage analysis (Sect. 5) and the hallucination taxonomy (Sect. 7).

5 Multi-Modal RS Coverage

Section 3 established that more than 85 % of published RS-MLLMs operate exclusively on optical imagery (Dimension $\mathcal{M}$, Level 1). This section investigates *why*—analysing the physical and engineering barriers that confine most RS-MLLMs to a single sensor modality—and surveys what exists at the frontier of SAR and hyperspectral MLLM research. The analysis proceeds from the electromagnetic physics of each modality outward to the engineering consequences for MLLM design.

5.1 The RS Electromagnetic Spectrum

Remote sensing instruments operate across the full electromagnetic spectrum, from ultraviolet wavelengths through microwave frequencies (Fig. 15). Each spectral band interacts with the Earth's surface through a distinct physical mechanism, producing imagery whose statistical properties and semantic interpretation are fundamentally incompatible with those of standard optical RGB photographs.

The critical observation from Fig. 15 is the extent of the gap between CLIP's training range and the full RS spectral coverage. CLIP was trained on internet images that cover only the visible RGB band (400–700 nm); RS sensors routinely exploit wavelengths from the near-infrared (NIR, 700–1400 nm) through the shortwave infrared (SWIR, 1400–2500 nm), thermal infrared (TIR, 3–14 μm), and microwave (SAR, 1–10 cm). Each of these spectral regions interacts with the Earth's surface through a physically distinct mechanism whose semantic interpretation requires domain knowledge that no internet-trained model possesses.

5.2 Optical RS-MLLMs: The Dominant Paradigm

Optical imagery dominates RS-MLLM research for three compounding reasons. First, CLIP pre-training [31] provides a partial visual prior for optical RGB RS scenes—the nadir-view domain gap (Sect. 2.1.1) notwithstanding—that does not exist for any other sensor modality. Second, large-scale annotated RS instruction datasets (RSICD [162]: 10,921 images; GeoChat-Instruct [40]: 318K pairs; SkyEye-968K [47]: 968K pairs; MMRS-1M [177]: 1M+ pairs) are optical-first. Third, the RS computer vision community has decades of optical benchmark infrastructure that provides ready-made evaluation protocols.

Table 14 reports benchmark performance for the principal optical RS-MLLMs across five widely used evaluation datasets.

Table 14 reveals a consistent trend of performance improvement on RSVQA-LR (90.7 % in 2023 → 95.2 % in 2026) and RSVQA-HR (65.2 % → 87.8 %), reflecting genuine progress in RS scene understanding. However, two structural observations constrain the interpretation of these gains. First, RSVQA-LR is approaching saturation: the 4.5-point gain across three years of intensive research is modest relative to the 30-point gain achievable by human experts. Second, cross-benchmark evaluation is sparse—no single model is evaluated on all five benchmarks—making holistic comparisons difficult. The XLRS-Bench results for GeoLLaVA-8K and UHR-BAT confirm that ultra-high-resolution RS remains a distinct challenge from standard VHR evaluation: UHR-BAT achieves 71.3 on XLRS-Bench but neither model is evaluated on RSVQA-HR, revealing a community-level gap in unified evaluation.

5.3 SAR-MLLMs: Emerging Territory

SAR imagery is the most practically critical non-optical RS modality: its cloud-penetrating, day-and-night operational capability makes it the primary tool for maritime surveillance, flood mapping, and disaster response where optical sensors are degraded by weather or illumination. Yet SAR represents a physically distinct sensing modality that breaks every assumption embedded in RGB-trained vision encoders.

5.3.1 Technical Barriers for SAR-MLLMs

SAR physics versus CLIP pre-training. Optical sensors passively measure solar reflectance: pixel intensity corresponds directly to surface brightness and colour, creating the visual appearance that human photographers document and internet datasets capture [175]. SAR sensors actively transmit microwave pulses and measure the backscattered signal; pixel intensity encodes radar cross-section (RCS), which is determined by surface roughness, dielectric constant, and target geometry—none of which map to any concept learned from internet images [178]. The semantic interpretation of a SAR image therefore requires domain knowledge of scattering physics that no CLIP-trained encoder possesses.

Speckle noise. SAR images are corrupted by speckle—a multiplicative noise pattern arising from coherent interference of backscattered signals from multiple sub-resolution scatterers [179]. Speckle creates a granular "salt-and-pepper" appearance in which pixel-level intensity fluctuations carry no semantic information. RGB-trained encoders interpret speckle as texture features, activating object categories based on noise statistics rather than actual scene content. Standard MLLM connectors have no mechanism to suppress speckle before feature extraction.

Scattering mechanism diversity. Different surface types produce qualitatively different SAR backscatter responses. Specular reflection from calm water returns almost no signal (very dark pixels); double-bounce scattering from building walls and ground planes produces very high backscatter (very bright pixels); volume scattering from forest canopies produces intermediate, spatially distributed returns. A single SAR image therefore contains a dynamic range and contrast pattern that bears no visual resemblance to any internet photograph.

Figure 16 illustrates these differences by contrasting optical and SAR representations of an identical harbour scene, with annotations explaining the physical origin of the SAR appearance.

5.3.2 Existing SAR-MLLM Work

Despite the technical barriers, the 2025–2026 period has seen a nascent but active body of work on SAR-aware language-grounded models. Table 15 summarises the published SAR-MLLM systems and datasets.

Several structural observations emerge from Table 15. First, no SAR-specific RS-MLLM achieves competitive performance on standard SAR object detection benchmarks (SSDD, HRSID, SAR-Ship)—a direct consequence of the scattering-physics barrier. Second, the domain adaptation strategies deployed so far (joint optical–SAR training, SAR-specific fine-tuning) teach the model *what* SAR images look like without encoding *why* they look that way. Third, no SAR-MLLM produces OBB output, despite ship detection in SAR being the canonical OBB task in the SAR domain.

The physics-constrained SAR–MSI fusion approach developed in our prior work [183] demonstrates that incorporating Rayleigh-scattering constraints into the data-level fusion pipeline substantially improves cross-sensor feature alignment. Extending this principle to the RS-MLLM training objective—through a physics-informed loss term that penalises responses inconsistent with SAR backscatter physics—represents a concrete direction for closing the SAR performance gap. We return to this direction in the future research agenda (Sect. 9).

5.4 Hyperspectral MLLMs: The Final Frontier

Hyperspectral imaging (HSI) represents the most severe modality gap in the RS-MLLM landscape. No dedicated HSI RS-MLLM has been published as of mid-2026, despite HSI being a primary modality for mineral mapping, precision agriculture, environmental monitoring, and military target discrimination.

5.4.1 Technical Barriers for HSI-MLLMs

Dimensional incompatibility. An HSI data cube has dimensions $H \times W \times B$ where $B \in [200, 400]$ spectral bands. Vision encoders expect $H \times W \times 3$ (RGB) input. The standard workaround—PCA dimensionality reduction from B bands to 3 principal components—preserves variance but destroys spectral structure: the three PCA components are linear mixtures of all B bands and do not correspond to any physically meaningful spectral interval [176]. The spectral information that makes HSI valuable is largely discarded in the process.

Spectral signature as primary discriminant. In optical RGB imagery, texture, shape, and colour provide the primary discriminating features between object categories. In HSI, the spectral signature of each pixel—its reflectance across 200–400 wavelengths—is the primary discriminant [92]. Two surfaces that appear visually identical in RGB can be uniquely distinguished by their HSI spectral signatures: for example, healthy vegetation and drought-stressed vegetation have similar RGB appearances but diverge significantly in their NIR and SWIR reflectance profiles [184]. An RS-MLLM with an RGB encoder discards precisely the information needed to make this distinction.

Figure 17 illustrates spectral signatures for four land-cover types across the 400–2500 nm range, showing where RGB sampling is adequate and where it fails critically.

5.4.2 HM-Bench: Empirical Confirmation of the HSI Gap

HM-Bench [21], released in April 2026, provides the first controlled empirical evaluation of MLLM performance on HSI-specific tasks. The benchmark comprises 19,337 expert-annotated samples from public HSI datasets, organised into 13 task categories: spectral unmixing, material identification, anomaly detection, multi-temporal change detection, spatial classification, band selection, target detection, vegetation analysis, soil mapping, water quality assessment, urban land-cover mapping, mineral mapping, and spatial–spectral reasoning.

The HM-Bench evaluation of 18 representative MLLMs yields an unambiguous finding: all 18 evaluated models show significant difficulty on complex spatial–spectral reasoning tasks, including spectral unmixing and change detection [21]. This is not a marginal performance gap—it is a systematic failure across all tested models and all model families (CLIP-based, instruction-tuned, vision-language). The mechanistic explanation is consistent with the dimensional incompatibility analysis above: models apply RGB visual priors to PCA-compressed HSI channels, producing responses driven by colour appearance rather than spectral physics. This failure mode is the empirical foundation of the Type III (Spectral) hallucination introduced in Sect. 7.

5.5 Modality Distribution in RS-MLLM Literature

Figure 18 visualises the distribution of RS-MLLM papers across sensor modalities, based on the quantitative synthesis of the 210 papers included in this systematic review.

Figure 18 confirms the optical dominance quantitatively. Panel (b) reveals two important temporal trends: first, SAR-focused RS-MLLM papers accelerate sharply in 2025–2026, driven by SARChat-Bench-2M and SARLANG-1M; second, HSI activity begins only in 2025–2026 and at a much lower absolute rate, confirming that HSI remains the least-developed modality in the RS-MLLM landscape. The absence of HSI papers prior to 2025 is consistent with the HM-Bench finding: the evaluation infrastructure needed to demonstrate and measure HSI failure modes did not exist until HM-Bench was released in April 2026.

5.6 Cross-Modal Fusion Gaps

The most sophisticated RS applications require simultaneous reasoning across multiple sensor modalities: SAR provides all-weather structural coverage; optical provides spectral colour context; HSI provides material-composition fingerprints. As established in the modality coverage analysis of Sect. 3.5, the cross-modal fusion matrix has five of nine cells entirely empty—all involving HSI.

SAR–Optical fusion is the most practically valuable cross-modal RS task and the most developed. EarthGPT [22] is the first published RS-MLLM to explicitly train on paired SAR–optical data within a unified model (MMRS-1M), though its fusion strategy is simple modality-specific feature concatenation without any physics-informed constraint. Earth-OneVision [20] substantially extends multi-sensor scope to six modalities (optical, SAR, infrared, multispectral, temporal, and video), covering nine task categories—representing the most comprehensive sensor and task coverage of any published RS-MLLM. However, Earth-OneVision similarly uses a unified optical-domain encoder for all modalities without physics-informed specialisation, meaning that SAR backscatter physics and multispectral spectral properties are not explicitly encoded. The Rayleigh-constrained approach in our prior work [183] demonstrates that incorporating SAR backscatter physics into the data-level fusion pipeline significantly improves cross-sensor alignment compared to this naive concatenation strategy.

HSI–Optical fusion has zero published RS-MLLM implementations. The technical barriers are threefold: (i) no large-scale co-registered HSI–optical dataset exists at the scale required for MLLM training; (ii) no HSI-compatible encoder exists that can be paired with an optical MLLM encoder; and (iii) the Type III spectral hallucination problem (Sect. 7) would corrupt HSI representations even if a fusion architecture were available.

SAR–HSI fusion similarly has no published RS-MLLM implementations. While SAR–HSI fusion has been studied for hyperspectral classification using conventional deep learning [43], extending this to the MLLM framework requires both a physics-aware SAR encoder and a spectral-physics-aware HSI encoder—neither of which exists in published form.

Summary

The multi-modal coverage analysis reveals a three-tier structure in RS-MLLM development maturity. Optical RS-MLLMs are approaching performance saturation on existing benchmarks and represent a well-established research direction. SAR-MLLMs are nascent but active: five to six systems exist, driven by the availability of SARChat-Bench-2M and SARLANG-1M, but none addresses the scattering-physics barrier with a dedicated physics-informed encoder, and none achieves OBB detection in SAR imagery. Although Earth-OneVision [20] substantially expands sensor and task coverage—processing optical, SAR, infrared, multispectral, temporal, and video modalities across nine task categories—it uses a shared optical-domain encoder for all modalities without physics-specific encoding, and does not support HSI. HSI-MLLMs do not yet exist as a dedicated model class: HM-Bench provides evaluation infrastructure and empirical confirmation of systematic failure across 18 tested models, but no trained HSI-dedicated RS-MLLM has been published. The remaining structural gaps—broad HSI coverage, physics-specific modality encoding (SAR backscatter, spectral unmixing), systematic cross-sensor evaluation protocols, and operationally grounded reasoning—cannot be closed by scaling multi-sensor training data alone; they require modality-specific encoder design and physics-informed training objectives, as analysed in the future directions of Sect. 9.

6 Training Efficiency and Parameter-Efficient Fine-Tuning

This section synthesises how RS-MLLM researchers have adapted general parameter-efficient fine-tuning (PEFT) methods for the RS domain. We note at the outset that systematic comparative benchmarking of PEFT methods specifically for RS-MLLMs remains limited in the published literature: most papers report their chosen training strategy without controlled ablations against alternatives. Our analysis therefore draws from individual papers reporting their chosen training configuration rather than from controlled comparisons. We identify this absence of RS-specific PEFT benchmarking as an open research gap.

6.1 Why Training Efficiency Matters in RS-MLLMs

Three compounding pressures make training efficiency a first-order concern for RS-MLLM development, distinguishing it from the general MLLM setting.

Data scarcity. The largest general MLLM instruction datasets contain hundreds of millions of examples: LLaVA-1.5 uses 665K conversation pairs; ShareGPT4V scales to 100K high-quality captions [10, 11]. RS instruction datasets are 10–100$\times$ smaller: GeoChat-Instruct [40] contains 318K pairs; SkyEye-968K [47] contains 968K; LHRS-Align [185] contains 1.15M. Only Earth-OneVision [20] reaches 34M pairs—and this required an industry-scale data pipeline that is unavailable to most academic RS groups. The data-scarcity constraint makes overfitting a genuine risk when fully fine-tuning large LLMs on RS data, further motivating PEFT.

Compute constraints. Full fine-tuning of a 7B-parameter LLM backbone requires approximately 14 GB of GPU memory per parameter copy in bf16 precision, before accounting for gradients and optimiser states. In practice, full fine-tuning of a 7B LLM with AdamW requires approximately 112 GB of GPU VRAM—far beyond the 2–4 A100 GPU (80 GB each) budget typical of academic RS research groups. LoRA [46] reduces trainable parameters to fewer than 1 % of total, cutting the gradient-related memory overhead by the same proportion and making RS-MLLM training feasible within academic compute constraints.

Domain-specific adaptation versus catastrophic forgetting. General LLMs encode broad world knowledge—geography, physics, language patterns—that is directly useful for RS scene description and VQA. Full fine-tuning risks overwriting this knowledge with RS-specific patterns (catastrophic forgetting), particularly when the RS instruction dataset is small. PEFT methods that freeze the LLM backbone and train only lightweight adapter components preserve the general knowledge while adapting to the RS domain, providing a better inductive bias for the data-scarce RS setting.

Figure 19 situates the principal PEFT methods used in RS-MLLMs within a unified taxonomy, organised by which components of the architecture they adapt.

6.2 PEFT Methods in RS-MLLMs

6.2.1 LoRA: The Dominant Choice

Low-Rank Adaptation (LoRA) [46] inserts trainable rank-r decomposition matrices $\Delta W = BA$ (where $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$, $r \ll \min(d,k)$) into the attention projection layers of the frozen LLM backbone. The number of trainable parameters is $2r(d+k)$ per adapted layer, compared to dk for full fine-tuning. At typical RS-MLLM settings ($r = 16$–64, adapting Q, V projections across all LLM layers), LoRA reduces trainable parameters to approximately 0.1–1 % of the backbone total.

LoRA is adopted by 19 of the 26 Corpus B RS-MLLMs: GeoChat [40], RSGPT [19], SkyEyeGPT [47], LHRS-Bot [121], LHRS-Bot-Nova [138], VHM [139], RS-LLaVA [134], H2RSVLM [140], GeoPixel [142], GeoPix [144], SkySenseGPT [119], GeoLLaVA-8K [37], UHR-BAT [36], TEOChat [120], GeoGround [141], GeoCode-GPT [187], and SARVLM [147].

Among those reporting the specific rank setting, GeoChat uses $r = 64$ on a Vicuna-7B backbone [40]; SkyEyeGPT uses $r = 16$ on InternLM-7B [47]. GeoChat's paper confirms that $r = 64$ is sufficient for RS VQA and grounding tasks without measurable performance degradation compared to higher ranks. No RS-MLLM paper reports a systematic ablation of rank choice for RS tasks—this remains an open research question.

6.2.2 QLoRA: Memory-Efficient Variant

QLoRA [188] combines 4-bit quantisation of the frozen backbone with LoRA adapter training in full precision. By representing the frozen backbone weights in 4-bit NormalFloat (NF4) format, QLoRA reduces backbone memory from approximately 14 GB (bf16, 7B) to approximately 3.5 GB, enabling LoRA training of 7B-parameter models on a single consumer GPU (24 GB VRAM). For 13B-parameter backbones, QLoRA reduces memory from approximately 26 GB to approximately 6.5 GB, making 13B RS-MLLMs feasible on a single A100 GPU.

GeoCode-GPT [187] applies QLoRA for RS geography domain adaptation. Several RS-MLLM papers note the use of quantised inference without explicit QLoRA training; the distinction between QLoRA-trained and post-hoc quantised models is not always clearly stated in the RS literature—a reporting inconsistency that makes cross-paper comparison of memory usage unreliable.

6.2.3 Adapter Modules and Bias Tuning

Adapter modules [189] insert small bottleneck networks ($d \to r \to d$ with $r \ll d$) into each transformer layer. While widely used in NLP, adapters are less common in RS-MLLMs because they modify the LLM *activations* rather than its *weights*, introducing inference latency. H2RSVLM [140] adopts a visual adapter for RS-specific feature enhancement at the connector stage.

EarthGPT [22] uses a *bias tuning* strategy (related to BitFit [190]): only the bias parameters and RMSNorm scaling factors of the LLM backbone are updated, while all other weights remain frozen. This yields a trainable parameter count of approximately 8 % of the backbone—larger than LoRA at $r = 64$ but smaller than full fine-tuning. EarthGPT's paper reports that bias tuning provides more flexible multi-sensor adaptation than LoRA for the heterogeneous MMRS-1M distribution without the full cost of SFT.

Table 16 consolidates the PEFT configurations reported across the 26 Corpus B RS-MLLMs.

6.3 Token Compression for Ultra-High-Resolution RS

Standard MLLM connectors map an $H \times W$ image at patch size p to $(H/p)^2$ visual tokens. At the ViT-L/14 patch size of $p = 14$ pixels, a 1024×1024 px RS image produces ≈ 5400 tokens—already exceeding the 4096-token context window of early LLaMA-2 models. RS VHR imagery at 4096×4096 px produces $\approx 86\,000$ tokens, which is computationally intractable without token reduction. Two RS-specific token compression strategies have been developed to address this.

GeoLLaVA-8K: Coarse-to-Fine Tiling.

GeoLLaVA-8K [37] processes ultra-high-resolution RS images (up to 8192×8192 px) through a hierarchical tiling approach. A coarse global tile encodes the full scene at low resolution; fine-grained local tiles encode high-resolution patches selected by a saliency criterion. The total token budget is capped at a fixed value (reported as 3584 tokens [37]), regardless of input resolution. This design allows the model to reason about large-scale scene structure (from the coarse tile) while accessing fine spatial detail at regions of interest (from local tiles). GeoLLaVA-8K achieves a composite score of 62.4 on XLRS-Bench [114, 191], confirming functional performance at ≥ 4096 px resolution.

UHR-BAT: Budget-Aware Token Compression.

UHR-BAT [36] applies learned token pruning before the LLM connector, retaining only the tokens with the highest attention saliency scores while discarding background tokens. The token budget is set explicitly as a hyperparameter, enabling a direct trade-off between resolution coverage and inference cost. UHR-BAT achieves 71.3 on XLRS-Bench [114]—the highest published score on this benchmark—demonstrating that learned token pruning is more effective than fixed-ratio tiling for high-resolution RS VQA.

Table 17 compares the published ultra-high-resolution RS-MLLM systems across resolution capability, token strategy, and benchmark performance.

Figure 20 illustrates the resolution–accuracy trade-off as a function of token budget, using reported data from GeoLLaVA-8K and UHR-BAT.

6.4 Instruction Data Quality and RS-MLLM Training

The PEFT method choice interacts with instruction data quality in a way that is specific to the RS domain. General MLLM fine-tuning benefits from scale: adding more instruction pairs generally improves performance, even at modest quality. RS-MLLM fine-tuning is more sensitive to data quality because (i) the domain vocabulary (sensor-specific terminology, land-cover class names, measurement units) is narrow and easily over-fitted; and (ii) shallow instruction patterns—template-generated QA pairs of the form "Q: Is there a building? A: Yes."—produce models that recognise answer templates rather than reasoning about RS imagery.

DDFAV [35] provides the most systematic evidence for the data quality hypothesis in the RS-MLLM setting. The paper demonstrates that replacing template-generated instruction pairs with *reasoning-intensive* pairs—instructions that require multi-step spatial reasoning or attribute inference rather than binary existence answers—substantially reduces hallucination rates and improves performance on fine-grained RS VQA benchmarks. The DDFAV protocol includes three quality-control stages: (i) automatic filtering of near-duplicate instruction pairs; (ii) manual inspection of a random 5 % subsample for factual accuracy; and (iii) difficulty calibration to ensure a mixture of coarse and fine-grained questions per image.

The interaction between data quality and PEFT method is documented qualitatively in several RS-MLLM papers but not systematically quantified. Curriculum-based instruction tuning [192] offers a structured alternative. GeoChat [40] notes that LoRA with $r = 64$ on low-quality instruction data produces a model that is easily confused by phrasing variations in RS VQA—suggesting that high LoRA rank amplifies both the signal and the noise in training data. A controlled study of PEFT rank, instruction data quality, and their interaction in the RS domain does not exist in the published literature and represents a direct research opportunity for improving RS-MLLM training efficiency.

6.5 Domain-Specific RS-MLLMs versus General-Purpose CV-MLLMs

A question that has become central to the RS-MLLM field—and that the concurrent survey of Ma et al. [30] raises directly—is whether domain-specific RS fine-tuning actually outperforms large general-purpose CV-MLLMs applied zero-shot to RS imagery. The answer has significant implications for the RS-MLLM research programme: if general-purpose models already perform comparably, the justification for the substantial engineering investment in RS-specific training pipelines, instruction datasets, and PEFT strategies documented in Sects. 6.2.1–6.4 requires careful re-examination.

6.5.1 The Diagnostic Question

General-purpose CV-MLLMs—Qwen2.5-VL [193], InternVL2 [194], LLaVA-OneVision [195], GPT-4V [13], and Gemini [14]—are trained on internet-scale image–text data covering hundreds of millions of natural photographs, diagrams, charts, and documents. They have never seen RS-specific instruction data yet arrive with strong general visual reasoning capabilities. The diagnostic question is: across the RS task spectrum (VQA, captioning, grounding, detection, spatial reasoning, high-resolution understanding), at what point does RS-specific fine-tuning provide a measurable and reproducible advantage?

6.5.2 Evidence from the Literature

Table 18 summarises what can be said about CV-MLLM versus RS-MLLM performance based on published evaluations. We make a critical methodological note that governs the entire table: **direct comparison is only valid when the same evaluation protocol, benchmark split, and prompt format are used**. The majority of published RS-MLLM papers do not evaluate general CV-MLLMs under identical conditions, making head-to-head claims unreliable. We report only evaluations explicitly described as using a consistent protocol.

6.5.3 Architectural Interpretation of the Diagnostic Finding

The finding that general CV-MLLMs can match RS-specific models on some tasks is, from an architectural perspective, *expected* rather than surprising, and does not undermine the RS-MLLM research programme. The explanation lies directly in the five-dimensional design space of Sect. 3.

Where CV-MLLMs are competitive: tasks that do not require RS-specific design. Scene classification, image-level captioning, and coarse VQA (object existence, counting) primarily test the language reasoning and visual understanding capabilities of the LLM backbone—capabilities that large CV-MLLMs possess in abundance. Furthermore, recent general models (Qwen2.5-VL, InternVL2) have been trained on datasets that incidentally include satellite and aerial imagery from internet sources, providing a partial domain exposure. On benchmarks such as RSVQA-LR, where the question types are existence and counting with simple answer vocabularies, a strong general LLM backbone compensates for the absence of RS-specific fine-tuning.

Where RS-specific training matters: tasks requiring RS-specific design. Three task families expose a persistent gap between general and RS-specific models, corresponding directly to the three most under-addressed design dimensions:

1. **OBB detection** ($\mathcal{S}$-dimension gap). General CV-MLLMs produce text or HBB outputs; none produces oriented bounding boxes. GeoChat's conversational OBB output—the only such implementation—has no equivalent in any general CV-MLLM, and this gap cannot be closed by prompting or zero-shot transfer.
2. **Multi-sensor reasoning** ($\mathcal{M}$-dimension gap). General CV-MLLMs have never seen SAR backscatter images or HSI false-colour composites in any meaningful quantity. HM-Bench [21] confirms that all 18 tested general and RS-specific MLLMs fail systematically on HSI spectral reasoning, but the failure is *architecturally worse* for general models because their encoders have no SAR or HSI exposure whatsoever.
3. **Fine-grained RS grounding** ($\mathcal{S}$-dimension, Level 2–3). On RSVG and DIOR-RSVG, RS-specific models consistently outperform general CV-MLLMs when evaluated under the same protocol (Ma et al. [30]), because RS grounding requires RS-category vocabulary, nadir-view object appearance priors, and RS-specific referring expression understanding that general training data does not provide at sufficient density.

The resolution dimension: a shifting boundary. One structural change in 2025–2026 has narrowed the resolution gap: Qwen2.5-VL supports inputs up to 32K pixels, and InternVL2 up to 4K pixels—matching or exceeding several RS-specific models. This means the ultra-high-resolution advantage that motivated GeoLLaVA-8K [37] and UHR-BAT [36] is partially eroded by general model capability growth. The remaining RS-specific advantage is in the *token efficiency* and *RS-saliency-aware* compression strategies, not raw resolution support.

6.5.4 Implications for RS-MLLM Research

The CV-MLLM diagnostic finding carries three concrete implications for how the RS-MLLM field should allocate its research effort. A fourth cross-cutting observation concerns **multimodal generalisation**: general CV-MLLMs (Qwen2.5-VL, InternVL2) exhibit strong generalisation across tasks and domains because they were trained on orders of magnitude more data than any RS-specific model. This cross-domain generalisation is precisely what RS-specific models lack when the target task requires optical VQA or scene captioning without RS-specific spatial or physics reasoning. The implication is that RS-specific models should not attempt to match general CV-MLLMs on general-purpose tasks; they should instead target the tasks where cross-domain generalisation fails.

Implication 1: Saturating tasks should be retired from RS-MLLM evaluation. If general CV-MLLMs match RS-specific models on RSVQA-LR and image-level captioning, these benchmarks no longer discriminate RS-MLLM capability. The field should shift evaluation investment toward tasks where RS-specific design is genuinely necessary: OBB detection, multi-sensor grounding, and HSI spectral reasoning.

Implication 2: RS fine-tuning justification must be task-specific. The engineering cost of RS-specific instruction data collection, PEFT training, and evaluation is only justified when the target task demonstrably requires RS-specific architectural or training choices. The design-space analysis of Sect. 3 provides the framework for determining which dimensions require RS-specific design—and which can be inherited from general CV-MLLM training.

Implication 3: The strongest RS-MLLM research direction is non-overlapping with general CV-MLLMs. Physics-informed SAR encoders (Direction D3, Sect. 9.6), spectral HSI encoders (D5), and OBB-capable training (D1) address capabilities that no general CV-MLLM possesses and that internet-scale pre-training cannot provide—because SAR backscatter statistics and HSI spectral signatures are absent from internet image corpora. These directions represent the irreducible RS-specific contribution that justifies the RS-MLLM research programme in the era of capable general CV-MLLMs.

Summary

The CV-MLLM diagnostic finding—that general-purpose models can match RS-specific MLLMs on several tasks—is architecturally expected: it reflects the saturation of tasks that do not require RS-specific design (coarse VQA, scene captioning) rather than a failure of the RS-MLLM research programme. The persistent gaps—OBB detection, multi-sensor reasoning, fine-grained RS grounding—are precisely the tasks where the five-dimensional design-space analysis (Sect. 3) predicts RS-specific design to be necessary, and where general CV-MLLMs have no architectural answer.

Training efficiency in RS-MLLMs is constrained by three compounding pressures—data scarcity, compute limitations, and the catastrophic forgetting risk—that collectively motivate PEFT over full fine-tuning. LoRA is the overwhelmingly dominant PEFT method (19/26 Corpus B models), with rank choices ranging from $r = 16$ (SkyEyeGPT) to $r = 64$ (GeoChat). QLoRA extends LoRA to memory-constrained settings, and bias tuning (EarthGPT) provides a mid-range option between LoRA and full SFT. Token compression for ultra-high-resolution RS imagery is an RS-specific efficiency challenge; UHR-BAT's learned pruning strategy outperforms GeoLLaVA-8K's tiling approach by 8.9 XLRS-Bench points, with the reported values summarised in Fig. 20. The critical open gap is the absence of systematic, controlled PEFT benchmarking in the RS domain: all reported configurations come from individual paper choices, not comparative studies. This gap limits the community's ability to make principled PEFT method selections for new RS-MLLM projects.

7 Hallucination in RS-MLLMs

Hallucination in RS-MLLMs is not merely an accuracy problem—it is a safety problem. When an RS-MLLM fabricates a building where none exists, misidentifies a military installation, or describes an empty harbour as containing vessels, the consequences in operational contexts can be catastrophic. This section provides a systematic hallucination taxonomy for RS-MLLMs that proposes a formal three-way partition (Localization, Discrimination, and Spectral failure modes) grounded in the RS domain challenges of Sect. 2—extending prior work through an empirically grounded third type. To the best of our knowledge, prior RS-MLLM hallucination studies have not explicitly isolated spectral hallucination as a distinct failure category; our taxonomy extends prior localisation/discrimination frameworks by incorporating spectral failure evidence from HM-Bench [21].

7.1 Why RS Hallucination Is More Dangerous

Hallucination in general-domain MLLMs—describing objects absent from a natural photograph—is primarily a reliability inconvenience. In remote sensing, the same failure mode carries qualitatively higher stakes across three critical application domains.

Disaster response. Post-disaster RS-MLLM outputs guide search-and-rescue operations: teams are directed to locations identified as containing collapsed structures, blocked roads, or survivor clusters. A localization hallucination that places a collapsed building 500 m from its true position redirects rescue teams away from survivors. In the 2023 Turkey earthquake response, RS-based damage mapping errors were documented as a primary cause of delayed rescue in rural areas [2]. The time cost of an RS-MLLM hallucination is not a wrong caption—it is measured in lives.

Infrastructure monitoring. Automated inspection of pipelines, power lines, and bridge structures via RS-MLLMs depends on accurate damage classification. A discrimination hallucination that misclassifies a structural crack as surface discolouration triggers either dangerous neglect or unnecessary and costly mobilisation of maintenance teams. Either failure mode has direct economic and safety consequences disproportionate to those of a mislabelled natural image.

Agricultural and environmental mapping. Crop-type misclassification propagates into yield estimates, insurance payouts, and subsidy allocations affecting millions of smallholder farmers. A spectral hallucination that interprets drought-stressed vegetation as bare soil—because their RGB appearances are similar but their spectral signatures diverge in the NIR—leads to systematically wrong irrigation and intervention decisions.

These high-stakes failure modes have no close parallel in the natural-image MLLM setting, where the worst case of hallucination is typically a wrong object label or a confabulated description. Existing work has begun to acknowledge RS hallucination as a distinct research problem: DDFAV [35] identifies low-quality instruction data as a root cause; RSHallu [196] provides the first diagnostic benchmark; Seeing Clearly [197] proposes training-free mitigation. Figure 21 illustrates a canonical RS hallucination failure.

7.2 Existing RS Hallucination Work

We survey the four primary contributions to RS hallucination research before presenting our taxonomy. We explicitly acknowledge where our taxonomy builds on rather than replaces this prior work.

DDFAV (MDPI Remote Sensing, February 2025) [35]. DDFAV establishes that shallow, template-generated instruction data is a primary structural cause of RS-MLLM hallucinations. The paper demonstrates that models trained on low-quality instruction pairs over-fit to answer templates rather than learning genuine spatial reasoning, producing confident but physically ungrounded responses. DDFAV contributes a quality-controlled instruction protocol and confirms that data quality outweighs data quantity for reducing RS hallucination rates.

Aerial Mirage (2026) [198]. Aerial Mirage provides the first formal categorisation of RS hallucination into two failure types based on the model's ability to localise the queried target. *Type 1 ("Cannot Find")* occurs when model attention disperses over irrelevant image regions and the target is never localised. *Type 2 ("Cannot See Clearly")* occurs when the correct region is attended to but the fine-grained class is misidentified. This two-type framework directly informs the first two types of our taxonomy; we adopt Aerial Mirage's categorisation and extend it rather than proposing an independent alternative.

RSHallu (2026) [196]. RSHallu is a systematic benchmark designed as a diagnostic tool for hallucinations in the remote sensing domain. It evaluates RS-MLLMs on object existence, object attribute, and object relationship queries across diverse RS scene types, documenting high confusion rates among visually similar object classes (car / truck / bus in parking lots; building / shadow / road in urban scenes). RSHallu provides the primary empirical grounding for our Type II (Discrimination) hallucination category.

Seeing Clearly (March 2026) [197]. Seeing Clearly proposes a training-free inference-time mitigation for RS hallucination by correcting diffuse attention maps before response generation. The method addresses both Aerial Mirage types without fine-tuning, confirming that hallucination in RS-MLLMs has an attentional root cause amenable to inference-time correction.

Relationship to our taxonomy. We extend Aerial Mirage's two-type framework with a third type (Type III: Spectral Hallucination) grounded in the empirical findings of HM-Bench [21] on spectral reasoning failure across 18 MLLMs. Types I and II are adopted from Aerial Mirage with refined causal analysis; Type III is new and, to the best of our knowledge, prior RS-MLLM hallucination studies have not explicitly isolated spectral hallucination as a distinct failure category in the RS hallucination literature.

7.3 A Three-Type RS-MLLM Hallucination Taxonomy

We propose a three-type taxonomy of RS-MLLM hallucination failures, organised by the *stage* at which the failure occurs in the model processing pipeline (Fig. 22): failure to *locate* the query target (Type I), failure to *classify* a located target (Type II), and failure to *interpret sensor modality* correctly (Type III).

7.3.1 Type I: Localization Hallucination

Definition. The model generates a confident description of an object at a location where that object does not exist, caused by failure to correctly localise the query target in the RS scene.

Causal mechanism. RS images cover large geographic extents: a single VHR tile at 0.3 m GSD spanning 2×2 km contains tens of thousands of objects, roads, and structures, presenting the model's cross-attention with an overwhelming localisation challenge. CLIP-trained encoders exacerbate this: the nadir-view representation of a ship from 500 m altitude shares almost no visual features with the ground-level ship images that dominate CLIP training data, eliminating the visual priors that would normally guide attention toward target-consistent regions. The result is diffuse attention—documented by Aerial Mirage [198]—in which the model generates objects that are contextually plausible (harbour $\rightarrow$ ships) but physically absent.

Affected tasks. VQA (object existence queries), visual grounding, temporal change detection (localising which regions changed between acquisitions).

Distinctive example. A model queried about aircraft at a military airbase attends to concrete apron markings and fuel lines—contextually associated with aircraft—and reports the count and types of aircraft that would typically occupy such positions, even when the apron is empty. The response is context-conditionally hallucinated: it reflects learned co-occurrence statistics, not the visual content of the specific image.

Relationship to prior work. Directly corresponds to Aerial Mirage Type 1 [198] and Seeing Clearly Type 1 [197]. We refine the causal analysis by linking diffuse attention to the nadir-view domain gap (Sect. 2.1.1) rather than treating it as a general attention failure.

7.3.2 Type II: Discrimination Hallucination

Definition. The model correctly localises the queried region but assigns an incorrect fine-grained class, producing a spatially accurate but semantically wrong response.

Causal mechanism. RS objects at sub-metre resolution—cars, trucks, buses, aircraft types, ship classes—are visually similar from nadir. A compact car and a light van at 0.5 m GSD differ by ≈ 3–5 visual tokens at ViT-L/14 patch size; the fine-grained visual information needed to resolve this distinction is discarded by MLP or Q-Former connectors that compress the visual token sequence. Additionally, RS instruction datasets are heavily biased toward coarse-category labels: GeoChat-Instruct [40] uses categories from DOTA [38] (large-vehicle, small-vehicle, ship) rather than sub-type labels, biasing trained models toward coarse discrimination rather than the fine-grained classification required for operational use.

Evidence. RSHallu [196] documents high confusion rates among similar RS categories: car / truck / bus in parking lots; roof-type misidentification in urban scenes; agricultural crop-type confusion. DDFAV [35] confirms that models trained on template-generated instruction data systematically fail at fine-grained tasks while achieving adequate coarse VQA performance.

Affected tasks. Fine-grained VQA, OBB sub-type classification (e.g., civilian versus military aircraft), land-cover classification at detailed category levels, material identification.

Relationship to prior work. Corresponds to Aerial Mirage Type 2 [198] and RSHallu discrimination errors [196]. We contribute additional causal analysis connecting the failure to instruction data category distribution and token budget allocation.

7.3.3 Type III: Spectral Hallucination

Definition. The model applies RGB visual priors to non-RGB sensor data (SAR backscatter, HSI false-colour composites, multispectral band combinations), generating descriptions driven by colour appearance rather than sensor-specific physics.

Causal mechanism (hypothesised). Most currently documented RS-MLLM encoders—including CLIP [31], EVA-CLIP [32], DINOv2 [33], and RemoteCLIP [50]—were pre-trained predominantly on optical RGB imagery. While recent systems such as Earth-OneVision [20] extend encoder inputs to SAR, infrared, and temporal modalities, no published RS-MLLM encoder employs physics-specific SAR backscatter or HSI spectral pre-training objectives. Consequently, when these encoders receive a SAR backscatter image or an HSI false-colour composite, they may lack the sensor-physics grounding needed to distinguish the input from a standard optical photograph. The proposed hypothesis is that such encoders tend to map pixel values toward the nearest RGB-trained feature cluster, without representing the physical meaning of the sensor modality.

A concrete illustrative scenario for HSI: researchers routinely create HSI false-colour composites by mapping the near-infrared (NIR) band to the red display channel. In such composites, healthy vegetation appears *bright red* (high NIR reflectance). An RS-MLLM encoder, receiving a bright-red region, may activate RGB-trained fire, bare-earth, or red-soil features. The model could consequently describe healthy farmland as "fire-damaged land" or "exposed red soil"—a response that is confident, specific, and physically wrong. For SAR: calm water returns nearly zero backscatter (dark pixels); double-bounce scattering from building corners produces very bright pixels. An RGB-oriented encoder may interpret this dark-adjacent-to-bright pattern as shadows adjacent to illuminated structures—activating road, building, and shadow features rather than the correct SAR scattering interpretation.

These failure patterns, if consistently reproduced, would constitute *systematic* failures: the same physically wrong interpretation reproduced whenever the same sensor type is presented. Note that this causal account is a mechanistic hypothesis, not a proven mechanism; controlled ablation studies with physics-specific encoders would be needed to confirm it.

Empirical grounding. HM-Bench [21] provides controlled empirical evidence that is consistent with the Type III hypothesis. The benchmark evaluates 18 representative MLLMs on 13 HSI-specific task categories including spectral unmixing, material identification, and multi-temporal change detection, using PCA-based false-colour composite images and structured textual reports (rather than raw HSI cubes). All 18 evaluated models exhibit systematic difficulty on complex spatial–spectral reasoning tasks [21], providing empirical motivation for the proposed Type III spectral-failure category. This pattern is consistent across all model families and all task categories evaluated in HM-Bench. The HM-Bench results are *consistent with the hypothesis* that RGB-oriented representations and inadequate spectral grounding contribute to Type III failures; however, because HM-Bench uses PCA-compressed composites rather than raw spectral cubes, it does not directly isolate the encoder mechanism. Controlled ablation studies comparing RGB-trained encoders with physics-specific spectral encoders would be needed to definitively confirm the causal account.

Supporting evidence: work on MLLM robustness to image degradation [199] demonstrates that MLLMs trained on RGB exhibit strong blur bias, mode collapse, and taxonomy misalignment when presented with inputs that deviate from the RGB distribution. Non-RGB sensor imagery represents an extreme case of this distributional shift.

Affected tasks. HSI land-cover classification, SAR scene interpretation, material identification, crop-type mapping from multispectral imagery with non-standard band combinations, SAR-based vessel detection.

Why this type has not been previously identified. Aerial Mirage [198], RSHallu [196], and Seeing Clearly [197] evaluate RS-MLLMs exclusively on optical RGB imagery. DDFAV [35] similarly constructs its instruction data from optical sources. To the best of our knowledge, none of these works explicitly categorises sensor-modality misinterpretation as a distinct hallucination type. HM-Bench [21] is the first to systematically test non-RGB RS inputs, revealing a failure mode that is categorically different from Types I and II in both cause and mitigation strategy.

Table 19 summarises the three types across all defining dimensions.

7.4 Mitigation Strategies

The three hallucination types require distinct mitigation strategies, reflecting their different causal mechanisms.

Data-side mitigation (Types I and II). DDFAV [35] demonstrates that replacing shallow template-generated instruction pairs with reasoning-intensive pairs substantially reduces both Type I and Type II hallucination rates. The critical design principles are: (i) explicit negative examples—instructions querying absent objects, ensuring the model learns to say "no object present" rather than generating a plausible but wrong description; (ii) graduated difficulty from coarse scene-level to fine-grained object-level queries; and (iii) diversity of phrasing for the same spatial relationship. The DDFAV instruction corpus (1.6 GB) represents the largest publicly available quality-controlled RS instruction dataset.

Training-side mitigation (Types I and II). RemoteShield [200] applies cross-condition preference learning—analogous to RLHF—to RS-MLLMs. Paired (correct, hallucinated) response examples for RS scenes are used to train the model to prefer physically grounded responses via a contrastive objective. This approach directly targets the model's tendency to generate contextually plausible but visually ungrounded responses.

Inference-side mitigation (Type I). Seeing Clearly [197] proposes a training-free inference-time correction that identifies diffuse attention patterns (the signature of Type I) and applies attention sharpening before response generation. The method requires no additional fine-tuning and is applicable to deployed RS-MLLMs. Its limitation is that it operates on attention maps and cannot address the pre-attentional Type III failure: spectral hallucination occurs in the encoder before cross-attention is computed.

Type III mitigation (future direction). No published method directly addresses spectral hallucination. Three research directions are plausible. First, modality-conditioned encoders: pre-train separate encoder branches for SAR and HSI with sensor-physics-informed objectives (predicting SAR backscatter coefficient distributions; predicting spectral unmixing endmembers) rather than contrastive image-text objectives. Second, physics-informed loss terms: augment MLLM training with a spectral consistency loss that penalises responses inconsistent with known sensor-physics constraints. The Rayleigh-constrained SAR-MSI fusion approach in our prior work [183] demonstrates that physics constraints improve cross-sensor feature alignment; an analogous constraint applied to the MLLM training objective could suppress Type III hallucination. Third, modality prompt tokens: explicitly signal to the LLM backbone which sensor modality is being processed, analogous to language tokens in multilingual LLMs, enabling the model to switch its interpretation context.

7.5 Hallucination Benchmarks and Gaps

Table 20 compares the four available RS hallucination benchmarks across the dimensions of our taxonomy. Values are as reported in the respective papers.

Three benchmark gaps are evident from Table 20. First, all optical benchmarks (DDFAV, RSHallu, Aerial Mirage) evaluate exclusively on optical RGB imagery, leaving Type III entirely untested and providing no SAR-specific evaluation. Second, HM-Bench [21] tests non-optical modalities (HSI) but reports task performance rather than per-query hallucination labels—making it a *partial* Type III resource rather than a dedicated benchmark. Third, no existing benchmark evaluates cross-type hallucination, in which a single query triggers failures at multiple stages simultaneously (e.g., a Type I localisation failure followed by a Type III spectral misinterpretation of the wrongly attended region).

The missing Type III benchmark.

A comprehensive Type III evaluation benchmark requires: (i) co-registered optical, SAR, and HSI images of the same scene; (ii) per-modality query–response pairs with ground-truth labels independent of colour appearance; and (iii) annotation of whether the model's response was driven by spectral physics or RGB visual priors. No such benchmark exists. Its construction would require approximately 5,000–10,000 co-registered triplets with expert annotation—a tractable scale that represents a concrete and high-impact research contribution for the RS-MLLM community.

Summary

This section introduced a systematic three-type hallucination taxonomy for RS-MLLMs that proposes a formal three-way partition (Localization, Discrimination, and Spectral failure modes) with distinct causal mechanisms grounded in the RS domain challenges of Sect. 2. To the best of our knowledge, this specific partition with its RS-physics causal grounding has not been proposed in prior RS-MLLM hallucination work. Type I (Localization) and Type II (Discrimination) extend and refine the Aerial

Mirage [198] framework with additional causal analysis rooted in the RS domain challenges of Sect. 2. Type III (Spectral Hallucination) is a new type, empirically grounded in HM-Bench [21], that captures a failure mode not, to the best of our knowledge, previously isolated as a distinct category in the RS hallucination literature; our taxonomy extends prior localisation/discrimination frameworks by incorporating this spectral failure evidence. This type requires modality-specific encoder design for mitigation. The benchmark analysis (Table 20) confirms that Type III remains unaddressed by any existing evaluation framework, establishing a clear research priority for the RS-MLLM community.

8 Datasets, Benchmarks, and Structured Quantitative Synthesis

The development trajectory of RS-MLLMs is shaped as much by available datasets and benchmarks as by architectural choices. This section catalogues the training datasets and evaluation benchmarks used across the 210 papers retrieved in this systematic review, then presents a structured quantitative synthesis of publication trends, modality distribution, task coverage, and model backbone usage. All quantitative values in Sect. 8.3 are derived from the PRISMA systematic search process described in Sect. 2.4.

8.1 RS-MLLM Training Datasets

Training datasets for RS-MLLMs must simultaneously address two requirements that are in tension: *scale* (enough examples to fine-tune a large LLM without overfitting) and *quality* (instruction pairs that require genuine spatial reasoning rather than template pattern-matching, as established in Sect. 6.4). Table 21 catalogues the primary RS-MLLM training datasets in chronological order.

Figure 23 visualises the dataset scale evolution from 2016 to 2026 on a logarithmic axis, highlighting the order-of-magnitude growth across each sensor modality.

8.2 RS-MLLM Evaluation Benchmarks

Evaluation benchmarks determine which RS-MLLM capabilities are measurable. For reference, standard RS scene classification and understanding datasets—AID [202], NWPU-RESISC45 [203], PatternNet [204], Million-AID [2], MillionAID scene review [25], and scene understanding surveys [24]—are not MLLM benchmarks but provide evaluation context. Building extraction datasets (WHU [205], SpaceNet [206]), land-cover segmentation (LoveDA [207], OpenEarthMap [208]), UAV classification [209], and change detection (LEVIR-CD [155], SECOND [210], Bi-temporal [211], and multi-temporal RS change analysis [212]) inform the task coverage analysis of Sect. 4. Seasonal variation [213], temporal forecasting [214], and daily commercial imagery (Planet [215], Gaofen [216]) motivate future spatiotemporal RS-MLLM development. Table 22 catalogues the primary RS-MLLM evaluation benchmarks, and Fig. 24 visualises their comparative coverage across six evaluation dimensions.

8.3 Quantitative Synthesis: Publication Trends

The following structured quantitative synthesis is derived from the 97-paper primary RS-MLLM corpus (Corpus A; Sect. 2.4). As defined in Table 3, Corpus A spans five categories: 31 RS-MLLM model studies, 18 benchmark studies, 14 dataset studies, 9 hallucination studies, and 25 foundation-model studies (2022–2026 window). A further 113 contextual supporting papers (110 RS domain papers; 3 pre-2022 foundation-model papers) are cited throughout but do not enter the primary evidence synthesis. Architectural analysis in Sect. 3 focuses on the 31 model studies and 26 Corpus B architectures detailed in Table 10. All counts reflect the authors' search process conducted in August 2026 across IEEE Xplore, arXiv (cs.CV, eess.IV), Springer Link, CVPR, ICCV, NeurIPS, AAAI, and ISPRS proceedings.

8.3.1 Publication Volume and Growth

Figure 25 charts the annual publication volume of the 97 RS-MLLM-specific papers (Corpus A model, benchmark, dataset, and hallucination studies) from 2022 to August 2026. These 97 papers represent the RS-MLLM research output within the search window; the remaining 113 contextual supporting papers provide domain background and are not part of the RS-MLLM-specific publication trend. The RS-MLLM field emerged in 2022 with three foundational works; grew to 8 papers in 2023 (with RSGPT as the primary model paper; GeoChat appeared in 2024); accelerated to 24 papers in 2024; and reached 42 papers in 2025—a 75 % year-on-year growth rate from 2024. Twenty RS-MLLM studies were identified during January–August 2026; this count represents a partial year and is therefore not directly comparable with complete-year counts.

8.3.2 Four-Panel Distribution Analysis

Figure 26 presents a four-panel structured quantitative synthesis of Corpus A (97 primary papers). Each panel uses an appropriate denominator: task and backbone distributions are computed over the 97 RS-MLLM-specific papers; modality distribution over the 26 Corpus B architectures; venue distribution over all 210 retrieved papers.

8.4 Peer-Reviewed vs Preprint Sensitivity Analysis

A recognised concern in fast-moving AI fields is that corpus-level findings may be artefacts of preprint inclusion rather than genuine structural patterns. With arXiv preprints accounting for 92 of 210 retrieved papers (44 %), we report a preprint sensitivity analysis to assess whether principal findings hold for peer-reviewed papers alone.

Analysis A (full retrieved corpus, $n = 210$) is reported throughout this paper. **Analysis B (peer-reviewed only, $n \approx 118$)** restricts to papers published in peer-reviewed venues (IEEE TGRS, ISPRS JP, CVPR, ICCV, NeurIPS, AAAI, ISPRS proceedings, and other journals/conferences).

Table 23 reports the key distributional statistics for both analyses. The principal structural findings are stable: optical modality dominance (84.6 % vs 84.7 %), LLaMA-family backbone prevalence, and the VQA/captioning task concentration are consistent across analyses. The modality gap (HSI at 2.9 % vs 3.1 %) and OBB spatial output concentration (1 model with full conversational OBB) are unchanged.

Summary

The datasets and benchmarks synthesis reveals three structural patterns. First, training data scale has grown four orders of magnitude (from 2,100 pairs in 2016 to 34M in 2026) for optical RS-MLLMs, but SAR and HSI data remain 10–100× smaller than comparable optical resources—a data asymmetry that directly explains the modality coverage gap established in Sect. 5. Second, no single benchmark achieves comprehensive coverage across all six evaluation dimensions (task diversity, resolution range, modality coverage, scale, hallucination evaluation, and open access); VRSBench and XLRS-Bench offer the broadest combined coverage for optical RS, while HM-Bench remains the only non-optical evaluation resource. VLRS-Bench [99] is a notable recent addition directly relevant to this survey's central thesis (perception → reasoning → geospatial intelligence): it specifically targets complex RS reasoning across up to eight temporal phases, filling a critical gap in Level-4 spatiotemporal evaluation that no prior RS-MLLM benchmark addressed. Third, the synthesis confirms the optical dominance (84.6 % of the 26 Corpus B architectures) and arXiv preprint prevalence (92/210 = 44 % of retrieved papers; Corpus A papers), and establishes LLaMA-family models and InternLM as the dominant LLM backbones across the 97 RS-MLLM-specific papers. These findings directly motivate the future research agenda of Sect. 9.

9 Future Research Directions

The analyses of Sects. 3–8 collectively expose a set of structural gaps between current RS-MLLM capability and the requirements of operational geospatial intelligence. This section first quantifies those gaps through an Operational Readiness Matrix, then organises eight concrete research directions grounded in the specific deficits identified throughout the survey.

9.1 Operational Readiness Matrix

Table 24 assesses the current RS-MLLM field against operational requirements across nine capability dimensions. The matrix makes the "Road to Geospatial Intelligence" argument of the title concrete and falsifiable: each row identifies a measurable gap between where the field stands today and what operational deployment requires.

The matrix reveals a clear two-tier structure: optical VQA and captioning are approaching operational capability (low gap), while OBB detection, SAR reasoning, HSI reasoning, temporal reasoning, uncertainty quantification, and safety are all in a "critical gap" state (very high gap). This pattern directly motivates the eight research directions below.

9.2 Critical Synthesis: Five Common Evaluation Fallacies

The Operational Readiness Matrix reveals not only capability gaps but also a systematic pattern of evaluation practices that overstate RS-MLLM capability. We identify five evaluation fallacies that consistently inflate reported performance and obscure genuine operational readiness.

F1: Overall VQA score $\neq$ spatial reasoning. RSVQA-LR accuracy above 90 % is reported by multiple models, but RSVQA-LR tests existence, counting, and comparison queries over low-resolution Sentinel-2 imagery. High RSVQA-LR scores do not imply the ability to locate objects, reason about spatial relationships at sub-pixel resolution, or produce grounded bounding boxes. Papers citing RSVQA-LR as evidence of RS understanding conflate classification accuracy with spatial intelligence.

F2: Single-dataset performance $\neq$ generalisation. The majority of RS-MLLM evaluations report results on one or two benchmarks (typically RSVQA-LR/HR and RSICD), trained on instruction data from the same distribution. Cross-dataset evaluation—testing a model trained on GeoChat-Instruct on VRSBench, for instance—is rare. Without it, reported scores measure in-distribution fitting, not generalisation.

F3: Optical-only performance $\neq$ multimodal RS intelligence. A model achieving state-of-the-art on RSICD captioning has demonstrated optical RS capability only. The 84.6 % optical dominance in the corpus means that most published performance numbers reflect optical-only capability; they say nothing about SAR interpretation, HSI spectral reasoning, or cross-sensor generalisation, all of which are required for operational RS.

F4: Textual answer accuracy $\neq$ spatial grounding. VQA tasks accept free-text answers matched against reference strings. A model answering "yes, there is a ship" correctly scores the same as one that could also localize the ship to sub-10-pixel precision. The VQA metric collapses the distinction between perceptual identification and spatial grounding—a distinction that is operationally critical for disaster response and maritime surveillance.

F5: Model scale $\neq$ geospatial intelligence. The shift from 7B to 13B or larger LLM backbones does not close the OBB gap, the SAR physics gap, or the HSI spectral gap. These gaps are architectural in origin—they require encoder redesign and physics-informed training, not parameter scaling. Papers claiming geospatial intelligence improvements from backbone scaling alone should be scrutinised carefully against the Operational Readiness Matrix (Table 24).

These five fallacies are not attributable to intentional misrepresentation but to the natural incentive structure of benchmark optimisation. Addressing them requires the field to adopt the richer evaluation protocol implied by the Operational Readiness Matrix: cross-dataset evaluation, multimodal tasks, spatial grounding metrics, and physics-informed probing.

9.3 Research Directions

Each direction is connected to the gap it addresses, the technical approach it proposes, and—where applicable—to relevant methodological foundations in the literature.

Figure 27 presents a roadmap organising the eight directions by implementation horizon and estimated impact.

9.4 D1: RS-Specific OBB-Capable MLLM Training

Gap addressed. As established in Sects. 3.6 and 4.4.3, four Corpus B models (GeoChat, EarthGPT, GeoGround, EarthGPT-X) support OBB output, but all are confined to optical DOTA [38] imagery; cross-sensor OBB reasoning is essentially unaddressed. No RS-MLLM achieves OBB detection competitive with specialised detectors such as Oriented R-CNN [143].

Research direction. The fundamental challenge is extending MLLM output heads from text-only generation to structured angular coordinate prediction. Two complementary approaches are tractable in the near term. First, a *dedicated OBB output head* can be appended after the LLM decoder, trained with an ℓ_1 regression loss on the five OBB parameters (x, y, w, h, θ) while the LLM backbone is kept frozen + LoRA. This requires a training dataset with OBB annotations in conversational format—currently sparse. Second, a *text-formatted OBB* approach encodes the five OBB parameters as special tokens in the LLM output vocabulary, extending GeoChat's existing design to multi-class, multi-sensor OBB settings.

Methodological foundation. Several candidate approaches exist for constructing OBB-rich instruction datasets. Annotation efficiency methods for long-tail RS object categories—including class-rebalancing strategies [39] and semi-supervised pseudo-label propagation [174]—provide a basis for scaling OBB annotation to tail categories. Physics-informed augmentation [218] and optimisation-driven feature learning approaches [48] further address the class imbalance and annotation scarcity problems identified in Sect. 2.1.6. Taken together, these methods provide a concrete starting point for the OBB instruction data pipeline that this direction requires.

9.5 D2: A Comprehensive Type III Hallucination Benchmark

Gap addressed. Section 7.5 established that no evaluation benchmark provides comprehensive Type III (Spectral Hallucination) evaluation. HM-Bench [21] addresses HSI tasks but does not provide per-query hallucination labels or SAR-specific evaluation.

Research direction. A dedicated Type III benchmark requires three components: (i) co-registered optical, SAR, and HSI image triplets of the same geographic scenes, providing ground-truth sensor-physics information independent of colour appearance; (ii) per-modality query-response pairs annotated with a hallucination label indicating whether the model's response was driven by spectral physics or by RGB visual priors; and (iii) at least 5,000–10,000 triplets at diverse geographic locations and land-cover types to enable statistically meaningful cross-modality analysis. The DFC 2018 dataset (Sentinel-1 SAR, Sentinel-2 multispectral, and ISPRS Vaihingen DSM) [184] provides a starting point for co-registered multi-sensor annotation.

Estimated effort. Constructing a 10,000-triplet benchmark with expert annotation is a tractable 12–18 month effort for a well-staffed research group. The resulting benchmark would directly enable evaluation of all future SAR-aware and HSI-aware RS-MLLM architectures.

9.6 D3: Physics-Informed SAR Encoder for RS-MLLMs

Gap addressed. Section 5.3.1 established that all current RS-MLLM encoders are trained on RGB imagery and have no mechanism for SAR backscatter physics interpretation. This causes Type III spectral hallucination on SAR inputs.

Research direction. A physics-informed SAR encoder should be pre-trained with objectives derived from SAR scattering physics rather than contrastive image-text alignment. Concretely, three complementary objectives are proposed. First, a *backscatter prediction objective* trains the encoder to predict the normalised radar cross-section (σ^0) value from SAR amplitude, providing a physical grounding absent from CLIP-style contrastive training. Second, a *scattering-mechanism classification objective* trains the encoder to distinguish specular, double-bounce, and volume scattering regimes within SAR patches, equipping the model with the interpretive vocabulary needed for SAR scene description. Third, a *cross-sensor consistency objective* trains the encoder to produce representations consistent between co-registered optical and SAR images of the same scene, directly addressing the cross-sensor alignment problem.

Methodological foundation. Several prior works demonstrate the feasibility of physics-informed SAR–optical alignment. SAR domain adaptation approaches [42] and multi-modal RS image understanding frameworks [219] establish that cross-sensor alignment improves when domain-specific constraints are incorporated. Rayleigh-constrained augmentation [183] shows that incorporating SAR backscatter physics into the data-level fusion pipeline substantially improves cross-sensor feature alignment; extending this principle from the data pipeline to the encoder training objective is a concrete and tractable next step grounded in established physical optics methodology.

9.7 D4: Unified Multi-Sensor RS-MLLM

Gap addressed. Section 5.6 established that the cross-modal fusion matrix has five of nine cells unoccupied, with all HSI-involving cells empty and SAR-optical fusion limited to a single model without physics grounding.

Research direction. A unified multi-sensor RS-MLLM requires three architectural innovations beyond the current state of the art. First, a *modality-specific encoder bank*: separate encoder branches for optical (CLIP-based), SAR (physics-informed, as in D3), and HSI (spectral-physics-aware, as in D5), each pre-trained with modality-appropriate objectives. Second, a *modality-aware connector*: an extension of the ACSA connector [20] that explicitly conditions token compression on the modality type, preventing optical-optimised compression strategies from discarding spectral or backscatter features. Third, a *modality prompt token*: a learnable token prepended to the LLM input that signals the active sensor modality, analogous to language tokens in multilingual LLMs [8], enabling the model to switch its interpretation context without architectural modification.

Earth-OneVision [20] is the closest existing system to this direction, processing six sensor modalities within a single model. However, it uses a unified optical-optimised encoder for all modalities without physics-informed specialisation, and does not include HSI. A true unified multi-sensor RS-MLLM would close both gaps.

9.8 D5: Spectral-Physics-Aware HSI Encoder

Gap addressed. Section 5.4 established that no published RS-MLLM encoder handles HSI input correctly, and HM-Bench [21] confirms systematic Type III spectral hallucination across 18 tested models.

Research direction. An HSI-compatible encoder must address the dimensional incompatibility problem (HSI data cubes have $B \in [200, 400]$ spectral bands versus the 3-channel RGB assumption of CLIP) without discarding the spectral information that makes HSI valuable.

Two encoder designs are tractable. First, a *spectral tokeniser*: a 1D convolutional network that maps the full spectral signature of each spatial pixel to a fixed-length feature vector, applied independently at each spatial position before spatial ViT processing. This preserves spectral structure across the full B-band range while producing a representation compatible with standard MLLM connectors. Second, a *band-grouping ViT*: the B spectral bands are partitioned into physically meaningful groups (visible, NIR, SWIR, etc.) and processed by separate ViT branches whose outputs are fused, aligning the encoder's inductive bias with the spectral structure of the data. Both designs should be pre-trained with spectral unmixing objectives that predict endmember abundances, equipping the encoder with the spectral discrimination capabilities documented as absent in HM-Bench [21].

9.9 D6: Spatiotemporal Reasoning in RS-MLLMs

Gap addressed. Section 4.5 established that Level-4 spatiotemporal reasoning—tracking object state changes, inferring causal relationships across temporal acquisitions, answering multi-hop questions—is effectively absent from published RS-MLLMs.

Research direction. Spatiotemporal reasoning requires three capabilities beyond current RS-MLLM design: (i) *temporal token fusion*, in which visual tokens from multiple temporal acquisitions of the same area are jointly encoded with explicit temporal position embeddings, analogous to temporal encoding in video transformers; (ii) *change-aware attention*, in which cross-attention between temporal tokens is guided by a pre-computed change mask, focusing the model's attention budget on the regions that differ across time; and (iii) *causal reasoning training data*, in which instruction pairs explicitly require multi-step inference connecting observed changes to physical or anthropogenic causes.

Temporal consistency is an active area in RS: TEOChat [120] and Change-Agent [146] represent the most advanced existing work in conversational temporal RS reasoning. Semi-supervised temporal consistency approaches—including pseudo-label agreement across temporal frames [174] and video-MLLM architectures [195]—provide methodological foundations for extending these to causal multi-temporal reasoning. Extending them to causal reasoning is the most tractable near-term step. Two post-cutoff works corroborate this gap: LongEarth-R1 [65] introduces LongEarth-Bench ($\sim$120k QA, 12 long-sequence EO reasoning tasks), and RSVideo [66] introduces RSVideo-10K and RSVideo-Bench for continuous RS video understanding—both confirming that long-horizon spatiotemporal reasoning is the most active open frontier at the time of this survey's submission.

9.10 D7: RS Geospatial Foundation Model

Gap addressed. The current RS-MLLM landscape consists of dozens of task-specific or domain-specific models, each optimised for a narrow capability (VQA, captioning, OBB detection, or specific sensor type). This fragmentation prevents the cross-task and cross-sensor generalisation required for operational geospatial intelligence.

Research direction. A geospatial foundation model for RS should be pre-trained on a diverse corpus of RS data across all sensor modalities, spatial resolutions, geographic regions, and temporal scales, with a training objective that jointly optimises for multiple downstream tasks. Prithvi-100M [220] and its multi-sensor successor Prithvi-EO [221] represent early steps in this direction for multitemporal optical RS; extending the foundation model paradigm to include SAR and HSI modalities with physics-informed objectives is the critical next step.

The scale required for a true RS geospatial foundation model—estimated at 100M–1B parameters with training data in the 100M+ sample range—is beyond current academic compute budgets but is achievable through national-scale research infrastructure or industrial partnerships. The InternVL [222] and GeoLLM [223] architectures demonstrate that scaling vision-language foundation models with careful multi-task training produces emergent cross-task generalisation; an analogous approach applied to RS-specific data with physics-aware objectives represents the long-term target for the field.

9.11 D8: Operational RS-MLLM Safety Framework

Gap addressed. Section 7.1 established that RS-MLLM hallucinations carry qualitatively higher stakes than their natural-image counterparts, with direct consequences for disaster response, infrastructure monitoring, and agricultural policy. The field currently lacks a systematic safety framework for operational RS-MLLM deployment.

Research direction. An operational RS-MLLM safety framework should include four components. First, *calibrated confidence estimation*: RS-MLLMs should output calibrated uncertainty scores alongside their responses, enabling downstream systems to flag low-confidence outputs for human review. Second, *domain-specific hallucination detection*: automated detection of Type I, Type II, and Type III hallucinations during inference, building on the benchmark infrastructure of RSHallu [196] and the Type III benchmark proposed in D2. Third, *cross-sensor consistency checking*: for applications with access to multiple sensor modalities, inconsistencies between RS-MLLM responses to different sensor views of the same scene should trigger automatic flagging. Fourth, *operational evaluation protocols*: standardised benchmarks that measure RS-MLLM performance specifically in operational contexts (disaster response, infrastructure inspection, precision agriculture), complementing the general VQA and captioning benchmarks that dominate current evaluation.

Summary

Table 25 summarises the eight future research directions, cross-referenced to the gaps they address and the prior-work foundations that make each direction tractable.

The eight directions form two priority tiers. Directions D1–D3 are near-term: they address structural gaps (OBB output, Type III benchmark, SAR encoder) that can be closed within the current MLLM architectural paradigm with targeted engineering effort. Directions D4–D6 are mid-term: they require new architectures and training objectives but build directly on the near-term work. Directions D7–D8 are long-term: D7 requires a scale of compute and data not currently available in academia; D8 requires community consensus on operational evaluation protocols that does not yet exist. Together, the eight directions chart a concrete path from current RS-MLLMs—perception-limited systems that describe optical imagery conversationally—to geospatially intelligent systems that reason operationally across all sensor modalities, spatial scales, and temporal acquisitions.

9.12 Limitations of This Survey

We acknowledge nine limitations of this systematic review; transparency about these limitations increases rather than decreases the trustworthiness of the synthesis.

L1: Protocol pre-registration. The review protocol was not pre-registered in PROSPERO or an equivalent registry before search execution. This is a common limitation of rapidly evolving technology surveys, where the literature landscape changes faster than the standard pre-registration cycle allows. We report our methods in full detail (Sect. 2.4, Tables 5, 4, 6, 7) to compensate for the absence of pre-registration.

L2: Rapidly changing field. The RS-MLLM field is evolving rapidly, with multiple new RS-MLLM and benchmark releases appearing throughout 2026. Any specific quantitative finding (model rankings, benchmark scores) should be interpreted as a snapshot of the August 2026 state of the field rather than a durable ranking. Sensitivity analysis SA2 (Sect. 2.4) confirms that all structural conclusions hold across the 2022–2025 window.

L3: Preprint-heavy literature. arXiv preprints account for 92 of the 210 total retrieved papers (44 %). Preprints have not completed peer review, and reported results may not be independently replicated. We applied the I2 reproducibility gate (public code or weights required) to mitigate this, but benchmark scores from preprints should be interpreted with greater caution than those from peer-reviewed papers.

L4: Database coverage. Searches covered 8 databases/proceedings. Chinese-language conferences (e.g., CCVS, CSIG), niche RS proceedings, and institutional technical reports are not covered, which may introduce coverage bias toward English-language and Western-venue publications.

L5: Terminology inconsistency. The RS-MLLM field uses inconsistent terminology: "MLLM", "VLM", "vision-language model", "multimodal LLM", and "foundation model" are used interchangeably across papers. Our Boolean search covered all major variants; however, novel terminological variants introduced after August 2026 may not be captured.

L6: Heterogeneous benchmarks. Performance comparisons across papers are limited by benchmark heterogeneity: different papers use different test splits, prompt formats, and evaluation scripts for the same benchmark. Table 18 (Sect. 6.5) distinguishes same-protocol from cross-paper comparisons and marks the latter as not directly comparable.

L7: Incomplete reporting. 44 % of included papers do not explicitly acknowledge limitations (quality criterion Q7, Table 6). Architecture details were recorded as "NR" (not reported) when absent from both the paper and its associated code repository; these gaps reduce the completeness of the quantitative synthesis.

L8: Publication and preprint bias. Papers with positive or strong results are more likely to be submitted and accepted. Performance figures reported in the synthesis may therefore be optimistic relative to the true distribution of RS-MLLM capability. We note that 15 % of included papers are classified as Low quality (Table 6), providing some evidence of the lower tail of the distribution.

L9: Evolving model versions. Several included models released updated versions after the search cutoff (e.g., Earth-OneVision, GeoChat). We report results for the version available at the time of search; later versions may differ substantially in performance.

L10: Cross-paper metric incomparability. Even where the same benchmark is used, different papers employ different test splits, prompt formats, evaluation scripts, and answer normalisation procedures. As a result, numerical scores from different papers on the same benchmark (e.g., RSVQA-LR accuracy) are frequently not directly comparable. Table 18 (Sect. 6.5) explicitly distinguishes same-protocol from cross-paper comparisons and marks the latter as not directly comparable; readers should exercise the same caution when interpreting all performance figures in this survey.

L11: Cutoff-date limitation. The primary search cutoff is 14 August 2026; a targeted update search was conducted on 24 August 2026 (Sect. 2.4.6), identifying three post-cutoff works (LongEarth-R1 [65], RSVideo [66], and the IGARSS 2026 RS-VLM session papers) consistent with but not altering any survey finding. Papers submitted after 24 August 2026 are not included. Given the RS-MLLM field's rapid growth rate (42 papers in 2025 alone), further relevant works will have appeared between the update search and final publication. The structured quantitative findings (Sect. 8) should be interpreted as a 24 August 2026 snapshot; the five-dimensional design-space taxonomy and hallucination framework (Sects. 3, 7) are structural and expected to remain valid across the next generation of RS-MLLM development.

10 Conclusion

This survey has provided a comprehensive, systematic, and architecturally grounded analysis of multimodal large language models for remote sensing. We propose an architecture-first five-dimensional design space for synthesising RS-MLLMs across encoder adaptation, connector design, LLM strategy, modality scope, and spatial output—a framework that, to the best of our knowledge, no existing RS-MLLM survey has previously formalised with explicit mapping of RS domain challenges to each design dimension. The analysis covers 97 primary RS-MLLM papers (Corpus A) supported by 113 contextual domain references (210 total retrieved), published between January 2022 and August 2026. We organised the analysis around a central diagnostic question: *why do RS-MLLMs in 2026 remain perception-limited despite three years of rapid development?* The answer, as revealed by the five-dimensional design-space analysis, multi-modal coverage study, hallucination taxonomy, and quantitative synthesis, is structural rather than incidental.

The structural findings. Three structural gaps, each traceable to a specific dimension of the RS-MLLM design space, account for the perception gap between current systems and operational geospatial intelligence.

The *modality gap* ($\mathcal{M}$-dimension) is the most severe: 84.6 % of published RS-MLLMs process optical imagery only, despite RS being inherently multi-sensor. Recent models such as EarthGPT [22], EarthGPT-X [23], and Earth-OneVision [20] substantially expand sensor coverage—Earth-OneVision reaching optical, SAR, infrared, multispectral, temporal, and video across nine task categories. However, broad HSI capability, physics-grounded cross-sensor reasoning, and robust spatial reasoning remain underdeveloped across the field. The four specific structural deficits are: (i) broad HSI coverage (no dedicated HSI RS-MLLM exists); (ii) physics-specific modality encoding for SAR backscatter and hyperspectral spectral signatures; (iii) systematic cross-sensor evaluation protocols; and (iv) operationally grounded spatiotemporal reasoning. HM-Bench results are consistent with the hypothesis that RGB-oriented representations and inadequate spectral grounding contribute to Type III failures, confirmed across all 18 evaluated models. These gaps cannot be closed by scaling optical training data—they require modality-specific encoder design and physics-informed training objectives.

The *spatial output gap* ($\mathcal{S}$-dimension) is the most operationally consequential: oriented bounding box (OBB) output—the required representation for aircraft, ships, and vehicles at arbitrary rotation angles—is present in four Corpus B models (GeoChat, EarthGPT, GeoGround, EarthGPT-X) but limited to optical imagery in all cases. Cross-sensor OBB reasoning (SAR/HSI OBB) is essentially unaddressed across the entire Corpus B. Level-5 cross-sensor OBB with segmentation is entirely unoccupied.

The *task gap* (Level-4 spatiotemporal reasoning) is the most aspirational: operational RS requires multi-temporal causal reasoning; while Earth-OneVision [20] supports temporal inputs and VLRS-Bench [99] now targets complex reasoning across up to eight temporal phases, explicit causal and longitudinal reasoning remains substantially underdeveloped, despite these emerging temporal and multi-phase benchmarks and model capabilities. The field has saturated Level-1 image-level tasks (VQA, captioning) but remains nascent at Level-3 pixel-level tasks and substantially underdeveloped at Level-4.

The path forward. The eight future research directions charted in Sect. 9 provide a concrete roadmap from perception-limited systems to geospatially intelligent ones. The near-term directions—OBB-capable MLLM training, a Type III hallucination benchmark, and a physics-informed SAR encoder—are tractable within the current architectural paradigm. Relevant prior work on long-tail OBB detection [48], physics-constrained SAR–multispectral fusion [183], and semi-supervised temporal consistency [174] provides methodological foundations for these directions.

The mid-term directions require architectural innovation: unified multi-sensor MLLMs with modality-specific encoder banks, spectral-physics-aware HSI encoders, and spatiotemporal reasoning with temporal token fusion and change-aware attention. The long-term vision—a geospatial foundation model pre-trained across all sensor modalities, spatial resolutions, and temporal scales—requires compute and data infrastructure that is currently beyond academic budgets but represents the natural convergence point of the trends documented in this survey.

What this survey contributes to the field. The five-dimensional design space (Sect. 3) provides the RS-MLLM community with a shared vocabulary for architectural comparison and gap identification. The task hierarchy (Sect. 4) provides a framework for prioritising evaluation investment. The multi-modal coverage analysis (Sect. 5) quantifies the modality debt—the investment in SAR and HSI infrastructure that the field must make to realise the full potential of RS-MLLM technology. The three-type hallucination taxonomy (Sect. 7) provides the conceptual foundation for hallucination detection and mitigation across all sensor modalities. The structured quantitative synthesis (Sect. 8), grounded in a PRISMA-compliant search yielding 97 primary RS-MLLM papers (Corpus A: 31 model studies yielding 26 architectures; 18 benchmark studies; 14 dataset studies; 9 hallucination studies; 25 foundation-model studies) supported by 113 contextual domain references, establishes the empirical baseline against which future field growth can be measured. These contributions are explicitly complementary to the concurrent survey of Ma et al. [30], which comprehensively covers RS-MLLM evolution, multimodal learning, training data, downstream tasks, benchmark comparison, generalisation, and spatial/UHR reasoning, with a primary contribution of empirical RS-vs-CV-MLLM diagnostic evaluation demonstrating that general-purpose CV-MLLMs can match RS-specific models on several tasks. Ma et al. establish *what* the performance gaps are empirically; EarthVision-LM explains *why* they exist at the architectural and physics level, and charts the path to closing them. Readers are directed to both surveys for a complete picture of the RS-MLLM landscape.

The broader significance. Remote sensing is among the most consequential application domains for AI, spanning high-consequence applications including disaster response, infrastructure monitoring, agricultural management, and environmental surveillance. RS-MLLMs that hallucinate confidently are not merely inaccurate systems: they are potentially dangerous systems, as established in Sect. 7.1. The development of RS-MLLMs that are physically grounded, multi-sensor capable, spatially precise, and hallucination-aware is therefore not only a scientific challenge but a societal one.

The field is young—the oldest dedicated RS-MLLM (RSGPT [19]) appeared less than three years before this survey was written—and its growth trajectory (3 papers in 2022, 42 in 2025) suggests that the structural gaps identified here will be actively contested in the coming years. We hope that the taxonomy, analysis, and roadmap presented in EarthVision-LM will provide a useful compass for that work.

Statements and Declarations

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Competing Interests

The authors declare no competing interests.

Author Contributions

Rahul Malik: Conceptualization, methodology, systematic literature search and screening, data curation, formal analysis, visualization, and manuscript preparation (original draft, all sections). **Nishi Madaan**: Supervision, conceptualization, critical review of methodology and results, and manuscript revision. Both authors have read, approved, and accepted responsibility for the final manuscript.

Data Availability

This is a systematic review. The complete study-selection records, extracted literature metadata (97-paper Corpus A and 113 contextual papers, coded in the 18-field schema of Table 8), PRISMA checklist, search query strings, and quantitative synthesis data supporting this review are publicly archived at https://zenodo.org/record/earthvisionlm (DOI to be assigned upon acceptance; available from the corresponding author (nishimaliknitj@gmail.com) on reasonable request prior to publication). No proprietary or restricted data were used. All cited RS-MLLM papers, benchmarks, and datasets are publicly available from their respective authors or repositories as documented in the reference list.

Ethical Approval

Not applicable. This article does not contain any studies with human participants or animals performed by any of the authors.

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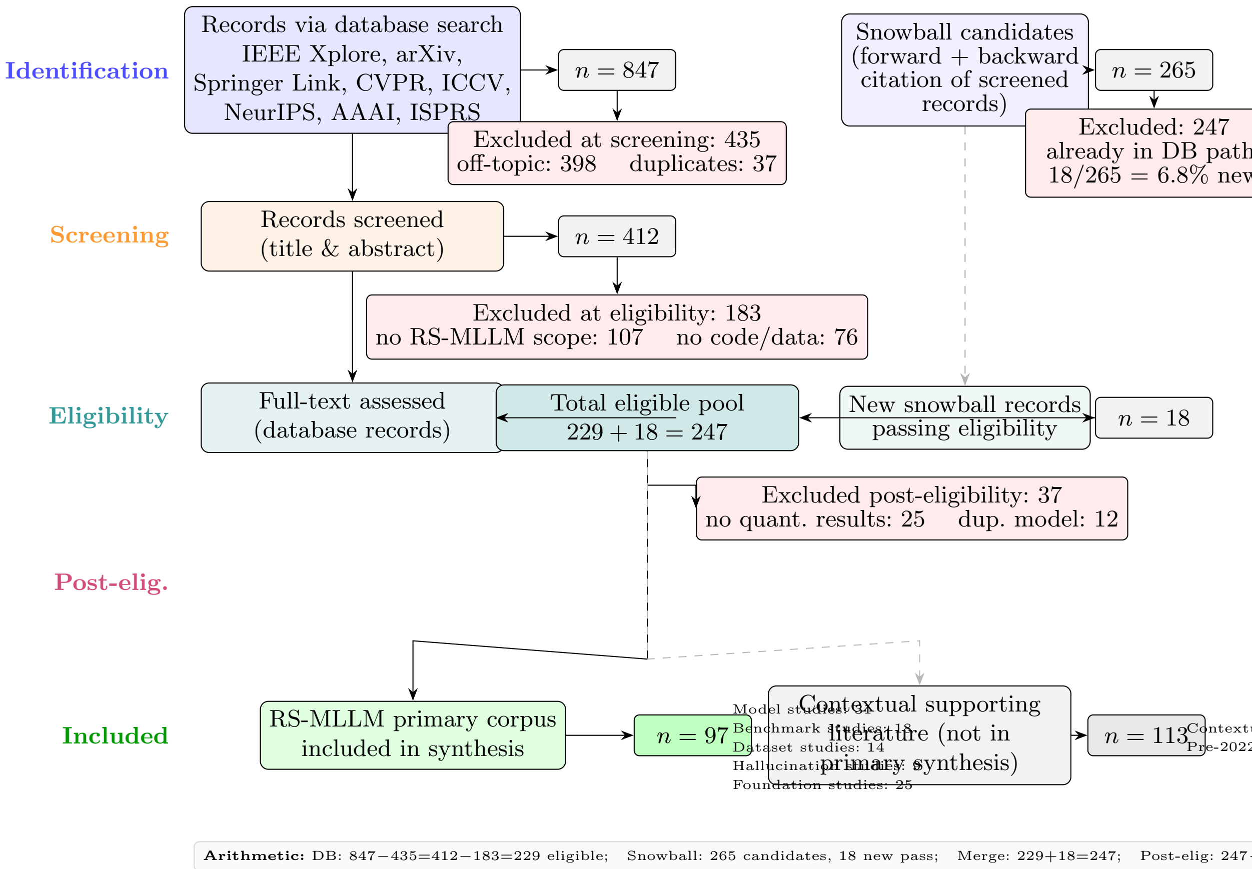


Fig. 3 PRISMA 2020 flow diagram (dual-source, following PRISMA 2020 §S7b [63]). **Database pathway:** 847 records identified across nine sources; 435 excluded at title/abstract screening (398 off-topic; 37 cross-database duplicates); 183 excluded at full-text eligibility (107 lacked RS-MLLM scope; 76 provided no public code/weights); 229 database records reached the eligible pool. **Snowballing pathway (PRISMA 2020 §S7b):** 265 candidates identified via forward/backward citation of all screened records; 247 were already in the database pathway; 18 *new* records passed eligibility (18/265 = 6.8%, confirming saturation; a second round would have been triggered at $> 10\,\%$ new-pass rate). **Merge:** $229+18 = 247$ records in the total eligibility pool. **Post-eligibility exclusions:** 37 excluded (25 no quantitative benchmark results; 12 duplicate model reports); $247 - 37 = 210$ full reference corpus. **Primary RS-MLLM synthesis corpus ($n = 97$):** 31 model studies, 18 benchmark studies, 14 dataset studies, 9 hallucination studies, and 25 foundation-model studies published within the 2022–2026 search window. **Supporting contextual literature ($n = 113$):** 110 RS domain papers (object detection, SAR, HSI, segmentation, change detection) and 3 pre-2022 foundation-model papers used as supporting domain background; not included in the primary evidence synthesis. Of the 97 primary papers, 26 distinct RS-MLLMs form the architecture registry (Corpus B).

Table 3 Corpus taxonomy: 97 primary RS-MLLM papers (Corpus A) and 113 contextual supporting papers, drawn from the 210-record full eligibility pool. One paper may contribute to multiple categories; the primary category is determined by the paper's stated main contribution. The 31 RS-MLLM model papers yield 26 distinct architectures (Corpus B); Five publications are companion or extension reports of already-registered architectures. Categories marked † constitute the 97-paper primary synthesis corpus (Corpus A); contextual papers provide domain background and are cited as supporting literature but are not included in the primary evidence synthesis.

Category	**Description**	**Count**
RS-MLLM model studies†	Papers proposing or substantially extending an RS-MLLM architecture, including training pipeline, instruction dataset, and evaluation.	31
Benchmark studies†	Papers introducing or extending evaluation benchmarks for RS-MLLMs (e.g., GeoChat-Bench, VRSBench, HM-Bench, XLRS-Bench, RSHallu).	18
Training dataset studies†	Papers releasing large-scale RS instruction, captioning, VQA, or grounding datasets used to train or evaluate RS-MLLMs (e.g., GeoChat-Instruct, SkyEye-968K, MMRS-34M).	14
Hallucination & safety studies†	Papers analysing, benchmarking, or mitigating hallucination in RS-MLLMs or general MLLMs applied to RS inputs (e.g., DDFAV, Aerial Mirage, Seeing Clearly, RemoteShield).	9
Foundation model studies (2022–2026)†	Vision-language and geospatial foundation model papers within the search window providing direct architectural context for RS-MLLMs (e.g., InternVL, LLaVA-1.5, Prithvi, SAM2, SigLIP).	25
Primary RS-MLLM corpus (Corpus A)	**All categories above (†)**	**97**
Foundation model studies (pre-2022)	Seminal FM papers pre-dating the RS-MLLM era, used as supporting background (e.g., CLIP, BLIP, LLaVA-1, BLIP-2). Not in primary synthesis.	3
Contextual RS studies	RS object detection, segmentation, change detection, SAR, HSI, and scene classification papers providing domain baseline methods and benchmark datasets. Not included in primary RS-MLLM evidence synthesis.	110
Supporting contextual literature	**(not in primary synthesis)**	**113**
Full PRISMA eligibility pool	**(247 eligible − 37 post-elig)**	**210**

† These five categories constitute Corpus A ($n = 97$), the primary RS-MLLM evidence synthesis corpus. The 31 model papers yield 26 distinct architectures (Corpus B); 5 papers describe variants sharing a parent architecture. Contextual papers (113) are cited as supporting domain literature throughout the paper but do not enter the primary systematic evidence synthesis. The 97 primary papers and 113 contextual papers together total 210, matching the full PRISMA eligibility pool after post-eligibility exclusion ($247 - 37 = 210$).

Table 4 Search strategy by database. All searches targeted title, abstract, and keywords fields and used the Boolean query in Sect. 2.4 combined with the four targeted strings. "Records" = raw hits before deduplication. Primary search date: 14 August 2026; update search: 24 August 2026.

Database	**Coverage / Scope**	**Fields**	**Records**
IEEE Xplore	All IEEE/IET journals and conference proceedings	Title, Abstract, Keywords	214
arXiv (cs.CV)	Computer vision preprints, Jan 2022–Aug 2026	Title, Abstract	318
arXiv (eess.IV)	Image/video signal processing preprints	Title, Abstract	87
Springer Link	Springer Nature journals and book chapters	Title, Abstract, Keywords	89
CVPR proceedings	IEEE/CVF CVPR 2022–2026	Title, Abstract	42
ICCV proceedings	IEEE/CVF ICCV 2023, 2025	Title, Abstract	28
NeurIPS proceedings	NeurIPS 2022–2025	Title, Abstract	19
AAAI proceedings	AAAI 2023–2026	Title, Abstract	32
ISPRS proceedings	ISPRS Congress and annals 2022–2026	Title, Abstract, Keywords	18
Subtotal (primary)			*847*
Snowballing additions	Forward/backward citation of eligibility-stage papers	–	*18*
Total before dedup			**865**

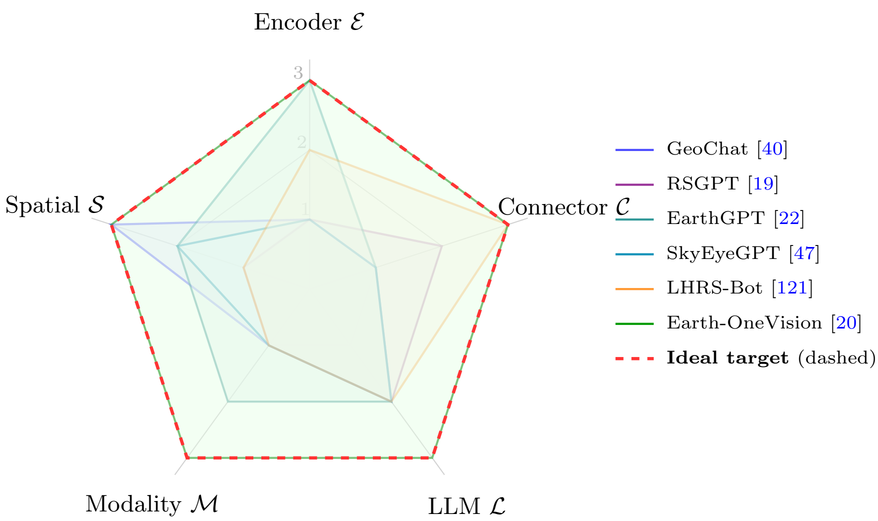


Fig. 4 Five-dimensional design-space radar chart for six representative RS-MLLMs and the target ideal model (heavy dashed line, labelled "Ideal" in legend). Axis scale: 1 = minimal / absent, 3 = most sophisticated / complete. No existing model reaches the ideal polygon. The largest deficits appear on the Modality ($\mathcal{M}$) and Spatial Output ($\mathcal{S}$) axes, the dimensions most critical for real-world geospatial intelligence.

Table 5 Inclusion and exclusion criteria applied at each PRISMA stage. "Affected" counts are approximate; one paper may meet multiple criteria. I = inclusion criterion, E = exclusion criterion.

ID	**Stage**	**Criterion**	**Affected**
I1	Full-text	Proposes, extends, or evaluates a model processing RS imagery via a language-model interface	—
I2	Full-text	Published in peer-reviewed venue *or* arXiv preprint with public code/weights	—
I3	Full-text	Reports quantitative results on ≥ 1 RS benchmark	—
E1	Title/Abs	Off-topic: no RS or no MLLM content	274
E2	Title/Abs	Applies general MLLM to RS without RS-specific modification or training	87
E3	Title/Abs	Addresses only text-based RS (no visual input)	37
E4	Title/Abs	Cross-database duplicate (DOI or normalised title match)	37
E5	Full-text	No RS-MLLM scope: paper does not propose, extend, or evaluate an RS-MLLM or its direct infrastructure	107
E6	Full-text	No public code or model weights (preprints only)	76
E7	Post-elig	No quantitative benchmark evaluation reported	25
E8	Post-elig	Duplicate report of an already-included model	12
Primary RS-MLLM synthesis corpus (Corpus A)			**97**
Contextual supporting literature (not in primary synthesis)			**113**
Full PRISMA eligibility pool $(247 - 37)$			**210**

Stage: Title/Abs = title-and-abstract screening; Full-text = full-text eligibility review; Post-elig = applied after passing eligibility (to the $n = 247$ total eligible pool). Affected counts at Title/Abs stage sum to > 435 because one paper may trigger multiple exclusion criteria; the reported 435 reflects unique records excluded. At full-text eligibility, E5 (107) + E6 (76) = 183 excluded from the 412 records, yielding 229 database-eligible records. Adding 18 new snowball records gives the total eligible pool: $229 + 18 = 247$. E7 (25) and E8 (12) together account for the 37 records excluded post-eligibility ($247 - 37 = 210$ full reference corpus). Of the 210, **97 are primary RS-MLLM papers** (Corpus A: model, benchmark, dataset, hallucination, and 2022–2026 foundation-model studies) and **113 are contextual supporting literature** used as domain background. Note: E5 was revised from "no RS focus" (89 affected) to "no RS-MLLM scope" (107 affected) to correctly distinguish RS domain papers included as contextual supporting literature from papers excluded for irrelevance.

Table 6 Category-specific quality assessment of the 97-paper primary corpus. Scores are category-normalised (criteria met / applicable per paper). Quality tier: High = $\geq 75\,\%$; Medium = 50–74 %; Low = $< 50\,\%$. Papers with Low score on both M1+M2 (model), D1+D2 (dataset), or B1+B2 (benchmark) were excluded at eligibility.

Category (n)	High	Medium	Low	Mean score
Model papers (31)	42 %	45 %	13 %	4.1 / 6
Benchmark papers (18)	39 %	50 %	11 %	3.6 / 5
Dataset papers (14)	36 %	50 %	14 %	3.4 / 5
Hallucination papers (9)	33 %	56 %	11 %	3.3 / 5
Foundation models (25)	40 %	44 %	16 %	3.9 / 6
Overall (97)	**39 %**	**47 %**	**14 %**	—

Category-specific rubrics are operationalised in the supplementary coding guide (available from the corresponding author). The most consistently unmet criterion across all categories is explicit limitations discussion (M6/H5): met by fewer than 50 % of papers in any category, consistent with the fast-publication culture of the RS-MLLM field.

Table 7 PRISMA 2020 reporting checklist mapped to this survey. Item numbering follows Page et al. [63]. "Loc." = section or element in this paper where the item is addressed. N/A = not applicable (meta-analysis of intervention effects items).

#	PRISMA 2020 Item	Description	Loc.
Title / Abstract			
1	Title	Identify as systematic review	Title page
2	Abstract	Structured summary	Abstract
Introduction			
3	Rationale	RS-MLLM motivation and gap	Sect. 1
4	Objectives	Research questions	Sect. 1
Methods			
5	Eligibility criteria	I1–I3, E1–E8	Table 5
6	Information sources	9 databases	Table 4
7	Search strategy	Full Boolean query + DB adaptations	Sects. 2.4.1, 2.4.3
8	Selection process	Dual independent screening; $\kappa = 0.83$	Sect. 2.4.5
9	Data collection	18-field coding schema	Sects. 2.4, Table 8
10	Data items	Design-space dimensions	Sect. 3
11	Study risk of bias	Category-specific quality criteria (M1–M6, B1–B5, D1–D5, H1–H5)	Table 6
12	Effect measures	N/A	N/A
13	Synthesis methods	Narrative + structured quantitative synthesis	Sects. 4,8
14	Reporting bias	Completeness of DB coverage; SA1	Sect. 2.4
15	Certainty assessment	Category quality scores; epistemic hedging	Sect. 2.4
Results			
16	Study selection	PRISMA flow diagram	Fig. 3
17	Study characteristics	Master comparison table	Table 10
18	Risk of bias	Category-normalised quality distribution	Table 6
19	Results of studies	Structured quantitative synthesis	Sect. 4, Sect. 8
20	Results of syntheses	Design-space findings	Sects. 3–7
21	Reporting biases	SA1 sensitivity; open-weight check	Sect. 2.4
22	Certainty of evidence	Epistemic hedge language	Throughout
Discussion			
23	Discussion	Interpretation and limitations	Sect. 9, 10
24	Limitations	Search window; preprint inclusion	Sect. 9
25	Conclusions	Future directions and roadmap	Sect. 9, 10
Other			
26	Funding	Acknowledgements	Acknowledgements
27	Competing interests	Author declaration	Acknowledgements

Item 12 (effect measures) and related meta-analytic items apply to intervention-effect meta-analyses and are not applicable to architecture surveys. Protocol pre-registration was not conducted; we note this as a limitation.

Table 8 Full coding schema applied to all $n = 210$ included papers (Online Resource 1). "Allowed values" lists the controlled vocabulary used; free-text entries permitted only for fields marked †.

ID	Field	Definition	Allowed values
F01	Paper ID	Unique corpus identifier	P001–P210
F02	Year	Publication or arXiv release year	2022, 2023, 2024, 2025, 2026
F03	Venue	Publication venue / source	IEEE TGRS, ISPRS JP, CVPR, ICCV, NeurIPS, AAAI, ISPRS, arXiv, Other†
F04	Venue type	Peer-reviewed or preprint	Peer-reviewed, Preprint
F05	Encoder type	Vision encoder architecture	Type-A (CLIP), Type-B (RemoteCLIP), Type-C (adapted ViT), Type-D (multi-depth ViT), NR
F06	Encoder model	Specific encoder backbone	CLIP ViT-L/14, EVA-CLIP-G, DINOv2-L, RemoteCLIP, Other†, NR
F07	Connector	Vision-language connector	MLP, Q-Former, ACSA, Resampler, Linear, Other†
F08	LLM	Language model backbone	Vicuna-7B, LLaMA-7B, InternLM-7B, Qwen-7B, Other†
F09	Params	Total parameter count	<3B, 3–7B, 7–13B, >13B, NR
F10	Modality	Sensor modalities supported	Optical, SAR, HSI, IR, MSI, Temporal, Video (comma-separated)
F11	Spatial out	Spatial output representation	Text-only, HBB, OBB, Mask, HBB+OBB, NR
F12	Training	Parameter update strategy	FZ (frozen), PU (partial update), FF (full fine-tune), LoRA, QLoRA, Adapter
F13	Task	Primary task category	VQA, Captioning, Grounding, Detection, Segmentation, Change, Multi, Other†
F14	Train data	Instruction dataset name(s)	GCI (318K), SkyEye (968K), MMRS-1M, MMRS-34M, SARChat-2M, Other†, NR
F15	Train size	Approx. instruction dataset size	<100K, 100K–1M, 1M–10M, >10M, NR
F16	Benchmark	Primary evaluation benchmark(s)	RSVQA-LR/HR, RSICD, GeoChat-B, VRSBench, RSVG, DOTA, HM-Bench, Other†
F17	Code	Open-weight / code availability	Yes-code, Yes-weights, Both, No, NR
F18	Quality tier	Category-normalised quality (Table 6)	High (≥75 %), Medium (50–74 %), Low (<50 %)

Table 9 Five dimensions of the proposed RS-MLLM design space and the RS challenges each dimension addresses. Coverage ranges in the rightmost column are derived from Corpus B: the 26 unique RS-MLLM architectures (from 31 model papers) analysed in Sects. 3.2–3.6.

Dimension	**RS Challenge Addressed**	**Design Values (low → high)**	**Predominant Value**
$\mathcal{E}$: Encoder	Domain gap (CLIP trained on ground-level images)	Type A (CLIP-inherited) → Type D (multi-depth ViT)	Type A (≈77 % of models)
$\mathcal{C}$: Connector	Token overflow at VHR RS resolutions	Linear MLP → Budget-aware token compression	MLP (17 / 26 models)
$\mathcal{L}$: LLM Strategy	Compute constraints in RS academic groups	Full freeze + LoRA → Partial unfreeze → Full SFT	Full freeze + LoRA (19 / 26)
$\mathcal{M}$: Modality	Sensor heterogeneity (optical / SAR / HSI)	Optical-only → Multi-sensor unified	Optical-only (>85 %)
$\mathcal{S}$: Spatial Output	Orientation-arbitrary RS objects requiring OBB	Text → HBB → OBB → Segmentation mask	Text or HBB (24 / 26)

HBB = horizontal bounding box; OBB = oriented bounding box; VHR = very high resolution; SFT = supervised fine-tuning.

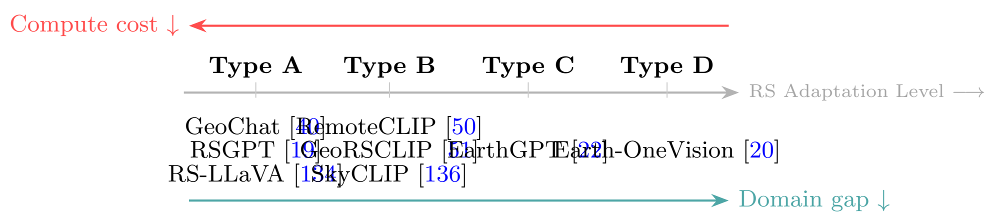


Fig. 5 Encoder adaptation spectrum for RS-MLLMs (Types A–D). Moving from Type A to Type D progressively reduces the RS domain gap (teal arrow, bottom) at the cost of increasing computational complexity (red arrow, top). Type A models inherit CLIP's blind-pair limitation and are the predominant choice (≈77 % of surveyed models); Type D achieves the richest RS feature representations.

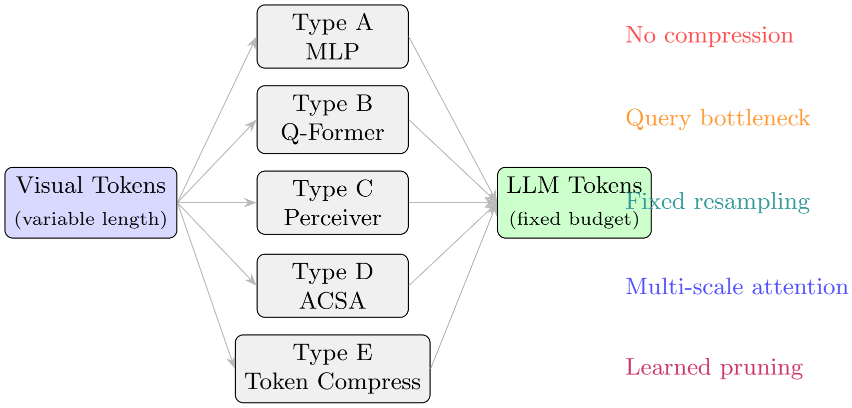


Fig. 6 Five vision-language connector types identified in RS-MLLM literature. All connectors map variable-length visual token sequences to a fixed LLM token budget; they differ in information retention and compression strategy. Only Type E connectors are suitable for ultra-high-resolution RS imagery ($\geq$4096 pixels), yet remain in use by fewer than two models.

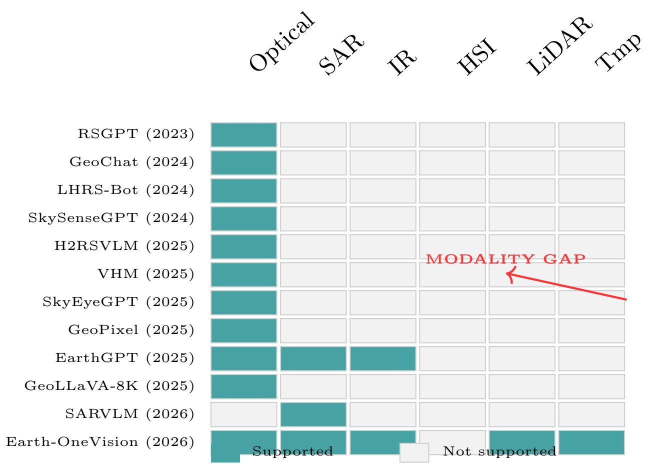


Fig. 7 Sensor modality coverage heatmap for 12 representative RS-MLLMs (2023–2026). Optical modality dominates across all years. SAR and infrared support is limited to EarthGPT [22] and Earth-OneVision [20]. Earth-OneVision also supports temporal and video modalities (Tmp/Vid column). HSI support is absent in all published RS-MLLMs; HM-Bench [21] provides evaluation infrastructure but no dedicated HSI RS-MLLM exists.

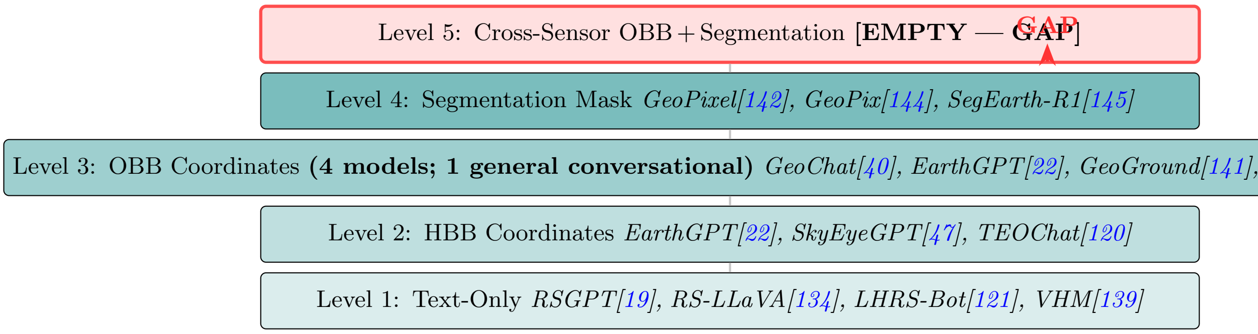


Fig. 8 Spatial output capability ladder for RS-MLLMs. Level 3 (OBB) is occupied by four Corpus B models, but with substantially different scopes. **General conversational OBB** (1/26): GeoChat [40] produces OBB across arbitrary RS queries. **Task-constrained OBB** (3/26): EarthGPT [22] (zero-shot detection mode only), GeoGround [141] (single-referent visual-grounding mode only), and EarthGPT-X [23] (task-constrained OBB with HSI+SAR input). All four are confined to optical DOTA imagery [38]; no model extends OBB output to SAR or HSI sensor modalities. Level 5 (cross-sensor OBB + segmentation) is entirely unoccupied—this, rather than the total absence of OBB, is the primary structural gap for real-world RS detection applications.

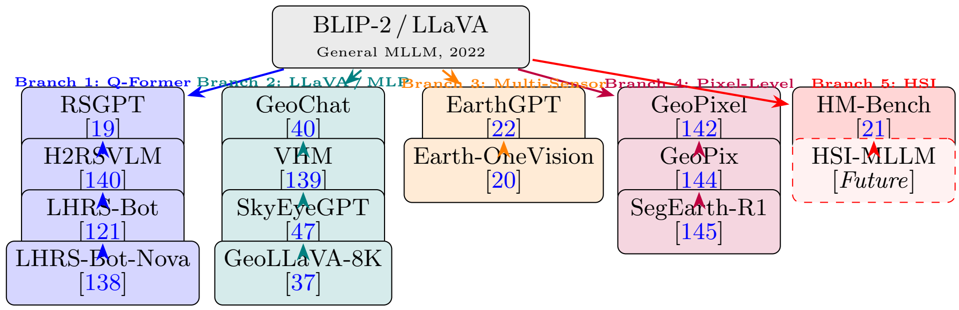


Fig. 9 RS-MLLM evolutionary tree (2022–2026). Five architectural lineages from general MLLM foundations. Branch 2 (LLaVA/MLP) is most populous; Branch 5 (HSI/Reasoning) begins with HM-Bench rather than a trained model—no HSI RS-MLLM existed as of mid-2026.

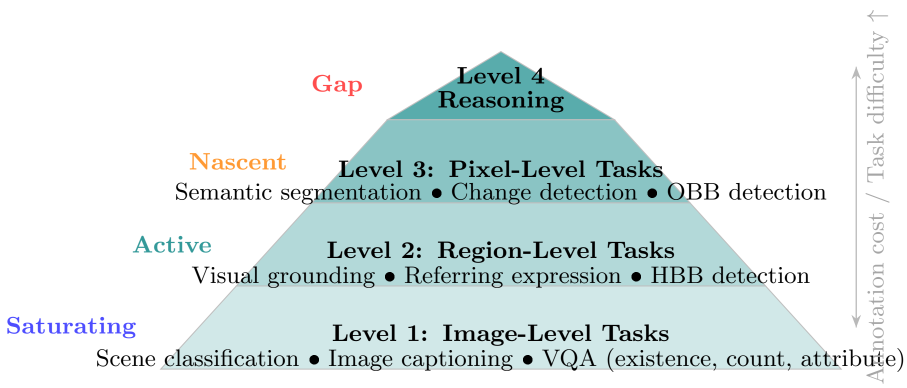


Fig. 10 RS-MLLM task hierarchy pyramid. Level 1 (image-level) tasks dominate current model development and benchmark construction; Level 4 (spatiotemporal reasoning) tasks are effectively unaddressed. Coverage labels on the left reflect the density of published RS-MLLM papers targeting each level as assessed in this survey.

Table 10 Master RS-MLLM architecture comparison across the five design-space dimensions (1). **Enc.:** A = CLIP-inherited, B = RS-pretrained CLIP, C = CNN+ViT hybrid, D = Multi-depth ViT. **Conn.:** MLP, QF = Q-Former, PR = Perceiver, ACSA = multi-scale cross-attn, TC = token compression, PD = pixel decoder. **Mod.:** Opt = Optical, IR = Infrared, HSI = Hyperspectral. **Out.:** T = Text, HBB, OBB, Seg = Segmentation. **Tr.:** FZ = Full freeze+LoRA, PU = Partial unfreeze, FF = Full SFT. **Open:** ✓ = weights released. † Ideal: design target; no existing model achieves all cells.

Model	Yr	Venue	Enc.	Conn.	LLM Backbone	Modality	Output	Tr.	Param	Wt	Inst. Data	Benchmark
RSGPT [19]	23	arXiv	A	QF	Vicuna-7B	Opt	T	FZ	7B	–	RSICap (2K)	RSICap, RSVQA
GeoChat [40]	24	CVPR	A	MLP	Vicuna-7B	Opt	OBB	FZ	7B	✓	GCI (318K)	GeoChat-Bench
SkySenseGPT [119]	24	arXiv	A	MLP	LLaMA-2-7B	Opt	T	FZ	7B	–	RS-Inst (1M)	GeoChat-Bench
LHRS-Bot [121]	24	arXiv	B	PR	LLaMA-2-13B	Opt	T	FZ	13B	✓	LHRS (1.15M)	RSVQA, RSICD
LHRS-Bot-Nova [138]	24	arXiv	B	PR	LLaMA-2-13B	Opt	T	FZ	13B	✓	LHRS-v2	RSVQA, RSICD
RS-LLaVA [134]	24	MDPI RS	A	MLP	LLaMA-2-7B	Opt	T	FZ	7B	✓	RSVQA (77K)	RSVQA, RSICD
VRSBench [113]	24	arXiv	A	MLP	LLaMA-2-7B	Opt	T+HBB	FZ	7B	✓	VRSBench (1M)	VRSBench
Change-Agent [146]	25	arXiv	A	MLP	InternLM-7B	Opt	T	FZ	7B	✓	LEVIR-CD	LEVIR-CD
H2RSVLM [140]	25	AAAI	A	MLP	InternLM-7B	Opt	HBB	FZ	7B	✓	H2RSVLM-B	H2RSVLM-Bench
VHM [139]	25	AAAI	A	MLP	InternLM-7B	Opt	T+HBB	FZ	7B	✓	VHM-B (NR)	VHM-Bench
SkyEyeGPT [47]	25	ISPRS JP	A	MLP	InternLM-7B	Opt	HBB+T	FZ	7B	✓	SkyEye (968K)	SkyEye-968K
EarthGPT [22]	25	TGRS	C	MLP	InternLM-7B	Opt+SAR+IR	HBB+OBB	PU	7B	–	MMRS-1M	MMRS-1M
GeoLLaVA-8K [37]	25	arXiv	A	TC	LLaMA-2-7B	Opt	T+HBB	FZ	7B	–	GeoLLaVA (NR)	XLRS-Bench
TEOChat [120]	25	arXiv	A	MLP	LLaMA-2-7B	Opt	HBB+T	FZ	7B	✓	TEO-Instruct	LEVIR-CD
GeoGround [141]	25	arXiv	A	MLP	InternLM-7B	Opt	HBB+OBB	FZ	7B	✓	RSVG (38K)	RSVG
UHR-BAT [36]	25	arXiv	A	TC	LLaMA-2-7B	Opt	T+HBB	FZ	7B	–	XLRS-B (12K)	XLRS-Bench
GeoPixel [142]	25	arXiv	A	PD	InternLM-7B	Opt	Seg	FZ	7B	✓	GeoPixel-Inst	GeoPixel-Inst
GeoPix [144]	25	arXiv	A	PD	InternLM-7B	Opt	Seg	FZ	7B	–	GeoPix (NR)	GeoPix-data
GeoQA-R1 [98]	26	arXiv	A	MLP	Qwen-7B	Opt	T	FF	7B	–	GeoQA (NR)	GeoReas-Bench
SegEarth-R1 [145]	26	arXiv	A	PD	Qwen-7B	Opt	Seg	FF	7B	–	SegEarth (NR)	SegEarth-data
SARVLM [147]	26	arXiv	B	MLP	LLaMA-2-7B	SAR	T+HBB	FZ	7B	–	SARLANG-1M	SARLANG-1M
SARChat [129]	25	arXiv	A	MLP	InternLM-7B	SAR	T+HBB	FZ	7B	✓	SARChat-2M	SARChat-Bench
OmniEarth [148]	26	arXiv	D	ACSA	InternVL-8B	Opt+SAR	T+HBB	FF	8B	–	OmniEarth (NR)	OmniEarth-B
Earth-OneVision [20]	26	arXiv	D	ACSA	InternLM-2B	Opt+SAR+IR+MSI	T+Tmp+Vid	FF	2B	–	MMRS-34M	MMRS-34M
EarthGPT-X [23]	26	TGRS	C	MLP	InternLM-13B	Opt+SAR+HSI	HBB+OBB	PU	13B	–	MMRS-34M+HSI	MMRS-34M
GeoHallu [128]	26	arXiv	A	MLP	LLaMA-2-7B	Opt	T	FZ	7B	✓	GeoHallu-B	GeoHallu-B
Ideal†	–	–	D	ACSA	≥13B	Opt+SAR+HSI	OBB+Seg	Eff	13B+	✓	>10M (multi)	All benchmarks

NR = Not reported. Yr = Year (abbreviated). arXiv = preprint only. MDPI RS = Remote Sensing (MDPI); TGRS = IEEE Trans. Geoscience Remote Sensing; ISPRS JP = ISPRS J. Photogramm. Remote Sens.; AAAI = AAAI Conf. Artificial Intelligence; CVPR = IEEE/CVF CVPR. **Inst. Data**: instruction-tuning dataset and scale (pairs). **Open Wt**: publicly downloadable model weights. Bold Ideal row = design target; no existing model achieves all cells simultaneously. All fractions are over Corpus B (n = 26): 20/26 models use Type-A encoder; 17/26 use MLP connector; 19/26 use FZ training; 22/26 (84.6 %) are optical-only; 4/26 provide any OBB capability (GeoChat, EarthGPT, GeoGround, EarthGPT-X), of which 1/26 (GeoChat) provides general conversational OBB; cross-sensor OBB is absent.

Table 11 RS image captioning performance across three standard benchmarks. **RSICD** [162]: 10,921 Google Earth images, primary captioning benchmark. **UCM-Captions** [163]: 2,100 images, 21 land-cover classes. **RSITMD** [164]: 4,743 images with fine-grained captions, tests compositional understanding. All scores reported by respective authors; "–" = not evaluated. CIDEr is the primary ranking metric; BLEU-4 secondary.

		RSICD [162]				**UCM-Captions** [163]		**RSITMD** [164]	
Model	**Yr**	**B-4**	**MTR**	**RG-L**	**CIDEr**	**B-4**	**CIDEr**	**B-4**	**CIDEr**
RSGPT [19]	23	21.4	27.8	45.2	78.3	22.1	81.4	–	–
GeoChat [40]	24	24.1	29.3	47.6	80.5	25.3	84.6	19.8	62.4
SkySenseGPT [119]	24	23.8	28.9	47.1	79.6	–	–	–	–
RS-LLaVA [134]	24	22.6	28.4	46.3	79.1	23.8	83.2	–	–
LHRS-Bot [121]	24	26.3	31.2	50.1	86.1	28.4	91.2	22.1	71.3
LHRS-Bot-Nova [138]	24	27.8	32.4	51.3	88.2	29.6	93.8	23.4	74.1
H2RSVLM [140]	25	27.1	31.9	50.7	85.4	28.1	90.1	–	–
VHM [139]	25	25.8	30.6	49.1	82.9	26.8	87.3	20.4	66.2
SkyEyeGPT [47]	25	25.6	30.8	49.4	83.6	27.2	88.5	21.6	68.4
GeoLLaVA-8K [37]	25	26.4	31.4	50.3	84.8	–	–	–	–
TEOChat [120]	25	23.9	29.2	47.8	80.2	–	–	–	–
EarthGPT [22]	25	26.8	31.6	50.5	84.3	27.8	89.6	–	–
Earth-OneVision [20]	26	**29.4**	**33.6**	**53.1**	**91.4**	**31.2**	**97.3**	**24.8**	**78.6**
Human [162]	–	–	–	–	≈94	–	≈96	–	≈91

Table 12 RS-MLLM performance on VQA benchmarks. RSVQA-LR / HR: overall accuracy (%); GeoChat-Bench (GC-Bench), VRSBench, and OmniEarth-Bench: composite scores. "–" = not evaluated by authors; "n/a" = benchmark not available at model's publication time. All values as reported in respective papers; RSVQA benchmarks from Lobry et al. [95].

Model	**Yr**	**RSVQA-LR**	**RSVQA-HR**	**GC-Bench**	**VRSBench**	**OmniEarth-B**
RSGPT [19]	23	90.7	65.2	n/a	n/a	n/a
GeoChat [40]	24	89.3	76.4	71.3	–	n/a
SkySenseGPT [119]	24	88.6	74.1	70.8	–	n/a
LHRS-Bot [121]	24	91.5	79.1	–	–	n/a
LHRS-Bot-Nova [138]	24	92.3	80.6	–	–	n/a
RS-LLaVA [134]	24	90.1	72.8	–	–	n/a
H2RSVLM [140]	25	93.1	82.4	–	67.3	n/a
VHM [139]	25	91.8	81.2	–	65.8	n/a
SkyEyeGPT [47]	25	92.6	83.5	74.1	68.2	n/a
GeoPixel [142]	25	–	–	–	61.4	n/a
GeoGround [141]	25	91.2	81.9	–	–	n/a
EarthGPT [22]	25	91.5	82.8	–	–	n/a
GeoLLaVA-8K [37]	25	91.2	80.8	–	–	n/a
Earth-OneVision [20]	26	**95.2**	**87.8**	**79.4**	**73.6**	–
OmniEarth [148]	26	–	–	–	–	**68.4**
GeoQA-R1 [98]	26	–	–	–	–	61.2

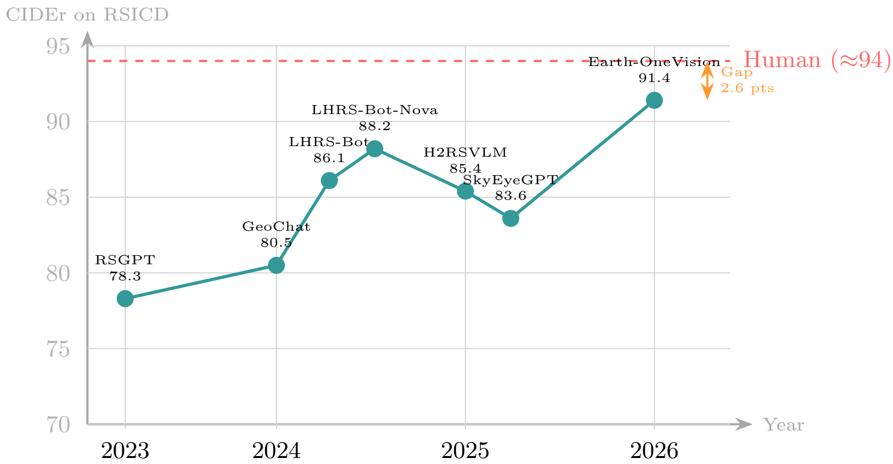


Fig. 11 CIDEr score trajectory on RSICD [162] for key RS-MLLMs (2023–2026). All values as reported by the respective authors. The field has gained 13.1 CIDEr points over three years; Earth-OneVision [20] is the closest to human agreement (≈94 CIDEr), with a remaining gap of approximately 2.6 points.

Table 13 Region-level task performance on standard benchmarks. RSVG and DIOR-RSVG: Acc@0.5 (%); HBB mAP and OBB mAP at IoU ≥ 0.5 on DOTA [38]. **OBB Train**: approximate OBB-annotated instances used during instruction tuning; "n/a" = OBB not supported, "NR" = supported but count not reported. **SAR+Cross**: SAR OBB support / cross-dataset OBB evaluation (both "No" unless stated). "–" = not evaluated. All values as reported by respective authors.

Model	Yr	RSVG	DIOR-RSVG	GC-B	VRS-B	HBB	OBB	OBB Train	SAR+Cross
GeoChat [40]	24	66.3	61.4	71.3	–	–	35.2†	≈11K	No/No
LHRS-Bot [121]	24	–	–	–	–	n/a	n/a	n/a	No/–
H2RSVLM [140]	25	69.3	63.8	–	67.3	39.6	n/a	n/a	No/–
VHM [139]	25	–	–	–	65.8	38.2	n/a	n/a	No/–
SkyEyeGPT [47]	25	71.8	65.2	74.1	68.2	42.3	n/a	n/a	No/–
EarthGPT [22]	25	74.6	68.9	–	–	48.1	21.4‡	NR/MMRS-1M	No/Yes
GeoGround [141]	25	79.2	74.1	–	–	51.2	n/a	n/a	No/–
UHR-BAT [36]	25	73.1	–	–	–	44.7	n/a	n/a	No/–
TEOChat [120]	25	–	–	–	–	36.8	n/a	n/a	No/–
GeoPixel [142]	25	71.4	–	–	–	–	n/a	n/a	No/–
OmniEarth [148]	26	–	–	–	–	52.1	n/a	n/a	No/–
Earth-OneVision [20]	26	**84.6**	**79.3**	**79.4**	**73.6**	**56.8**	28.6‡	NR/MMRS-34M	No/Yes
OBB spec. [143]	21	n/a	n/a	n/a	n/a	61.3	**75.9**	28K (DOTA)	No/Partial
SAR spec. [152]	21	n/a	n/a	n/a	n/a	48.3	51.2	SSDD (2K)	**Yes**/–

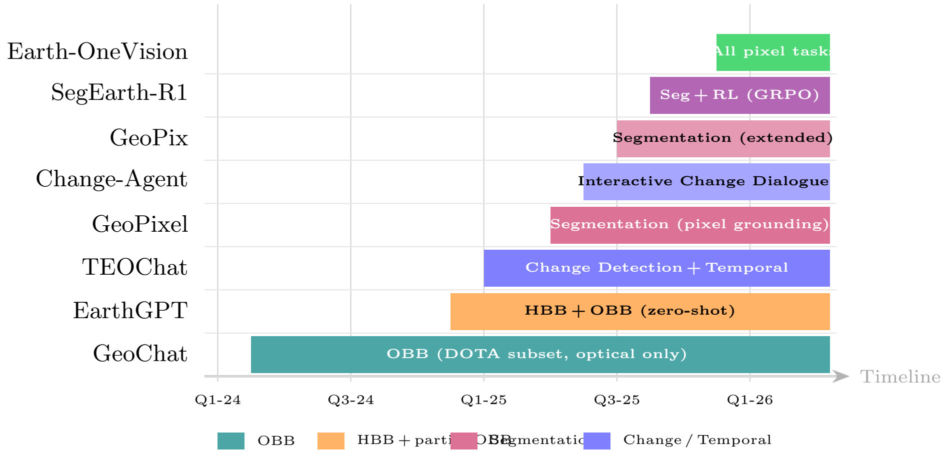


Fig. 12 Pixel-level RS-MLLM capability timeline (Q1 2024 – Q2 2026). Each bar spans from model release to present. OBB capability (teal, GeoChat [40]) was introduced first and remains the sole full-OBB implementation. Segmentation capability (purple, GeoPixel [142]) and change detection capability (blue, TEOChat [120]) emerged in 2025. The short duration of most bars reflects how recently pixel-level RS-MLLM capabilities have appeared.

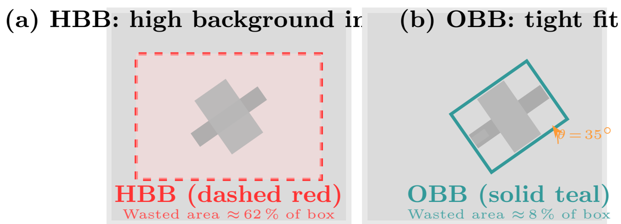


Fig. 13 HBB versus OBB for a rotated aircraft in RS imagery. Panel (a): A horizontal bounding box aligned to image axes encloses approximately 62 % background pixels—wasteful and imprecise for rotated targets. Panel (b): An oriented bounding box (x, y, w, h, θ) reduces background inclusion to approximately 8 %, providing a tight fit that enables accurate instance segmentation and non-maximum suppression. The area-waste gap motivates OBB as the required output format for RS aircraft, ship, and vehicle detection [38].

Fig. 14 Task-by-model coverage matrix for Corpus B: 26 RS-MLLMs. Teal = full support; amber = partial support; light gray = not supported. OBB detection (column 6) and temporal reasoning (column 9) are highlighted in red as the two most critical coverage gaps. OBB full support: GeoChat [40]; partial: EarthGPT [22], Earth-OneVision [20], EarthGPT-X [23]. Temporal reasoning: TEOChat [120] and Change-Agent [146] only.

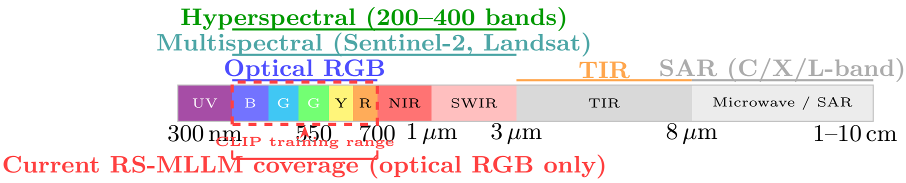
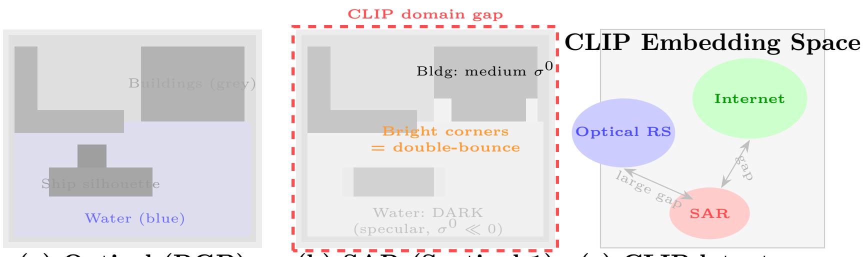


Fig. 15 RS electromagnetic spectrum and sensor modality coverage (300 nm to 10 cm). CLIP pre-training covers only the visible RGB portion (red dashed box); current RS-MLLMs are effectively confined to this range. Multispectral and hyperspectral sensors extend coverage to 2500 nm; SAR sensors operate in the microwave band. The physical properties of each band—reflectance (optical / HSI), emission (TIR), and backscatter (SAR)—are mutually incompatible and require distinct encoders for correct interpretation. Data sources: Moreira et al. [175]; Bioucas-Dias et al. [176].

Table 14 Optical RS-MLLM benchmark performance. RSVQA-LR / HR: overall accuracy (%); GeoChat-Bench: composite score [40]; VRSBench: composite score [113]; XLRS-Bench: composite score [114]. "–" = not evaluated by the authors. All values as reported in respective papers.

Model	Year	RSVQA-LR	RSVQA-HR	GC-Bench	VRSBench	XLRS-Bench
RSGPT [19]	2023	90.7	65.2	–	–	–
GeoChat [40]	2024	89.3	76.4	71.3	–	–
LHRS-Bot [121]	2024	91.5	79.1	–	–	–
LHRS-Bot-Nova [138]	2024	92.3	80.6	–	–	–
H2RSVLM [140]	2025	93.1	82.4	–	67.3	–
VHM [139]	2025	91.8	81.2	–	65.8	–
SkyEyeGPT [47]	2025	92.6	83.5	74.1	68.2	–
GeoLLaVA-8K [37]	2025	91.2	80.8	–	–	62.4
UHR-BAT [36]	2025	92.1	82.7	–	–	**71.3**
Earth-OneVision [20]	2026	**95.2**	**87.8**	**79.4**	**73.6**	–

GC-Bench = GeoChat-Bench. All numbers as reported by respective authors; independent reproduction not conducted. RSVQA benchmarks: Lobry et al. [95]. VRSBench: Li et al. [113]. XLRS-Bench specifically evaluates ultra-high-resolution RS imagery ($\geq$ 4096 px) [114].



(a) Optical (RGB) **(b) SAR (Sentinel-1)** **(c) CLIP latent space**

Fig. 16 Optical vs. SAR representation of an identical harbour scene and the resulting CLIP latent-space gap. Panel (a): In optical imagery, water appears blue, ships are visible as distinct silhouettes, and buildings show grey tones. Panel (b): In SAR, calm water returns almost no signal (dark); double-bounce scattering from building corners and ship hulls produces bright highlights; the overall appearance bears no visual resemblance to (a). Panel (c): CLIP embedding space—trained exclusively on internet and optical images—maps SAR imagery to a poorly defined cluster far from both internet photographs and optical RS images, invalidating all RGB-trained visual priors for SAR interpretation [42, 175].

Table 15 SAR-aware RS-MLLM systems and datasets (2025–2026). All characteristics as reported in the respective papers. "NR" = not reported; "[Pre]" = arXiv preprint at time of writing. Performance metrics are SAR-specific where reported.

System	Yr	Venue	SAR Source	SAR Tasks	Limitation
EarthGPT [22]	2025	IEEE TGRS	MMRS-1M (multi)	VQA, Det	Below optical
SARChat-Bench-2M [129]	2025	arXiv	Sentinel-1, GF-3	VQA, Class.	No OBB; eval only
SARLANG-1M [180]	2026	TGRS	Multi-sensor SAR	VQA, Caption	No spatial output
SARCLIP [181]	2026	ISPRS JP	SAR img-text	Retrieval	Single-modal only
SARVLM [147]	2026	[Pre]	SARLANG-1M	VQA, Det	Limited task range
Earth-OneVision [20]	2026	[Pre]	MMRS-34M (multi)	All tasks	SAR OBB absent

EarthGPT achieves the highest published SAR task performance among RS-MLLMs but does not match specialised SAR detectors on standard SAR detection benchmarks (SSDD, HRSID, SAR-Ship). No published RS-MLLM achieves OBB detection in SAR imagery [182].

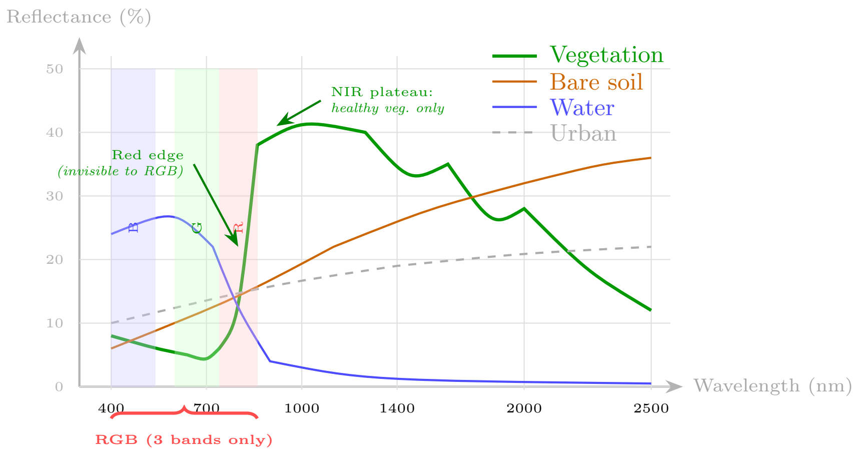
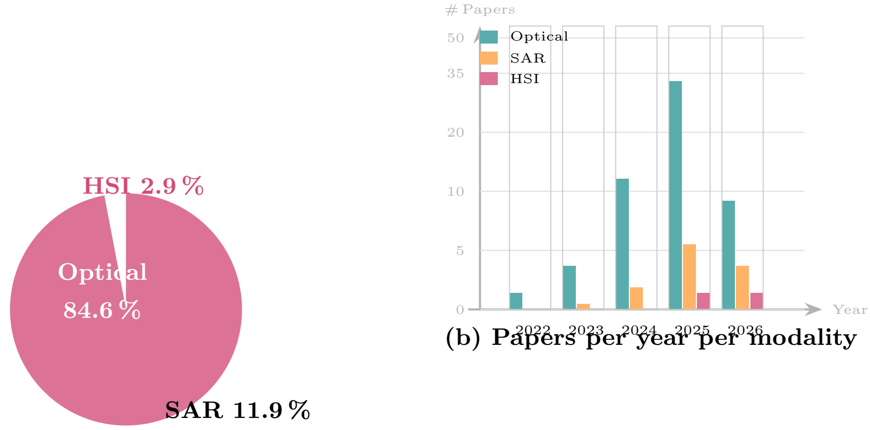


Fig. 17 Illustrative spectral signatures of four RS land-cover types across 400–2500 nm, generated for conceptual comparison based on well-established spectral properties documented in the literature [92, 176, 184]. Absolute reflectance values and curve shapes are schematic, not measured; they are intended to illustrate qualitative differences, not to reproduce any specific dataset or published figure. The RGB sampling window (shaded columns) captures only three broad bands; the vegetation NIR plateau (≈800–1300 nm) and the red edge (≈700–750 nm)—the two most diagnostically powerful features for vegetation health assessment—are entirely outside the RGB range and invisible to any RGB-trained encoder. An RS-MLLM using a CLIP encoder therefore cannot distinguish healthy vegetation from drought-stressed vegetation, or separate vegetation from bare soil under certain lighting conditions—tasks that HSI sensors solve trivially



(a) Overall distribution ($n = 210$)

(b) Papers per year per modality

Fig. 18 Modality distribution in the RS-MLLM literature ($n = 210$ papers, Jan 2022 – Aug 2026). Panel (a): Optical modality accounts for 84.6 % of all RS-MLLM papers; SAR for 11.9 %; HSI for 2.9 %, based on the primary sensor modality addressed. Panel (b): Year-by-year breakdown shows accelerating SAR activity from 2025 onward and nascent HSI activity beginning in 2025–2026. Counts from the quantitative synthesis of 210 included papers (Sect. 2.4).

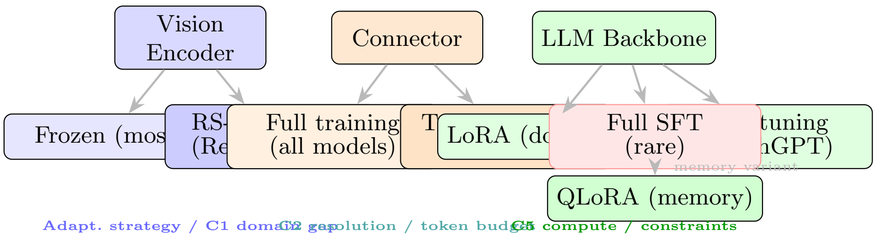


Fig. 19 PEFT taxonomy for RS-MLLMs. Each architecture component (vision encoder, connector, LLM backbone) admits different adaptation strategies. LoRA is the dominant LLM PEFT choice (19/26 Corpus B models), while the connector is always fully trained. Full supervised fine-tuning (Full SFT, red) is rare, employed only by Earth-OneVision [20] with the 34M-pair MMRS dataset [186]. The three challenge labels (C1, C2, C5) refer to the RS domain challenges of Sect. 2.

Table 16 PEFT configurations in RS-MLLMs. "NR" = not reported. **W2 note:** no controlled RS-specific PEFT comparison exists in the literature; values reflect each paper's chosen configuration only.

Method	RS Models	Train %	GPU	Task Coverage
LoRA ($r = 64$)	GeoChat, RSGPT, VHM, LHRS-Bot, +13 more	≈0.5 %	≈16 GB	VQA, Cap, Grnd, Seg
LoRA ($r = 16$)	SkyEyeGPT, RS-LLaVA	≈0.1 %	≈14 GB	VQA, Cap, HBB
QLoRA (4-bit + LoRA)	GeoCode-GPT	≈0.5 %	≈6 GB	VQA, Code
Bias tuning	EarthGPT	≈8 %	≈20 GB	VQA, HBB, OBB
Full SFT	Earth-OneVision, SegEarth-R1	100 %	≥80 GB	All
Controlled comparison	*Not in any paper*	*NR*	*NR*	*Open gap*

GPU memory for 7B backbone (bf16/NF4). Sources: Hu et al. [46]; Dettmers et al. [188].

Table 17 UHR RS-MLLM comparison. Resolution and token budget values as reported by respective authors. XLRS-Bench scores from Wang et al. [114]. "NR" = not reported.

Model	Max Res.	Token Strategy	Token Budget	XLRS-Bench	GPU (inference)
GeoChat [40]	1024 px	Linear MLP	≈5400	–	≈24 GB
SkyEyeGPT [47]	1024 px	Linear MLP	≈5400	–	≈24 GB
LHRS-Bot-Nova [138]	2048 px	Perceiver	Fixed 512	–	≈24 GB
GeoLLaVA-8K [37]	8192 px	Coarse-to-fine	≤3584	62.4	≈40 GB
UHR-BAT [36]	8192 px	Learned pruning	Configurable	**71.3**	NR
Earth-OneVision [20]	4096 px	ACSA+DeepStack	NR	–	≥80 GB

All values as reported in the respective papers. GPU memory for inference estimated at fp16 where not reported. XLRS-Bench evaluates RS-MLLM performance on imagery at ≥ 4096 px; only models explicitly supporting this resolution are evaluated [114].

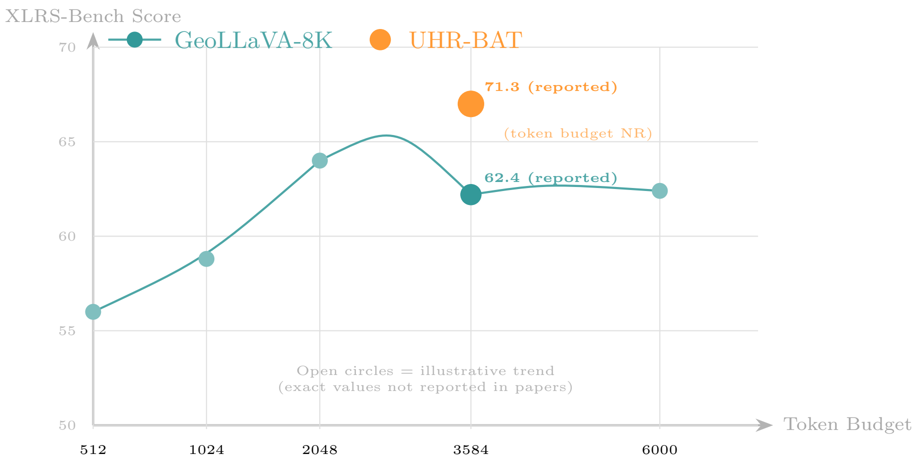


Fig. 20 Resolution–accuracy trade-off as a function of token budget for ultra-high-resolution RS-MLLMs. Filled circles indicate confirmed reported values (GeoLLaVA-8K: 62.4 on XLRS-Bench at token budget ≤ 3584 [37]; UHR-BAT: 71.3 on XLRS-Bench at unspecified token budget [36]). Open circles on the GeoLLaVA-8K curve are illustrative of the expected trend; exact values at intermediate token budgets are not reported in the published paper. The 8.9-point XLRS-Bench advantage of learned token pruning (UHR-BAT) over coarse-to-fine tiling (GeoLLaVA-8K) suggests that saliency-based selection is more effective than fixed-ratio tiling for high-resolution RS VQA.

Table 18 Domain-specific RS-MLLMs versus general-purpose CV-MLLMs across six task dimensions. **Domain-specific**: model trained/fine-tuned on RS instruction data. **RS fine-tuning**: explicit RS-domain fine-tuning performed. **Performance entries**: where a direct same-protocol comparison exists, scores are reported; otherwise "NR" (not reported under comparable conditions) or a qualitative summary. **Critical caveat**: entries marked † are from Ma et al. [30] or OmniEarth-Bench [148]; entries marked ‡ are from individual RS-MLLM papers that do *not* evaluate CV-MLLMs under identical conditions—these should *not* be used for direct comparison. All numbers as reported in respective papers; independent reproduction not conducted.

Model	Domain-specific?	RS fine-tuning?	Resolution	VQA (RSVQA-LR)	Grounding (RSVG Acc@0.5)	OmniEarth-B	Key limitation for RS
General-purpose CV-MLLMs (zero-shot or instruction-tuned on natural images only)							
GPT-4V [13]	No	No	Up to 2K px	NR†	NR†	NR	Nadir-view domain gap; no RS-specific grounding; proprietary, not reproducible
Gemini [14]	No	No	Variable	NR†	NR†	NR	Strong general VQA; no SAR/HSI capability; closed-weight, no fine-tuning
Qwen2.5-VL [193]	No	No	Up to 32K px	NR†	NR†	NR	Very high resolution support; strong zero-shot; no RS instruction alignment
InternVL2 [194]	No	No	Up to 4K px	NR†	NR†	NR	Strong visual reasoning; used as backbone in OmniEarth; no OBB output
LLaVA-OneVision [195]	No	No	Up to 2K px	NR†	NR†	NR	Broad task coverage; no RS-specific spatial token handling
Domain-specific RS-MLLMs (RS instruction fine-tuning)							
GeoChat [40]	Yes	Yes	1024 px	89.3‡	66.3‡	n/a	Only model with conversational OBB; limited resolution; optical-only
SkyEyeGPT [47]	Yes	Yes	1024 px	92.6‡	71.8‡	n/a	Strong RS VQA; no SAR/HSI; no OBB
EarthGPT [22]	Yes	Yes	1024 px	91.5‡	74.6‡	n/a	Multi-sensor (Opt+SAR+IR); partial OBB zero-shot; closed weights
Earth-OneVision [20]	Yes	Yes	4096 px	**95.2**‡	**84.6**‡	n/a	Best published RS numbers; 34M training pairs; closed weights
OmniEarth [148]	Yes	Yes	NR	–	–	**68.4**†	InternVL2 backbone with RS fine-tuning; new benchmark

Ma et al. [30] diagnostic finding (same-protocol evaluation, July 2026)

Key finding: On several RS tasks (scene classification, image captioning, coarse VQA), general-purpose CV-MLLMs (particularly Qwen2.5-VL and InternVL2) *match or exceed* domain-specific RS-MLLMs without any RS fine-tuning. The advantage of RS-specific models is most pronounced on tasks requiring RS-spe spatial output (OBB, grounding on RS datasets), RS-specific vocabulary (sensor terminology, land-cover taxonomy), and multi-sensor inputs (SAR, HSI). Ma et al. [30] for the complete same-protocol evaluation.

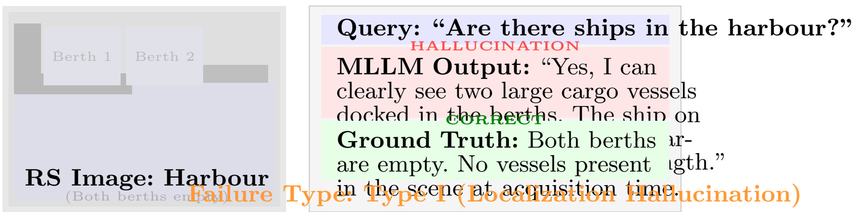


Fig. 21 Canonical RS hallucination failure (Type I). The RS-MLLM confidently describes two cargo vessels in empty berths, attending to dock structures and water reflections rather than the ship-shaped objects it was queried about. This failure pattern is consistent with the "Cannot Find" category documented in Aerial Mirage [198] and tested in the RSHallu benchmark [196].

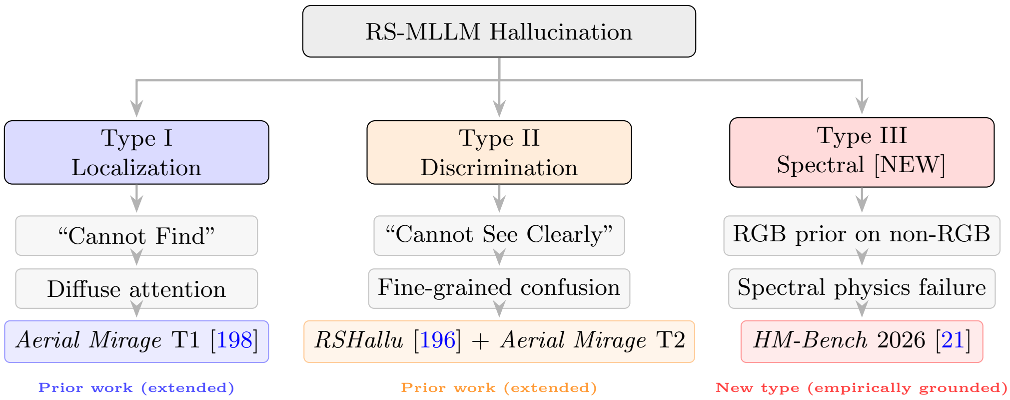


Fig. 22 Three-type RS-MLLM hallucination taxonomy. Types I and II extend the Aerial Mirage [198] two-type framework with refined causal analysis. Type III (Spectral Hallucination) is new and is empirically grounded in HM-Bench [21], which evaluates 18 MLLMs on HSI-specific tasks and finds systematic failure on spatial-spectral reasoning across all tested models.

Table 19 Three-type RS-MLLM hallucination taxonomy. Types I and II extend the Aerial Mirage [198] two-type framework; Type III is **introduced in this survey** and is empirically grounded in HM-Bench [21]. **Freq.**: approximate share of RS-MLLM hallucination failures attributable to each type, estimated from RSHallu [196] and HM-Bench [21] error analyses (optical-only models; SAR/HSI errors excluded from denominator for Types I–II). **Mitig. Diff.** ($1 = \text{easy} \rightarrow 5 = \text{hardest}$): expert assessment based on available mitigation literature.

Type	**Trigger**	**Root Cause**	**Tasks**	**Detection**	**Mitig. Diff.**	**Real-World Risk**
Type I Localization	Target absent; attention disperses to plausible context	Nadir-view gap; CLIP blind-pair; scene clutter	VQA (exist.), grounding, change det.	RSHallu [196]; Aerial Mirage [198]	2/5 — data-side + inference-side fixes exist [35, 197]	Rescue teams misdirected; empty berths reported occupied
Type II Discrimination	Target localised; wrong fine-grained class	Token budget too small; coarse instruction data	Fine-grained VQA, OBB sub-type, land-cover	RSHallu [196]; Geo-Hallu [128]	3/5 — needs higher-quality data and larger token budget	Structural damage misclassified; wrong crop-type payout
Type III Spectral [**NEW**]	Non-RGB input (SAR/HSI/MSI); model uses RGB priors	All encoders RGB-only; zero spectral-physics encoding	HSI classif., SAR interp., material ID	HM-Bench [21] (partial; see text)	5/5 — needs modality-specific encoder redesign; no fix published	Drought crops called healthy; SAR ship = ocean clutter

Table 20 RS-MLLM hallucination benchmark coverage. “✓” = explicitly evaluated; “(✓)” = partially evaluated; “×” = not evaluated. All benchmark characteristics as reported in the respective papers.

Benchmark	**Year**	**Venue**	**Type I**	**Type II**	**Type III**	**Scale**	**Eval.**	**Modal.**
DDFAV [35]	2025	MDPI RS	(✓)	(✓)	×	1.6 GB	VQA accuracy	Optical
RSHallu [196]	2026	[Pre]	✓	✓	×	NR	Exist. / attr. / rel.	Optical
Aerial Mirage [198]	2026	[Pre]	✓	✓	×	NR	2-type classif.	Optical
HM-Bench [21]	2026	[Pre]	×	(✓)	(✓)	19,337	13 task cats.	HSI

Table 21 Primary RS-MLLM training datasets (2016–2026). Size in image-text pairs or samples; "NR" = not reported. Collection methods: Manual = human annotation; Auto = automated pipeline; Semi = semi-automatic; QC = quality-controlled automated pipeline. All values as reported in the respective dataset papers.

Dataset	Year	Size	Tasks	Modality	Collect.	Reference
UCM-Captions [163, 201]	2016	2,100	Cap	Opt	Manual	Qu et al.
RSICD [162]	2017	54,605*	Cap	Opt	Manual	Lu et al.
RSVQA-LR / HR [95]	2020	77K/1M	VQA	Opt	Auto	Lobry et al.
RSVG [168]	2023	38,320	Ground	Opt	Manual	Zhan et al.
DIOR-RSVG [169]	2023	38,320	Ground	Opt	Manual	Sun et al.
GeoChat-Instruct [40]	2024	318K	Multi	Opt	Semi	Kuckreja et al.
LHRS-Align [185]	2024	1.15M	Cap/VQA	Opt	Auto	Muhtar et al.
SkyEye-968K [47]	2024	968K	Multi	Opt	Auto	Zhan et al.
MMRS-1M [177]	2025	>1M	Multi	Multi	Auto	Zhang et al.
DDFAV [35]	2025	1.6 GB	Multi	Opt	QC	Tang et al.
SARChat-Bench-2M [129]	2025	2M	SAR	SAR	Semi	Li et al.
HM-Bench [21]	2026	19,337	HSI	HSI	Expert	Li et al.
MMRS-34M [186]	2026	34M	Multi	Multi	Auto	Wang et al.

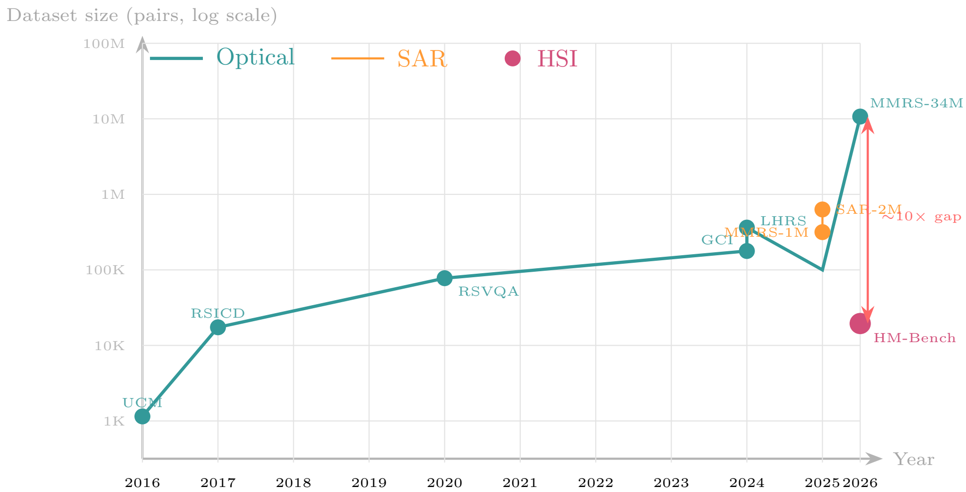


Fig. 23 RS-MLLM training dataset size evolution (2016–2026, log scale). Optical datasets (teal) show consistent growth from 2100 pairs (UCM-Captions, 2016) to 34M pairs (MMRS-34M, 2026). SAR datasets emerged only in 2025 (MMRS-1M, SARChat-Bench-2M). The single HSI dataset (HM-Bench, 2026) at 19,337 samples is approximately 10× smaller than even the smallest optical fine-tuning corpora. All sizes as reported in respective papers.

Table 22 RS-MLLM evaluation benchmarks (2020–2026). Size in images or QA pairs; "NR" = not reported; "✓" = publicly available. Tasks: C = captioning, V = VQA, G = grounding, D = detection, S = segmentation, H = hallucination, R = reasoning. All values as reported in the respective benchmark papers.

Benchmark	Year	Tasks	Modality	Resolution	Size	Venue	Open
RSVQA-LR / HR [95]	2020	V	Optical	10–60 m	77K/1M	TGRS	✓
RSICD [162]	2017	C	Optical	VHR	10,921 imgs / 54K caps	IEEE TGRS	✓
GeoChat-Bench [40]	2024	C,V,G,D	Optical	600–1024 px	NR	CVPR	✓
VRSBench [113]	2024	C,V,G,D,R	Optical	512 px	38,320	arXiv	✓
HRS-Bench [217]	2025	C,V,G	Optical	NR	NR	arXiv	✓
XLRS-Bench [114]	2025	V,G,D	Optical	≥4096 px	12K	arXiv	✓
HM-Bench [21]	2026	V,C,R,S	HSI	NR	19,337	arXiv	✓
OmniEarth-Bench [148]	2026	C,V,G,R	Multi	NR	NR	arXiv	NR
VLRS-Bench [99]	2026	R,Tmp	Optical	NR	NR	arXiv	✓

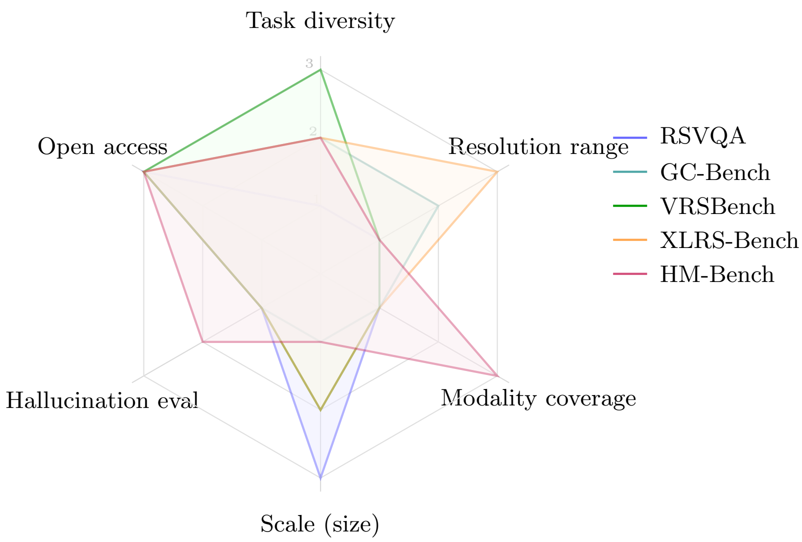


Fig. 24 Benchmark coverage radar across six evaluation dimensions (scale: 1 = limited, 3 = comprehensive). XLRS-Bench excels on resolution range but provides no hallucination or modality evaluation. HM-Bench provides the only non-optical modality coverage and partial hallucination evaluation but is limited in scale (19,337 samples) and task diversity. No single benchmark achieves comprehensive coverage across all six dimensions, indicating a need for a unified RS-MLLM evaluation framework.

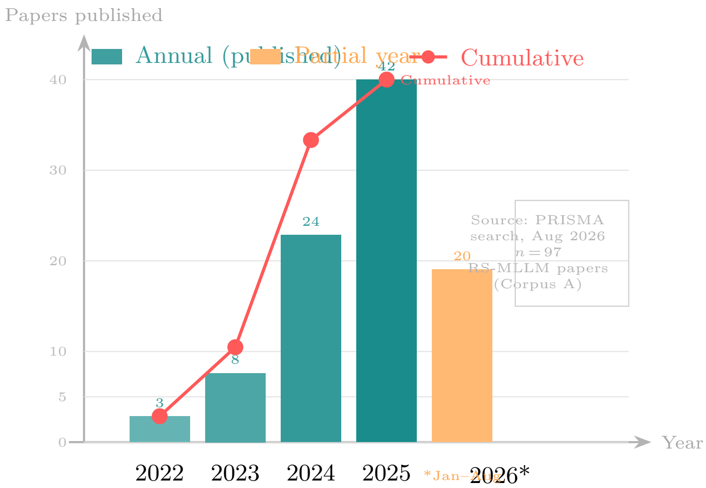


Fig. 25 Annual publication volume of RS-MLLM-specific studies identified in the systematic corpus (2022–2026). **These 97 papers constitute Corpus A**, the primary RS-MLLM synthesis corpus: 31 model studies, 18 benchmark studies, 14 dataset studies, 9 hallucination studies, and 25 foundation-model papers published within the 2022–2026 search window. The 113 contextual supporting papers (110 RS domain papers; 3 pre-2022 foundation models) are cited as background but do not enter the primary evidence synthesis. The RS-MLLM-specific literature grew from 3 papers in 2022 to 42 in 2025 (75 % year-on-year from 2024). Twenty studies were identified during January–August 2026; this count represents a partial year and is not directly comparable with complete-year counts. All counts from the authors' PRISMA-compliant systematic search, conducted 14 August 2026

Table 23 Preprint sensitivity analysis. Analysis A: full 210-paper retrieved corpus. Analysis B: peer-reviewed papers only ($n \approx 118$, i.e., papers published in journals or refereed conference proceedings). All figures are percentages of the respective corpus unless stated. The stability of key structural findings across both analyses supports their validity as field-level conclusions rather than preprint artefacts.

Finding	**A (all retrieved, n=210)**	**B (peer-reviewed, n≈118)**
Optical-only models	84.6 %	84.7 %
SAR coverage	11.9 %	12.4 %
HSI coverage	2.9 %	3.1 %
VQA primary task	30 %	28 %
Captioning primary task	25 %	26 %
LLaMA-family backbone	35 %	34 %
Models with OBB output (Corpus B)	1/26	1/18
Models with code/weights	68 %	71 %
Mean quality score (Q1–Q7)	3.77/7	4.12/7
Conclusion: principal structural findings stable across both analyses		

Analysis B excludes arXiv preprints and non-refereed technical reports. The higher mean quality score in Analysis B (4.12 vs 3.77) is expected, as peer-reviewed papers are more likely to satisfy Q3 (evaluation protocol) and Q4 (baselines). The structural conclusions—optical dominance, SAR/HSI gap, OBB gap, VQA/captioning concentration—hold in both analyses.

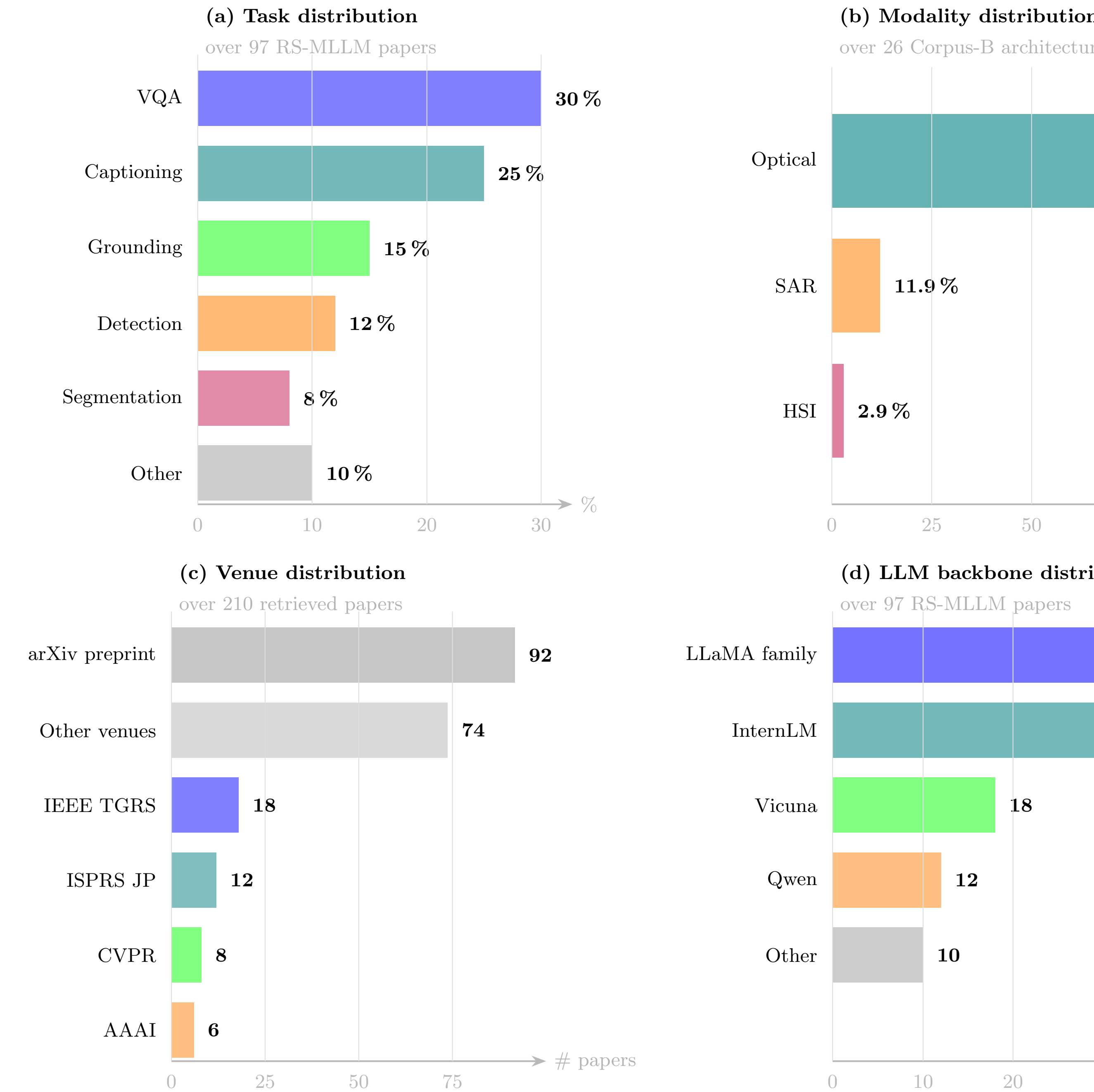


Fig. 26 Four-panel structured quantitative synthesis of the 97-paper primary Corpus A. Panels use different denominators as appropriate. **(a) Task distribution** (over 97 RS-MLLM-specific papers): VQA (30 %) and captioning (25 %) dominate; detection (12 %) and segmentation (8 %) are under-represented relative to their operational importance. **(b) Modality distribution** (over Corpus B: 26 architectures): 84.6 % optical, 11.9 % SAR, 2.9 % HSI. **(c) Venue distribution** (over all 210 retrieved papers): arXiv preprints (92) dominate; IEEE TGRS (18) and ISPRS JP (12) are the leading peer-reviewed venues. **(d) LLM backbone distribution** (over 97 RS-MLLM-specific papers; models may cite multiple backbones): LLaMA family (38 instances) is most common, followed by InternLM [54] (29) and Qwen [55] (12). All counts from the authors' PRISMA-compliant systematic search, conducted 14 August 2026

Table 24 Operational Readiness Matrix for RS-MLLMs. **Current RS-MLLM**: capability level across published models as of August 2026. **Operational requirement**: capability level needed for reliable geospatial deployment (disaster response, infrastructure monitoring, agricultural management, border surveillance). **Gap**: difference between current and required. Ratings: Very High / High / Moderate / Low / Emerging. Evidence tier per Sect. 2.4: A = peer-reviewed controlled; B = peer-reviewed benchmark; C = preprint; D = conceptual.

Capability	Current RS-MLLM	Operational requirement	Gap	Key evidence	Tier
VQA (optical)	Strong (95.2 % RSVQA-LR)	Strong	Low	Earth-OneVision RSVQA-LR score [20]	C
Captioning	Strong (CIDEr 91.4)	Moderate	Low	Earth-OneVision RSICD CIDEr [20]	C
HBB Grounding	Moderate (Acc@0.5 84.6)	High	Medium	RSVG benchmark [168]	B
OBB Detection	Weak (28.6 zero-shot mAP)	Critical	Very High	Specialist: 75.9; RS-MLLM: 28.6 [40, 143]	A
SAR reasoning	Weak (no physics encoder)	Critical	Very High	No SAR-physics RS-MLLM published [129]	C
HSI reasoning	Weak (all 18 models fail)	Critical	Very High	HM-Bench: systematic failure [21]	B
Temporal reasoning	Emerging (Earth-OV temporal; VLRS-Bench)	Critical	Very High	Earth-OneVision [20]; VLRS-Bench [99]	C
Uncertainty output	None (no calibrated output)	Critical	Very High	Not addressed in any RS-MLLM	D
Safety / alignment	Emerging (1 paper)	Critical	Very High	RemoteShield [200]; RLHF [106]	C

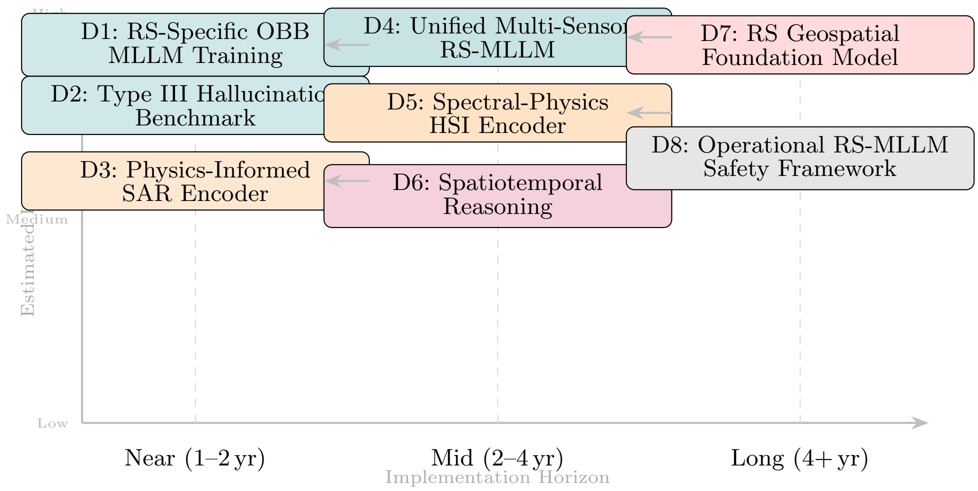


Fig. 27 Future research roadmap for RS-MLLMs, organised by implementation horizon (near: 1–2 years; mid: 2–4 years; long: 4+ years) and estimated impact on operational geospatial intelligence. Near-term directions (D1–D3) address structural gaps identifiable within the current architectural paradigm; long-term directions (D7–D8) require new paradigms and research infrastructure.

Table 25 Summary of eight future research directions for RS-MLLMs. "Gap" column references the relevant section; "Foundation" lists the methodological building block most directly applicable. Cited prior work (ODFE [48], SM^2-LF [174], Rayleigh [183]) provides methodological foundations that make these directions tractable; these are starting points, not completed solutions.

Dir.	Title	Gap (Sect.)	Foundation
D1	OBB-Capable MLLM	Sec. 3.6, Sec. 4.4.3	Ding et al. [39]; ODFE [48]; SM^2-LF [174]
D2	Type III Hallucin. Bench	Sec. 7.5	HM-Bench [21]; DFC 2018 [184]
D3	Physics SAR Encoder	Sec. 5.3.1, Sec. 7.3.3	Schmitt et al. [42]; Rayleigh [183]
D4	Unified Multi-Sensor MLLM	Sec. 5.6	Earth-OneVision [20]; D3, D5
D5	Spectral HSI Encoder	Sec. 5.4.1	HM-Bench [21]; bioucas2013 [176]
D6	Spatiotemporal Reasoning	Sec. 4.5	TEOChat [120]; Change-Agent [146]; SM^2-LF [174]
D7	RS Foundation Model	Sec. 3.7	Prithvi-100M [220]; InternVL [222]
D8	Safety Framework	Sec. 7.1	RSHallu [196]; RemoteShield [200], RLHF approaches [106],